

Spatial scale space from hemigroup axioms: the admissible cone, its corners and their implementation

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Abstract

Gaussian scale space is the classical answer to the question of which smoothing kernels can serve a multi-scale representation. A companion paper showed that on the line the answer is much larger when the semigroup requirement of the axioms is weakened to a cascade, a two-parameter hemigroup: the admissible kernels are the symmetric self-decomposable laws, each given by a Gaussian coefficient and a nonincreasing profile of displacement sizes. This paper describes that class for a reader who computes with it, in three steps. The first is its shape. The admissible exponents form a convex cone in which every member is a unique superposition of the Gaussian and the members whose profile is a unit step. Subordination maps the time-causal scale spaces of the same axioms injectively into the cone, and the image consists of the members whose radial extension is self-decomposable in every dimension. The second is the familiar families in it: the Matérn kernels of spatial statistics, the Student-t laws and the symmetric stable laws are located and compared by their moments, their tails and their behaviour at the origin. The third is computation. For signals on the line the cascade composes exactly over any finite set of scales, and the Matérn families of integer index are exactly those whose stages are cascades of identical first-order recursive filters, run forward and backward. The families computed in this way do not reach the stable members even in the limit, and the sampling of the signal is a separate approximation. A numerical example checks the statements and finds that the continuous prototypes of two recursive Gaussian filters in common use are the kernels of no such cascade. Locality and the non-creation of local extrema are outside the paper. Most of the results of the first two steps are machine-checked in Lean 4, and those of the third are proved in prose.

1 Introduction

A scale-space representation replaces a signal by a family of smoothed versions of it, one for each scale, so that structure can be read at the scale at which it lives. The classical theory says which smoothing to use. Its requirements are those of a measurement made without prior knowledge of the signal: linearity, covariance under translations and dilations, positivity, that a smoothing of a smoothing is again a member of the family, and a requirement of locality, in one of its forms: that the evolution in scale is generated by a differential operator, or that no new structure appears as the scale grows. Under them the kernel must be the Gaussian, and the scale space is the solution of the heat equation ([33]; [35]; [62] for the history). The last requirement has a precise form, the *semigroup* property $K_s K_t = K_{s+t}$ of the smoothing operators K_t , and it is stronger than its motivation. It asks that a smoothing of a smoothing be a member, and it asks in addition that the result depend only on the total amount of scale crossed. [44] showed what the axioms give once the requirements of locality are set aside: the kernels with transform $e^{-t|\omega|^\alpha}$, the symmetric stable laws for $0 < \alpha \leq 2$. Every one of them other than the Gaussian has infinite variance.

A companion paper, called the line paper below ([17]), kept the axioms of Pauwels et al. and weakened the semigroup to what its motivation says. The smoothing operators are indexed by pairs of scales. Call the smoothed signal at a scale the *record* at that scale. The operator $\Phi_{s,t}$ takes the record at scale s to the record at scale t , and the operators are required to compose,

$$\Phi_{s,t} \Phi_{r,s} = \Phi_{r,t}, \quad r \leq s \leq t,$$

a two-parameter *hemigroup*, or cascade. How the stage from s to t depends on the two scales is left to the other axioms. The main theorem of that paper says what they allow. In the parametrization of scale in which dilating the signal by λ multiplies the scale by λ , called canonical below, the kernel at scale t has the Fourier transform $e^{-F(t\omega)}$, with

$$F(\omega) = a\omega^2 + \int_0^\infty (1 - \cos \omega x) \frac{k(x)}{x} dx, \quad a \geq 0, \quad k \geq 0 \text{ nonincreasing},$$

and every such F occurs. The two data have a direct meaning. A symmetric kernel is the law of a random displacement of the point at which the signal is read. The coefficient a is the Gaussian part of that displacement, and the *profile* k describes its jumps: displacements of size about x occur at the rate $k(x) dx/x$. The Gaussian scale space is $k = 0$. A profile that is a pure power gives the stable kernels of Pauwels et al. The exponential profile $k(x) = 2\gamma e^{-x}$ gives the kernels with transform $(1 + t^2\omega^2)^{-\gamma}$, which for $\gamma > \frac{1}{2}$ are the Matérn covariance functions of spatial statistics. They have every moment finite, and no semigroup that is covariant under dilation contains them; in the order γ at a fixed range they do form a semigroup, the Bessel scale space discussed below. In probabilistic terms the admissible kernels are the symmetric *self-decomposable* laws: those of a random displacement that is, for every contraction ratio, a contracted copy of itself plus an independent remainder ([50], Def. 15.1, p. 90). The monotonicity of k is the whole restriction.

So the class is large, and the line paper stopped at its classification. For an imaging reader the gain is concrete. The larger class contains kernels with a finite variance and exponential tails that are computed exactly by cheap recursive filters, which the stable kernels are not. This paper describes the class, by the questions that a user of it would ask. What does the class look like as a whole? How does it relate to the time-causal theory, where the same weakening was carried out first? Which of its members are already known, and what distinguishes them? What equation does a member solve? And how is a member computed? §2 restates the line paper's axioms, notation and theorem in the form used here, so that this paper can be read on its own.

The shape of the class (§3). The admissible exponents F form a convex cone, and the cone has simple building blocks. The simplest profile is a unit step, displacements of every size up to a cutoff τ and none beyond, and its exponent is a classical special function, the entire cosine integral $\text{Cin}(\tau\omega)$ ([15], §6.2). A nonincreasing profile is a stack of unit steps, so every admissible exponent is a Gaussian term plus a superposition of these, and the superposition is unique (Proposition 3.3, Figure 2). The Gaussian and the unit-step members are the extreme rays of the cone, and the named families are slices through it. The kernel of a unit-step member has no closed form, but it satisfies a delay equation (Lemma 3.5, Figure 1). The equation shows that a hard cutoff in the profile leaves echoes in the kernel at the multiples of τ . The section ends with a practical criterion. An admissible kernel with no Gaussian part is bounded at the origin exactly when its profile starts above 1 (Corollaries 3.9 and 3.10, the second for a profile that is unbounded at the origin). For the Matérn family this is the known condition $\gamma > \frac{1}{2}$.

The relation to the time-causal theory (§4). The same weakening of the semigroup was first carried out for signals in time, where a measurement may use the past and not the future ([18], called the causal article below). There the kernels are the laws of random delays on the half-line. The two theories are connected by Bochner's subordination: run a Brownian motion for a random causal delay, and the displacement reached has a symmetric law. The section makes

this a map from the causal class into the spatial one, family to family, under which the causal scale is the square of the spatial one (Proposition 4.5, Figure 3). The map is injective, since the image determines its ancestor, and it is not onto. A kernel in its image has an exponent whose derivative is positive at every positive frequency, and the exponent of a unit-step member has stationary points, so the unit-step members have no causal ancestor (Proposition 4.6). The spatial theory is therefore strictly larger than the causal theory carried across. The image is the intersection of a decreasing chain of classes indexed by dimension, the exponents whose radial extension to $\mathbb{R}^d$ is again self-decomposable: a member has a causal ancestor exactly when it belongs to every class of the chain (Proposition 4.9, proved in prose on cited facts about laws in dimension d , which are listed before it). One explicit family shows that the first two inclusions of the chain are strict (Lemma 4.12, proved here and not machine-checked).

The named members (§5). Three classical families are located in the cone (Figure 5). They are called its corners here, in the sense that each is singled out by an extremal property; they are not extreme rays of the cone, which are the Gaussian and the unit-step members. The main result concerns the Matérn family. Four descriptions single it out: a profile that is a single exponential, a mixing measure that is a single atom with no Gaussian part, the transform $(1+t^2\omega^2)^{-\gamma}$ up to the choice of a range, and, for integer γ , stages whose transfer function is the γ -th power of one ratio of first-order factors (Theorem 5.4; the family is read in its canonical gauge). Its members of integer index are exactly the admissible families whose stages, over every finite ladder of scales, are cascades of identical first-order recursive sections, run forward and backward. One kernel from the origin of that form already forces the family, and the word that matters is identical: there are admissible families whose stages are cascades of first-order sections of different ranges. The Matérn members have every moment finite and exponential tails (Corollary 5.5). The Student-t laws are the image of one of the causal families under the bridge, with polynomial tails whose order is set by the degrees of freedom (Proposition 5.10). The symmetric stable laws are the members for which the stages depend only on the difference of their scales, and the heavy tails of all of them but the Gaussian are the price of that requirement (Proposition 5.12). Two general facts make the comparison possible. The number of finite moments of a kernel and the weight of its tail are read off the tail of the profile (Proposition 5.1). And all three families lie in one subcone, the profiles that are mixtures of exponentials, on which the bridge is a bijection (Proposition 5.7). For the Student-t family the membership rests on a theorem of Halgreen's that is taken as a hypothesis.

The evolution equation (§6). For a Schwartz signal, every member's scale space solves an equation $\partial_t u = \mathcal{A}_t u$. The generator is a Laplacian for the Gaussian coefficient plus, for every displacement size in the catalogue, the amount by which the average of the signal's two neighbours at that distance exceeds the signal (Proposition 6.2, Figure 6). The heat equation and the fractional heat equation of the α -scale spaces ([16]) are recovered for the Gaussian and the stable family. The Matérn family has a bounded generator, and a unit-step member has a single second difference (Proposition 6.4).

The algorithm (§7). The hemigroup is also how a member is computed. Over any finite ladder of scales, the record at each scale is obtained from the record at the previous one by one stage, and the result is the scale space itself, with no approximation in the scale direction (Proposition 7.1, Figure 7). For the Matérn family with integer γ a stage is γ pairs of first-order recursive filters, each pair with one filter run forward along the signal and one backward (Proposition 7.2). This is the kind of filter long used to approximate the Gaussian ([13, 65]), and the theory says which of those filters are themselves scale spaces (Proposition 7.3, read at the two designs in Remark 7.4). The members whose profile is a mixture of exponentials, the stable family among them and the Student-t family under the hypothesis named above, are limits of members with finitely many ranges, and a member with finitely many ranges and integer weights has such stages (Proposition 7.7). The families realized by such sections reach, in the limit, exactly the members whose Thorin measure consists of atoms of even integer mass,

finitely many in every bounded set, possibly with a Gaussian part. Every such limit has an exponential moment, so the stable and the Student-t members, which lack moments, are not among the limits, and this needs no hypothesis (Proposition 7.8). The kernels from the origin of the limits are the symmetric Pólya frequency functions of Schoenberg’s theorem (Remark 7.9). A numerical run closes the section (Example 7.12). It checks the statements on a sampled signal, it measures how far the members with such stages stay from a member like the Poisson scale space, and it finds that neither of the two recursive Gaussians in common use is the kernel of an exact cascade.

Relation to the literature. The families at the corners are classical, and the section on each says where. The Matérn kernels are the covariance functions of [63, 24, 47]. As a scale-space family composed along the smoothness and not along the range they are the Bessel scale space of [9], who also dilate the kernels at a fixed order; what the hemigroup adds there is the stage between two ranges, which makes the family in the range a cascade. The same authors’ relativistic scale spaces ([10]), with the exponent $\sqrt{\omega^2 + m^2} - m$, are a further non-Gaussian semigroup in imaging with exponentially decaying kernels; their scale parameter is not the dilation of the kernel, so they are covariant in a different sense from the one required here. In probability the Matérn kernels are the variance-gamma laws of [41, 22]. The mixtures of exponential profiles are the symmetric part of the generalized gamma convolutions of Thorin and Bondesson ([8]), and the Student-t corner rests on Halgreen’s theorem that the generalized inverse Gaussian laws belong to that class ([25]). The subordination bridge is Bochner’s, and its Bernstein-function calculus is that of [52], where Schoenberg’s radial theorem, used for the chain of classes by dimension, is also stated. The Gaussian variance mixtures with an infinitely divisible mixing law, the type G laws, are the class the image of the bridge sits inside ([42]; [56]). Self-decomposable laws and self-similar additive processes are the probability classes of [50, 49]. Hemigroups appear in the probability-on-groups literature ([28, 54, 27]), and the random integral representation of a self-decomposable law ([30]; [7] for hemigroups) is the closest in shape to the bridge, though it starts from no axioms on measurement. In the scale-space literature the nearest construction is Lindeberg’s time-causal limit kernel ([37]), an infinite cascade of first-order filters over a geometric ladder of time constants, of which the pyramid of §7.3 is the two-sided form. The α -scale spaces of [16] are the evolution equation of the stable corner, and the Poisson scale space of [20] is its member of index 1.

What this paper adds is the geometry of the class as a consequence of the axioms, and the bridge as a map between two theories with its strictness proved. It adds the characterization of the Matérn family by its exact implementation, the evolution equation of every member, and the implementation as a theorem about the cascade.

Scope. Locality of the evolution and the non-creation of local extrema across scale, the requirements from which the Gaussian axiomatics select one member, are not treated here. They are the subject of the next module, and nothing below depends on them. The paper stays on the line and in the L^1 signal model, and it makes no empirical claim about natural signals. Most of the results of §§3–6 are machine-checked in the Lean 4 proof assistant, and the results of §7 are proved in prose. §1.1 says which, and what they rest on. The remaining sections mention the formalization only where a statement or a step is cited and not proved. §8 collects what is settled and what is left open.

1.1 What is machine-checked

Most of the results of §§3–6 have been formalized in the Lean 4 proof assistant, on top of its mathematical library Mathlib, in the same development as the line paper’s. The exceptions are listed below. This subsection is self-contained, and a reader interested in the mathematics alone can pass to §2.

What is proved. Section by section:

- of §3, every proposition, lemma and corollary but the last: the Cin rays and the superposition, the Choquet structure in all its parts, the delay equation in both forms, the folding translation, and the behaviour at the origin with the boundedness threshold for a profile that is finite there;
- of §4, everything but the dimension filtration and its lemma: the bridge on exponents in both clauses, the bridge on families in its five clauses, that of the Student-t family being the mixture identity, with the self-decomposability of its delay law as a hypothesis, the strictness proposition in all four clauses, and the memberships of the Gaussian ray and the Matérn, Student-t and stable families, the last under its hypothesis;
- of §5, the Matérn theorem in its four clauses, the Matérn exponent, transforms and moments, the Thorin subclass in its representation, its bridge and its strictness, the two-sided reading of Thorin's exponent, the Student-t family in its kernel, its transform and its Thorin clause under its hypothesis, and the symmetric stable family in all its parts; and of the moments-and-tails proposition, the variance formula, the completely monotone case and the Gaussian exclusion;
- of §6, everything. The generator of the Thorin members uses the uniqueness of the Lévy–Khinchine pair, through the representation of §5.3; the rest of the section rests on Lean's logical foundations alone.

The headline theorem is assembled in one declaration, so that it is verified as it is printed.

What it rests on. Beyond Lean's logical foundations, the results above use the cited facts of the line paper and three more. The line paper's are Bochner's theorem in the forward direction, the symmetric Lévy–Khinchine representation at its converse and uniqueness clauses, and the regularity of a self-decomposable density. The three further facts are each cited with a page reference checked against the source where they are used.

1. *Bernstein's theorem, the existence equivalence only.* A function on the half-line is completely monotone if and only if it is the Laplace transform of a positive measure ([52], Thm. 1.4, p. 3). The uniqueness of the measure, which the source states with it, is proved and not cited (Proposition 2.9).
2. *The moment criterion for infinitely divisible laws and the tail floor for a bounded jump support* ([50], Thm. 25.3, p. 159, and Thm. 26.1, p. 168), the second at its divergence branch alone and at a lower bound for the radius.
3. *The behaviour of a self-decomposable density at the origin* ([50], Thm. 28.4, p. 191, and Thm. 53.8, pp. 410–411), three regimes admitted as three named facts at the source's own letter, over a two-sided profile and its two one-sided limits.

Further cited facts are used by statements that are proved in prose and not machine-checked. Proposition 4.9 uses Schoenberg's radial theorem ([52], Thm. 13.14, p. 212), in two of its clauses, with the Lévy–Khinchine representation and the polar criterion of self-decomposability in dimension d and the cited facts listed before it, and the third clause of Lemma 4.12 uses the same two facts in dimension d . Proposition 7.8 uses Bondesson's closure theorem for the generalized gamma convolutions ([8], Thm. 3.1.5, pp. 34–35), with the continuity theorem for Laplace transforms ([19], Vol. 2, §XIII.1, Thm. 2, p. 431), and the moment criterion a second time. Corollary 3.10 uses a further clause of the theorem on the density at the origin ([50], Thm. 28.4(ii), p. 191), which the development admits nowhere. Table 1 lists every cited fact with the statements that use it directly.

The three names of the fact on the density at the origin carry the same steps beyond their pages. The profile is any monotone two-sided profile and not the source's version of it that

Table 1: The cited facts and their direct uses, with the entry of the development’s ledger that carries each. A numbered proposition, lemma, theorem or corollary that is not listed uses no cited fact directly, apart from the regularity of a self-decomposable density, which enters through Proposition 2.8. Cited and used by no machine-checked proof: the closed form of the Matérn kernel and the Bessel asymptotics behind its tail (ledger entries A15 and A16, Proposition 5.3, from which Corollary 5.5 reads its exponential tails); Thorin’s definition of the extended generalized gamma convolutions (A17, Lemma 5.9); Halgreen’s theorem on the inverse-gamma law (A18, Remark 5.11). Cited in remarks, for orientation or attribution: the identification of the constant of Lemma 3.1 ([15], Remark 3.2), the background driving process ([50], Thm. 17.5), and Schoenberg’s theorem on Pólya frequency functions (A21, Remark 7.9). The rows below the rule are used only by statements proved in prose.

cited fact	entry	clause used	used directly by
Bochner’s theorem	A1	forward direction	Prop. 4.5
symmetric Lévy–Khintchine representation	A3	converse uniqueness	Prop. 4.5 Prop. 3.3; Thm. 5.4; Prop. 5.7; Prop. 5.10(3); Prop. 6.4
Bernstein’s theorem	A11	existence equivalence	Prop. 5.7; Prop. 5.1(2), last sentence; Prop. 5.10(3)
the moment criterion	A13	as stated	Prop. 5.1(1), which is the fact; Prop. 5.12
the tail floor	A14	divergence, reached radius	Prop. 5.1(2), last sentence
the density at the origin	A10	three regimes, at the letter	Prop. 3.7; Cor. 3.9
the density at the origin	A10	the clause at $k(0+) = \infty$	Cor. 3.10
the moment criterion	A13	exponential weight	Prop. 7.8(5)
Lévy–Khintchine in dimension d	A3	converse, uniqueness	Prop. 4.9; Lem. 4.12(3)
the polar criterion in dimension d	A7	both directions	Prop. 4.9; Lem. 4.12(3)
the radial theorem	A23	Bernstein from every dimension	Prop. 4.9(3), (4)
closure of the gamma convolutions, with continuity of Laplace transforms	A24	closure, vague convergence	Prop. 7.8(1), (4)

is continuous from one side, which differs from it on a countable set and gives the same law. The hypothesis is this paper’s cosine-transform specification of a symmetric law and not the source’s self-decomposable law, a passage through the Lévy–Khintchine representation already in the trust base. The conclusion is a two-sided comparison on a punctured neighbourhood with an existential representative, against the source’s comparison function with its slowly varying factor. That is less than the source’s asymptotic with its constant because the constant is not zero, which holds for a symmetric law. The passage between the source’s two-sided convention and this paper’s folded one is a theorem beside them (Lemma 3.6). An earlier version of the development had admitted the singular regime without the slowly varying factor, which made the admitted fact false; an external review found it, and the fact is now stated as the source states it ([50], (53.25) and (53.28), p. 411). In print the repair added the slowly varying factor and the comparison function of the threshold regime to the statement of Proposition 3.7. The threshold of Corollary 3.9 and every later result were unchanged.

What is not formalized. The following, in the scope of this paper, and the text names each where it occurs.

- The smoothness of the kernel when the profile is unbounded at the origin, proved in prose on a clause of a cited theorem (Corollary 3.10).
- The moment criterion at bounded jump support and the finiteness branch of the tail floor (Proposition 5.1, two clauses).

- The closed form of the Matérn kernel and its tail (Proposition 5.3).
- The Thorin subclass as Thorin’s own class, which is the reading of a definition (Lemma 5.9).
- The moment count of the Student-t family (Proposition 5.10(4)).
- The dimension filtration, a proposition proved in prose on cited facts about laws in dimension d , of which this development has none (Proposition 4.9).
- Its lemma, of which two clauses are stated in the development and not proved there, and the third is proved in prose (Lemma 4.12).
- The whole of the implementation section, §7, where no machine-checked proof is scheduled, as for the causal article’s, and the corollary of the Matérn theorem that rests on it (Corollary 5.5). Its statements are proved in prose from the statements of this paper, except Proposition 7.8, which is proved on cited facts.

Every other numbered proposition, lemma, theorem and corollary is machine-checked. Who has read the statements of this list is said in the statement on the use of AI at the end of the paper. The remarks are commentary. One of them describes a computation that is not carried out (Remark 3.8), and two carry the cited facts that ground hypotheses and not statements (Remarks 4.3 and 5.11).

Where to look. The development and its axiom ledger, with page-verified citations for every cited fact, are at the repository accompanying this paper, <https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-cone>, which is deposited with this version of the paper under the DOI printed on the first page. It is an export from the development repository restricted to what the statements of this paper rest on: the Lean modules, the trust boundary and the ledger, and the sources of the blueprint chapters behind §3–§7, in which the declaration behind each numbered statement is named in a `\lean` tag on that statement. The toolchain is pinned in-tree, so the verification is reproducible with one command, `lake build in Formalization/`. A second, `lake env lean CIAxiomGuard.lean`, prints the axiom usage of every named declaration, to be read against the trust-boundary file beside it. On the declaration `SpatialLine.matern_theorem`, the headline of §5, it prints Lean’s three logical axioms and the uniqueness clause of the Lévy–Khintchine representation, and nothing else:

```
'SpatialLine.matern_theorem' depends on axioms: [propext,
Classical.choice, Quot.sound,
SpatialLine.fourier_toolbox_levy_unique]
```

The strictness of the bridge, Proposition 4.6, prints the three logical axioms alone in all five of its declarations.

2 What the line paper settled

This paper continues the line paper [17], and is written to be read on its own. The main things the later sections use of that paper are restated in this section: its axioms, the form of its exponents, and its characterization theorem, together with the notation. A restated statement carries the line paper’s statement number in its title, and its proof is at that number. Further results are used and only cited, in the form “Lemma 7.1 of [17]” with the number of that paper’s first release. Four are used throughout: the Fourier toolbox (Bochner’s theorem and the Lévy–Khintchine representation, Proposition 2.3 of [17]), the quadratic growth of an exponent (Lemma 2.10 of [17]), the uniqueness of the transform of a measure (Proposition 2.7 of [17]),

and Lévy's continuity theorem (Proposition 2.8 of [17]). The others are used once or twice each: the differentiability of a self-decomposable exponent and its representation (Lemma 7.1 of [17], in §§6 and 7), the two members that are not stable (Proposition 3.3 of [17], in §§4 and 5), the zero set of an exponent and the absence of lattice kernels (Lemmas 2.11 of [17] and 6.2 of [17], in §4), and the additivity of the stage exponents with the nonvanishing of the transforms (Lemmas 5.1 of [17] and 4.2 of [17], in §7). One classical inequality is used from inside a proof of the line paper, that of Theorem 5.2 of [17], and is restated here because Lemma 3.5 rests on it. For a symmetric probability measure μ and $\eta > 0$,

$$\int \left(1 - \frac{\sin \eta x}{\eta x}\right) \mu(dx) = \frac{1}{\eta} \int_0^\eta (1 - \hat{\mu}(\omega)) d\omega, \quad 1 - \frac{\sin u}{u} \geq \frac{2}{3\pi^2} (1 \wedge u^2), \quad (2.1)$$

the first by Fubini's theorem and the second elementary. Together they give the truncation inequality $\mu\{|x| \geq 1/\eta\} \leq \frac{3\pi^2}{2} \eta^{-1} \int_0^\eta (1 - \hat{\mu})$. When $\hat{\mu} = e^{-F}$ with $F(\omega) \leq C\omega^2$ near the origin, the right-hand side is $O(\eta^2)$, since $1 - e^{-F} \leq F$, so $\mu\{|x| > R\} = O(R^{-2})$ and μ has a finite first absolute moment. The section ends with one elementary proposition of the line paper's preliminaries that the line paper did not print, because its uses are in this module; that one is proved.

2.1 Notation

Space runs over $\mathbb{R}$ with coordinate x , the wavenumber is ω , and scale is t . Table 2 is the notation table shared by the spatial modules, the line paper's Table 2 with the letters of the bridge added; the letters of §§5–7 are introduced where they are used.

Spaces. $L^1 = L^1(\mathbb{R}, dx)$, real, with positive cone L_+^1 and $\int f := \int_{\mathbb{R}} f(x) dx$; $M_+(\mathbb{R})$ is the finite positive Borel measures on $\mathbb{R}$, of total mass $\|\mu\| = \mu(\mathbb{R})$. *Actions.* Translations are $(T_a f)(x) = f(x - a)$. The reflection is $(Rf)(x) = f(-x)$, extended to measures as the pushforward under $x \mapsto -x$, and a measure is *symmetric* when $R\mu = \mu$. Dilations act on functions by $(D_\lambda f)(x) = \lambda^{-1} f(x/\lambda)$, $\lambda > 0$, which preserves mass, and on measures by the pushforward under $x \mapsto \lambda x$. *Convolution and transform.* $(\mu * f)(x) = \int f(x - y) \mu(dy)$ for $\mu \in M_+(\mathbb{R})$ and $f \in L^1$. Both actions are compatible with convolution. The transform of a symmetric measure is $\hat{\mu}(\omega) = \int \cos(\omega x) \mu(dx)$, real and even, and the transform of a stage's kernel is written $\hat{\mu}_{s,t}$. It is the restriction to symmetric measures of $\hat{\rho}(\omega) = \int e^{-i\omega x} \rho(dx)$, the convention for a finite measure, positive or signed, that is not symmetric. The sign matters in one place, the proof of Lemma 3.5, where a translate has the transform $e^{i\omega\tau} \hat{\nu}(\omega)$, and the uniqueness of the transform (Proposition 2.7 of [17]) is stated in the line paper for every finite Borel measure on $\mathbb{R}$ in this convention.

2.2 The axioms

The line paper's object is a family of operators indexed by pairs of scales. A pair of scales $s \leq t$ is a *stage*: the operator $\Phi_{s,t}$ of the stage takes the signal at scale s to the signal at scale t , and its kernel $\mu_{s,t}$, which the line paper's representation lemma supplies (its Lemma 4.1), is the law of the random displacement the stage adds. The axioms are these.

Definition 2.1 (symmetric cascade measurement family, Definition 3.1 of the line paper). A family of linear operators $\Phi_{s,t} : L^1 \rightarrow L^1$, $0 \leq s \leq t < \infty$, is a *symmetric cascade measurement family* if:

- (A1) *Boundedness.* Each $\Phi_{s,t}$ is a bounded operator on L^1 .
- (A2) *Translation covariance.* $\Phi_{s,t} T_a = T_a \Phi_{s,t}$ for every $a \in \mathbb{R}$.
- (A3) *Reflection symmetry.* $\Phi_{s,t} R = R \Phi_{s,t}$.

Table 2: Notation: the symbol, where it is defined, and the letter the causal article uses for the same object. The rows above the rule are the line paper’s Table 2; the rows below it are the letters of the bridge, §4, where the causal article’s Laplace variable is written σ , s being a scale here. Brownian motion is written W , the letter B being the symbol of the exponent.

object	here	the causal article
signal coordinate	x (space)	t (time)
transform variable	ω (Fourier)	s (Laplace)
scales in a stage	$r \leq s \leq t$	$x \leq y \leq z$
stage operator and its kernel	$\Phi_{s,t}, \mu_{s,t}$	$\Phi_{x,y}, \mu_{x,y}$
exponent of a stage	$g_{s,t}(\omega) = -\log \hat{\mu}_{s,t}(\omega)$	$g_{x,y}(s)$
kernel from the origin, accumulated exponent	$\mu_{0,t}, G(t, \omega) = g_{0,t}(\omega)$	$\mu_{0,x}, G(x, s)$
canonical gauge	$\chi(t)$ (§2.4)	$\chi(x)$
the family’s exponent and its pair	$F(\omega)$; Gaussian coefficient a , profile k	$F(s)$; drift b_0 , profile k
Lévy measure, tail measure of the profile	ν (2.2), $\varpi = -dk$ (2.4)	ν, ϖ
symbol of the exponent	$B(\omega) = \omega F'(\omega)$ (2.4)	$B(s) = sF'(s)$
dilation ratio, action, orbit coordinate	$\lambda, S_\lambda, c(\lambda)$	$\sigma, S_\sigma, c(\sigma)$
random displacement, random delay	$X_{s,t}$	$T_{x,y}$
function space	$L^1 = L^1(\mathbb{R}, dx)$	$X = L^1(\mathbb{R}, dt)$
translation, reflection	T_a, R	$\tau_a, —$
exponents by definition, by representation	ND_s, LE_s (§2.3)	BF_0, LE
standard Brownian motion, its law at time u	W, g_u	—
the causal exponent and its data	$F_1(\sigma)$; (b_0, k_1) (Def. 4.1)	$F(s)$; (b_0, k)
the random delay of the bridge	T_1 , with exponent F_1	$T_{0,1}$

(A4) *Positivity.* $\Phi_{s,t} L_+^1 \subseteq L_+^1$.

(A5) *Unit mass.* $\int \Phi_{s,t} f = \int f$ for every $f \in L_+^1$.

(A6) *Cascade (hemigroup).* $\Phi_{t,t} = \text{Id}$, and $\Phi_{s,t} \Phi_{r,s} = \Phi_{r,t}$ for all $r \leq s \leq t$.

(A7) *Continuity.* For every $f \in L^1$ the map $(s, t) \mapsto \Phi_{s,t} f$ is continuous from $\{0 \leq s \leq t\}$ to L^1 .

(A8) *Scale covariance.* There is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0, \infty)$ with $D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s, S_\lambda t} D_\lambda$ for all $0 \leq s \leq t$ and all $\lambda > 0$.

(ND) *Nondegeneracy.* $\Phi_{s,t} \neq \text{Id}$ whenever $s < t$.

A family satisfying the nine is called *admissible* throughout this paper. The word is also used of a pair (a, k) and of an exponent, in the senses fixed in §2.3 and after Theorem 2.5.

Hemigroup versus semigroup. The axiom that separates this class from the classical one is (A6). The scale-space axioms of [44] ask for a one-parameter family with $K_s K_t = K_{s+t}$, so that a stage depends on the difference $t - s$ of scales only: the family is a convolution semigroup, and the kernel at scale t is the law at time t of a process with stationary independent increments. Axiom (A6) asks only that the stages compose. The stage from r to t is the stage from s to t after the stage from r to s ; how the stage from s to t depends on the two scales is left free. The family is then the family of increment laws of a process with independent increments that need not be stationary in scale, and scale covariance (A8) fixes what those increments are: the process is self-similar, and its marginals are self-decomposable. That is the difference in the composition axiom, and it is the reason the class here has a profile k where the semigroup class has one index α (Corollary 2.6). The hemigroup form of the axioms comes from the time-causal article [18], whose stages run over time on the half-line; the line paper carried them to the reflection-symmetric line, and §4 is where the two meet.

2.3 Exponents and profiles

Under the axioms, every stage is a convolution, $\Phi_{s,t}f = \mu_{s,t} * f$, with a symmetric probability measure $\mu_{s,t}$ whose transform does not vanish, and the *exponent* of the stage is $g_{s,t} = -\log \hat{\mu}_{s,t}$ (the line paper's Lemmas 4.1 and 4.2). The exponents add along the cascade, and each is a symmetric continuous negative definite function. Such a function is a continuous even $\psi \geq 0$ with $\psi(0) = 0$ for which $e^{-\tau\psi}$ is the transform of a symmetric probability measure for every $\tau > 0$. They form the class ND_s of the line paper's Definition 2.2, whose members are exactly the functions below. This is the symmetric Lévy–Khintchine representation, clause (3) of Proposition 2.3 of [17], which asserts the representation, its converse and the uniqueness of the pair:

$$\psi(\omega) = a\omega^2 + \int_{(0,\infty)} (1 - \cos \omega x) \nu(dx), \quad a \geq 0, \quad \int_{(0,\infty)} (1 \wedge x^2) \nu(dx) < \infty, \quad (2.2)$$

with the pair (a, ν) unique. The line paper names the class by this representation, LE_s (its Definition 2.5), so that $\text{ND}_s = \text{LE}_s$, and this paper uses both names for the one class. In the representation a is the *Gaussian coefficient* and ν the *Lévy measure*. The measure ν lives on the positive half-line by a convention that matters for constants: the Lévy measure of a symmetric law on $\mathbb{R}$ is a symmetric measure ν_2 on $\mathbb{R} \setminus \{0\}$, and ν is its image under $x \mapsto |x|$, the two directions of a displacement of a given size counted together.

A symmetric law μ is *self-decomposable* (the line paper's Definition 2.12) if for every $0 < c < 1$ there is a law ρ_c with $\mu = (D_c\mu) * \rho_c$: a random displacement with law μ is, for every contraction ratio, a contracted copy of itself plus an independent residual. The characterization theorem below says that the kernels of an admissible family are of this kind, and this is the class that has a profile.

Proposition 2.2 (symmetric self-decomposable exponents, Proposition 2.13 of the line paper). *A symmetric infinitely divisible law with pair (a, ν) is self-decomposable if and only if its Lévy measure has the form $\nu(dx) = k(x)x^{-1}dx$ on $(0, \infty)$ with $k : (0, \infty) \rightarrow [0, \infty)$ nonincreasing, so that its exponent is*

$$\psi(\omega) = a\omega^2 + \int_0^\infty (1 - \cos \omega x) \frac{k(x)}{x} dx, \quad a \geq 0, \quad k \geq 0 \text{ nonincreasing}. \quad (2.3)$$

Self-decomposability imposes no restriction on the Gaussian coefficient a .

The function k is the *folded profile* of the law, or its profile for short. It is folded because ν is: a two-sided Lévy density written $h(|x|)/|x|$, the form of [50], Cor. 15.11, p. 95, has folded profile $k = 2h$, and the factor is the one place a constant is easily lost in carrying a statement from a source into this paper. The profile is the object of this paper. It is a nonincreasing function on the half-line, and the representation reads it as an intensity: displacements of size about x arrive at rate $k(x)x^{-1}dx$, so k is that rate per unit of $\log x$, and self-decomposability says that the intensity of a displacement never grows with its size. The Gaussian is the member with no profile, $k = 0$; every other member is described by the shape of k , and §3 describes the set of all shapes, §4 the shapes that come from a causal delay, §5 the shapes of the named families.

The integrability condition in (2.2), read for a profile, is the following criterion.

Lemma 2.3 (the integrability criterion for a profile, Lemma 2.15 of the line paper). *Let $k : (0, \infty) \rightarrow [0, \infty)$ be measurable and put $\nu(dx) := k(x)x^{-1}dx$. Then*

$$\int_{(0,\infty)} (1 \wedge x^2) \nu(dx) < \infty \quad \Longleftrightarrow \quad \int_0^1 x k(x) dx < \infty \quad \text{and} \quad \int_1^\infty \frac{k(x)}{x} dx < \infty.$$

Consequently, by Proposition 2.3 of [17](3), the right-hand pair of conditions is necessary and sufficient for (2.3) to define a member of ND_s .

A pair (a, k) with $a \geq 0$, $k : (0, \infty) \rightarrow [0, \infty)$ nonincreasing and the two integrals of the criterion finite is called *admissible*, and so is the exponent (2.3) it defines; profiles equal almost everywhere define the same exponent. The zero exponent is admissible as an exponent, and it is the one exponent that (ND) excludes as a family.

The representation reads a kernel as a Gaussian part plus a catalogue of jumps. The reading is the line paper's Remark 2.6, reproduced here because the cone section leans on it.

Remark 2.4 (the displacement dictionary). A symmetric kernel is the law of a *random displacement*: $\Phi_{s,t}f$ reports the signal at a randomly displaced position $x - X_{s,t}$ with $X_{s,t} \sim \mu_{s,t}$. Its transform $\hat{\mu}_{s,t}(\omega) = \mathbb{E} \cos(\omega X_{s,t})$ is the contrast that a sinusoidal probe of wavenumber ω retains after the random displacement, the stage's modulation transfer function (Remark 4.3 of [17]). The exponent $\psi = -\log \hat{\mu}$ adds along the cascade. In (2.2) the Gaussian coefficient a is the *diffusive part*, a Gaussian displacement of variance $2a$, the one term the time-causal axis deletes and the only one Gaussian scale space uses. The measure ν is a *catalogue of jump displacements*. Three objects describe it in this paper, and they are kept apart by name: the catalogue itself, the measure $\nu(dx) = k(x) dx/x$ on the half-line; the *profile* k , its density against dx/x ; and the *tail measure* $\varpi = -dk$ of §3, which says how much of each unit step the profile contains. Displacements of size about x arrive as a Poisson stream, that is at independent random moments with a constant rate, and $\nu(dx)$ is that rate; the streams of different sizes are independent, and each event of size x costs the probe the phase deficit $1 - \cos \omega x$. So a symmetric infinitely divisible displacement is a Gaussian part plus an independent sum of jumps of all sizes, and the representation says that every such displacement decomposes in this way.

The symbol. An admissible exponent, written F from here on, is C^1 away from the origin, and the line paper writes its derivative through the *symbol*

$$B(\omega) = \omega F'(\omega) = 2a\omega^2 + \int_{(0,\infty)} (1 - \cos \omega u) \varpi(du), \quad (2.4)$$

where $\varpi = -dk$ is the *tail measure* of the profile, the Stieltjes measure of the nonincreasing function k (the line paper's Lemma 7.1). The symbol is the rate at which the exponent grows along a dilation, $B(\omega) = \partial_\lambda F(\lambda\omega)|_{\lambda=1}$, and it is itself an exponent of the form (2.2), with pair $(2a, \varpi)$. It is the object the scale evolution is written in. In the canonical gauge of §2.4 the kernel at scale t has transform $e^{-F(t\omega)}$, whose logarithmic derivative in t is $-B(t\omega)/t$, one equation for every wavenumber. §6 makes that equation an operator on the signal.

2.4 The characterization theorem

The canonical gauge. Scale covariance (A8) says that dilating space by λ relabels the scales by a bijection S_λ , and it says nothing about what that bijection is. The line paper shows (its Proposition 6.6) that $c(\lambda) := S_\lambda 1$ is a continuous strictly increasing bijection of $(0, \infty)$ with $c(1) = 1$, and defines the *canonical gauge* as its inverse, $\chi := c^{-1}$, extended by $\chi(0) = 0$. In the coordinate $\chi(t)$ the relabelling is dilation itself, $\chi(S_\lambda t) = \lambda \chi(t)$, and the accumulated exponent is $G(t, \omega) = F(\chi(t)\omega)$ with $F := G(1, \cdot)$. A family is *in canonical gauge* when χ is the identity, that is when the kernel at scale λt is the dilation by λ of the kernel at scale t . Every admissible family is brought to canonical gauge by reparametrizing its scale, and the gauge is unique once $\chi(1) = 1$ is fixed. In canonical gauge the family is generated by one law: the kernel at scale t is the law of tX_1 , where X_1 is the random displacement from scale 0 to scale 1.

Theorem 2.5 (main theorem, Theorem 7.3 of the line paper). *A family $(\Phi_{s,t})_{0 \leq s \leq t}$ satisfies (A1)–(A8) and (ND) if and only if there are an increasing bijection χ of $[0, \infty)$, the gauge, and a function F of the form (2.3) with $F \not\equiv 0$, such that*

$$\Phi_{s,t}f = \mu_{s,t} * f, \quad \hat{\mu}_{s,t}(\omega) = \exp[-(F(\chi(t)\omega) - F(\chi(s)\omega))].$$

In the canonical gauge the kernel at scale t is the symmetric probability measure ϕ_t with $\hat{\phi}_t = e^{-F(t\cdot)}$, and the pair (χ, F) is unique up to the normalization $\chi(1) = 1$.

So an admissible family is, in its canonical gauge, an admissible pair (a, k) and nothing else: a Gaussian coefficient and a nonincreasing displacement profile. The word *admissible*, said of an exponent F in this paper, means of the form (2.3) with an admissible pair, and the *admissible cone* of §3 is the set of such exponents. The theorem rests on two cited facts, Bochner's theorem in the forward direction and the symmetric Lévy–Khintchine representation at its converse and uniqueness clauses (§1.1).

The semigroup class is the case where the stages depend only on the difference of their scales, and the theorem recovers it.

Corollary 2.6 (recovering the semigroup case, Corollary 7.4 of the line paper). *Assume in addition that the family is one-parameter, that is, $\Phi_{s,t}$ depends only on $t - s$. Then, after the normalization $g_{0,1}(1) = 1$,*

$$F(\omega) = |\omega|^\alpha \quad \text{for some } 0 < \alpha \leq 2 :$$

the kernels are exactly the symmetric stable densities, which are the members of the extended semigroup class of [44] with positive kernels. Here $\alpha = 2$ is the Gaussian, and $0 < \alpha < 2$ has Gaussian coefficient 0 and profile $k(x) = C_\alpha^{-1}x^{-\alpha}$ with $C_\alpha := \int_0^\infty (1 - \cos u) u^{-1-\alpha} du \in (0, \infty)$.

Two consequences of the theorem are used throughout: the admissible exponents form a cone, and the kernels have densities.

Lemma 2.7 (the admissible exponents form a convex cone, Lemma 7.6 of the line paper). *If F_1, F_2 are admissible in the sense of Theorem 2.5, of the form (2.3) with Gaussian coefficients $a_1, a_2 \geq 0$ and nonincreasing profiles k_1, k_2 , then so are $F_1 + F_2$, with data $(a_1 + a_2, k_1 + k_2)$, and cF_1 for $c \geq 0$, with data (ca_1, ck_1) . So the admissible exponents form a convex cone, and the correspondence with the pairs (a, k) is linear.*

Proposition 2.8 (regularity of the kernels from the origin, Proposition 7.9 of the line paper). *Let $(\Phi_{s,t})$ satisfy (A1)–(A8) and (ND). Then for every $t > 0$ the kernel $\mu_{0,t}$ is absolutely continuous, and its density is unimodal with mode at the origin.*

The proposition rests on two cited facts about self-decomposable laws: absolute continuity ([50], Thm. 27.13, p. 181) and unimodality ([64], Thm. 1, p. 523). The theorem does not use them. It is stated for measures, so that it rests only on Bochner's theorem and the Lévy–Khintchine representation. A reader who wants the density reads the theorem and the proposition together.

2.5 What the line paper takes on trust

The line paper rests its theorems on cited facts about positive definite functions and self-decomposable laws, and this paper inherits them where it reads its statements. §1.1 lists them, with what this paper adds, and the page each rests on is printed before the statement that uses it.

2.6 One proposition the line paper did not print

The one result of the line paper's preliminaries that is stated here and not cited is the uniqueness of the Laplace transform of a measure on the half-line, in the form that allows the measure to be infinite. It is used twice. In §5 it identifies the mixing measure of a profile that is a mixture of exponentials, its Thorin measure. In §4 it rules out a causal ancestor. The result is elementary once finiteness of the transform is used to reduce to finite measures, and its proof uses finiteness at *one* point of (τ_0, ∞) , which is weaker than the hypothesis stated.

Proposition 2.9 (uniqueness of the Laplace transform of a locally finite measure). *Let m, m' be Borel measures on $(0, \infty)$, not necessarily finite, and suppose that $\int e^{-\tau u} m(du) = \int e^{-\tau u} m'(du) < \infty$ for every τ in some interval (τ_0, ∞) . Then $m = m'$.*

Proof. Fix $\tau_1 > \tau_0$. Both transforms being finite at τ_1 , the measures $e^{-\tau_1 u} m(du)$ and $e^{-\tau_1 u} m'(du)$ are finite, and they have equal Laplace transforms on $(0, \infty)$ after the shift $\tau \mapsto \tau + \tau_1$; so it is enough to treat finite measures. For finite m , the map $u \mapsto e^{-u}$ carries $(0, \infty)$ homeomorphically onto $(0, 1)$, and $\int e^{-nu} m(du)$, $n \in \mathbb{N}$, is the n -th moment of the image measure on $[0, 1]$. A finite measure on a bounded interval is determined by its moments, the polynomials being dense in the continuous functions there. So the image measures agree, and so do m and m' . $\square$

3 The admissible cone: the Gaussian ray and the Cin rays

The line paper ends with a set: the admissible exponents, the functions F of the form (2.3), one for each admissible family in its canonical gauge. It shows that the set is a convex cone (Lemma 2.7) and stops. This section describes the cone. The description has four steps, one subsection each. The cone is generated by one family of simple members, the Cin rays, whose profiles are unit steps: a nonincreasing profile is a stack of unit steps, so every admissible exponent is a Gaussian term plus a superposition of Cin rays (§3.1, Lemma 3.1, Figure 1). The superposition is unique, so the Gaussian coefficient and the tail measure $\varpi = -dk$ of the profile are coordinates on the cone, and its extreme rays are the Gaussian ray and the Cin rays (§3.2, Proposition 3.3, Figure 2); none of the named families of the later sections is extreme, apart from the Gaussian endpoint of the stable family. The kernel of a ray has no closed form and satisfies a delay equation, the twin of the equation of the Dickman function (§3.3, Lemma 3.5). And what a kernel with no Gaussian part does at the origin is decided by one number, the value $k(0+)$ of its profile there: the kernel is bounded exactly when that value exceeds 1 (§3.4, Proposition 3.7, Corollary 3.9). A reader who wants the criterion alone can go to §3.4 directly, which uses the earlier subsections only for its examples.

The causal theory, on the temporal half-line, has a cone of the same shape. The comparison is Remark 3.4, and nothing in this section depends on it.

3.1 The unit-step profiles and the superposition

The function Cin is the entire cosine integral, $\text{Cin}(z) = \int_0^z (1 - \cos v) v^{-1} dv$ ([15], §6.2). The lemma first records the two facts about it that the section uses: it starts like $z^2/4$ and grows like $\log z$. It then gives the two facts about the unit-step profiles. Each is admissible with Cin as its exponent, and every admissible exponent is a superposition of them.

Lemma 3.1 (the Cin rays and the superposition). *Write $\text{Cin}(z) := \int_0^z (1 - \cos v) v^{-1} dv$ for $z \geq 0$, extended evenly to $\mathbb{R}$.*

1. (The rays.) *For $\tau > 0$ the profile $k = \mathbf{1}_{(0, \tau)}$ is admissible, with Gaussian coefficient 0, and its exponent is $\omega \mapsto \text{Cin}(\tau\omega)$. The function Cin is even and nondecreasing on $[0, \infty)$, with $\text{Cin}(z) \leq z^2/4$ and $\text{Cin}(z) = \frac{1}{4}z^2 + O(z^4)$ as $z \rightarrow 0$; and there is a finite constant C with $\text{Cin}(z) = \log z + C + O(1/z)$ as $z \rightarrow \infty$.*

2. (The superposition.) *Every admissible F , with Gaussian coefficient a and profile k , satisfies*

$$F(\omega) = a\omega^2 + \int_{(0, \infty)} \text{Cin}(\tau\omega) \varpi(d\tau), \quad \omega \in \mathbb{R}, \quad (3.1)$$

where ϖ is any positive measure on $(0, \infty)$ with $\varpi((x, \infty)) = k(x)$ for almost every $x > 0$; such a ϖ exists.

The measure ϖ is the tail measure of §2.3, the Stieltjes measure $-dk$ of the nonincreasing profile; the lemma builds it and says in which sense it inverts the tail. Remark 3.2 identifies the constant C , and the paper uses only that it is finite: what the coordinates of §3.2 need is the two growth rates z^2 and $\log z$.

Proof. (1) Evenness of Cin needs no stipulation: the integrand $v \mapsto (1 - \cos v)/v$ is odd, so the integral from 0 is even, and $\text{Cin}(-z) = \text{Cin}(z)$. Monotonicity on $[0, \infty)$ is the positivity of that integrand there. From $1 - \cos v \leq v^2/2$ we get $\text{Cin}(z) \leq \int_0^z v/2 dv = z^2/4$ for $z \geq 0$, and for $z < 0$ by evenness, so the bound holds on $\mathbb{R}$. For the expansion at the origin, $|\cos v - (1 - v^2/2)| \leq \frac{5}{96}v^4$ for $|v| \leq 1$ gives $|(1 - \cos v)/v - v/2| \leq \frac{5}{96}v^3$ on $(0, 1]$, and integrating over $[0, z]$,

$$\left| \text{Cin}(z) - \frac{1}{4}z^2 \right| \leq \frac{5}{96}z^4, \quad |z| \leq 1,$$

which is the asserted $\text{Cin}(z) = \frac{1}{4}z^2 + O(z^4)$. The constant $\frac{5}{96}$ is not sharp: the series gives $\text{Cin}(z) = \frac{1}{4}z^2 - \frac{1}{96}z^4 + O(z^6)$, so $\frac{1}{96}$ is the best constant, and the elementary $|\cos v - (1 - v^2/2)| \leq v^4/24$ integrates to it. It is kept because it is the constant of the cosine bound this proof calls, and because nothing reads the value: what the statement asserts is $O(z^4)$. For the expansion at infinity, split at $v = 1$:

$$\text{Cin}(z) = \text{Cin}(1) + \log z - \int_1^z \frac{\cos v}{v} dv, \quad z \geq 1,$$

using $(1 - \cos v)/v = v^{-1} - v^{-1} \cos v$ on $[1, z]$; and integrating by parts against the primitive $\sin v/v$, $\int_1^z v^{-1} \cos v dv = \sin z/z - \sin 1 + \int_1^z v^{-2} \sin v dv$. The last integrand is dominated by v^{-2} , so $\int_1^\infty v^{-2} \sin v dv$ converges absolutely and, with

$$C := \text{Cin}(1) + \sin 1 - \int_1^\infty \frac{\sin v}{v^2} dv,$$

one gets $\text{Cin}(z) - \log z - C = -\sin z/z + \int_z^\infty v^{-2} \sin v dv$, whence $|\text{Cin}(z) - \log z - C| \leq 2/z$ for $z \geq 1$. That is the expansion, with an explicit constant in the error.

For the ray, $k = \mathbf{1}_{(0, \tau)}$ is nonnegative and nonincreasing on $(0, \infty)$ and satisfies both integrability conditions of Lemma 2.3 at once, being bounded by 1 and supported in $(0, \tau)$: $\int_0^1 x k(x) dx \leq \frac{1}{2}$ and $\int_1^\infty k(x) x^{-1} dx \leq \tau$. So $(0, k)$ is a pair of the form (2.3), and its exponent is $\int_0^\tau (1 - \cos \omega x) x^{-1} dx = \text{Cin}(\tau\omega)$ by the substitution $v = \omega x$, valid for every real ω because both sides vanish at $\omega = 0$ and Cin is even.

(2) A positive measure ϖ on $(0, \infty)$ with $\varpi((x, \infty)) = k(x)$ for almost every $x > 0$ is the image of Lebesgue measure on $(0, \infty)$ under the generalised inverse $y \mapsto \sup\{u > 0 : k(u) > y\}$, restricted to $(0, \infty)$ to discard the atom the transform leaves at the origin when k is bounded. Only two properties of k are used: it is nonincreasing, and it tends to 0 at infinity — and the second is forced rather than assumed, since k nonincreasing and bounded below by a positive constant would make $\int_1^\infty k(x) x^{-1} dx$ diverge like the harmonic integral. What comes out is the sandwich $k(x+) \leq \varpi((x, \infty)) \leq k(x)$, whose two sides agree off the countably many discontinuities of k ; that is why the tail identity is asserted almost everywhere and not everywhere, and a right-continuous version of k is neither available nor needed.

Given such a ϖ , (3.1) is the layer-cake exchange

$$\int_{(0, \infty)} \text{Cin}(\tau\omega) \varpi(d\tau) = \int_0^\infty \frac{1 - \cos \omega x}{x} \varpi((x, \infty)) dx = \int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x},$$

in which $\text{Cin}(\tau\omega)$ is read as $\int_0^\tau (1 - \cos \omega x) x^{-1} dx$ and the exchange is Tonelli, for which ϖ is σ -finite: the tail measure of an unbounded profile puts infinite mass near the origin, but its restriction to $[\varepsilon, \infty)$ is finite for every $\varepsilon > 0$. Adding $a\omega^2$ gives (3.1). $\square$

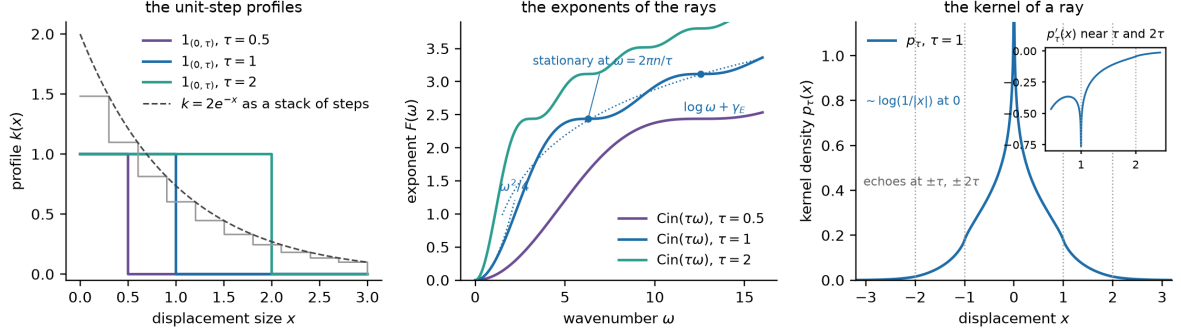


Figure 1: A Cin ray, three ways. Left: the unit-step profiles $\mathbf{1}_{(0,\tau)}$ for three steps, and a smooth nonincreasing profile drawn as the positive stack of unit steps it is, with the tail measure $\varpi = -dk$ as the weights (Lemma 3.1(2)). Middle: the exponents $\text{Cin}(\tau\omega)$ of the same three rays; for $\tau = 1$ the quadratic start $\frac{1}{4}\omega^2$ and the logarithmic asymptote $\log \omega + \gamma_E$ are dotted (Lemma 3.1(1), Remark 3.2), and the stationary points at $\omega = 2\pi n/\tau$ are marked; §4 uses them. Right: the kernel p_τ of the ray $\tau = 1$, computed numerically from its transform $e^{-\text{Cin}(\omega)}$. It is logarithmically singular at the origin, the threshold case of Proposition 3.7. The delay equation of Lemma 3.5 relates the kernel at x to its values at $x \pm \tau$, and the computed kernel shows echoes of the singularity at $\pm\tau$ and $\pm 2\tau$, marked by the dotted lines. The inset shows the first echo where it is visible, in the computed derivative: p'_τ has a logarithmic spike at τ and a kink at 2τ . Remark 3.8 describes the echoes and does not prove them. The kernel and its derivative are numerical, the transform having been inverted on a grid with a taper at high frequency.

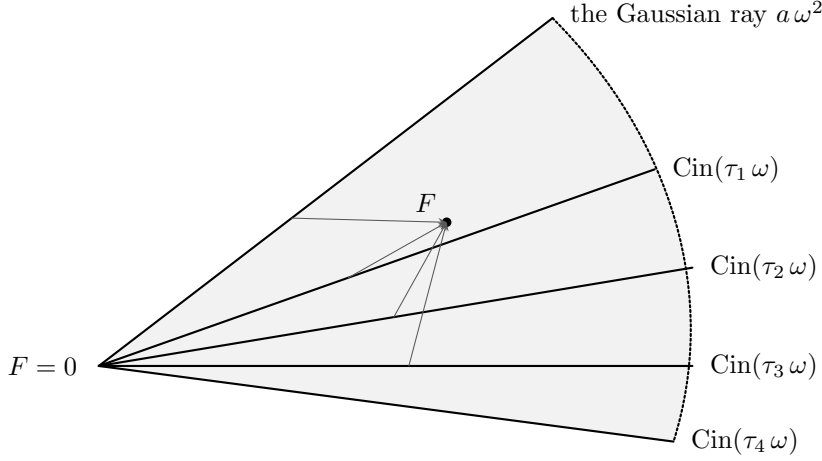
Remark 3.2 (the constant). The constant of Lemma 3.1(1) is Euler’s constant: $\text{Cin}(z) = \gamma_E + \log z - \text{Ci}(z)$ with Ci the cosine integral and $\text{Ci}(z) = O(1/z)$ ([15], (6.2.13) for the identity, and (6.2.20) with the expansions (6.12.3) and (6.12.4) for the rate). The identification is not used anywhere in this paper, which is why Lemma 3.1 asserts only that the constant is finite.

3.2 The Choquet structure

Choquet’s theorem says that a convex cone with a compact metrizable base is generated by its extreme rays, the rays that are not a sum of two others: every point is a positive superposition of extreme points ([46]). The superposition need not be unique. This paper says that a cone has a *Choquet structure* when the representation exists and is unique. The theorem is background here, and the proposition below does not use it. A convex cone with a Choquet structure has coordinates, the representing measure, and its geometry is the geometry of the extreme rays; a family cut out of the cone by some principle, such as the semigroup families of Corollary 2.6, is then described by which mixtures of the extreme rays it contains. The superposition (3.1) is the representation, explicitly, and the proposition below proves what Choquet’s theorem would assert about it, directly and with the coordinates named: the map $(a, \varpi) \mapsto F$ is a linear bijection, so the Gaussian coefficient and the tail measure of the profile are coordinates on the cone, and the extreme rays are the Gaussian ray and the Cin rays and no others.

The statement uses two terms. The pair (a, ν) of the Lévy–Khintchine form (2.2) is the *Lévy pair* of a law, and it is unique (Proposition 2.3 of [17](3)); when the Lévy measure ν has a density, that density is the *Lévy density*. For an admissible law the Lévy measure is $k(x)x^{-1}dx$ on $(0, \infty)$, so the folded Lévy density is $k(x)/x$, and for a Cin ray it is $x^{-1}\mathbf{1}_{(0,\tau)}(x)$: displacements of every size below τ at the rate dx/x , and none above.

Proposition 3.3 (Choquet structure of the admissible cone). *The superposition map of*



$F(\omega) = a\omega^2 + \int \text{Cin}(\tau\omega) \varpi(d\tau)$: a Gaussian term and a weighted sum of Cin rays, the weight $\varpi = -dk$ read off the profile

Figure 2: The admissible cone, schematically. The exponents F are the pairs (a, k) , a Gaussian coefficient and a nonincreasing profile, and they form a convex cone (Lemma 2.7). Its extreme rays are the Gaussian ray, the member with no jumps, and the Cin rays, one for each cutoff τ , the exponents of the unit-step profiles. Every member is a superposition of them in exactly one way, with the weights given by the profile: a nonincreasing profile is a stack of unit steps, and $\varpi = -dk$ says how much of each (Proposition 3.3). The drawing shows that superposition and nothing else; the cone has infinitely many dimensions, and where the named families and the image of the bridge lie in it is the subject of Figure 4 and of the chain in §5.3.

Lemma 3.1,

$$(a, \varpi) \mapsto F, \quad F(\omega) = a\omega^2 + \int_{(0, \infty)} \text{Cin}(\tau\omega) \varpi(d\tau),$$

is a linear bijection from $\{(a, \varpi) : a \geq 0, \varpi \text{ a positive Borel measure on } (0, \infty), \int [\tau^2 \wedge (1 + \log_+ \tau)] \varpi(d\tau) < \infty\}$ onto the cone of Lemma 2.7, with inverse $\varpi = -dk$, $k(x) = \varpi((x, \infty))$, and a the Gaussian coefficient. Consequently the extreme rays of the cone are the Gaussian ray $\omega \mapsto \omega^2$ together with the one-parameter family $\text{Cin}(\tau \cdot)$, $\tau > 0$, whose folded Lévy densities are $x^{-1} \mathbf{1}_{(0, \tau)}(x)$; and every admissible family is a unique superposition of Cin rays and Gaussian jitter, the term $a\omega^2$.

The proof has three parts. The domain condition on ϖ is the two integrability conditions of Lemma 2.3, read on the tail measure through the layer cake, which gives surjectivity. Injectivity is the uniqueness of the Lévy pair, with two measure-theoretic steps that carry an equality of tails at almost every point to an equality of measures. The extremality clauses are read at the profile, and the classification of *all* extreme rays is the one place the coordinates are used.

Proof. An admissible F has, by Theorem 2.5, a pair (a, k) of the form (2.3); a positive measure ϖ on $(0, \infty)$ with $k(x) = \varpi((x, \infty))$ almost everywhere exists by Lemma 3.1(2), and (3.1) is then the displayed representation. Conversely a pair (a, ϖ) in the stated domain gives back an admissible F : its profile is the tail $k(x) = \varpi((x, \infty))$, nonnegative and nonincreasing, and the two integrability conditions of Lemma 2.3 are the domain condition read through Tonelli,

$$\int_0^1 x \varpi((x, \infty)) dx = \int \frac{1}{2}(\tau \wedge 1)^2 \varpi(d\tau), \quad \int_1^\infty \varpi((x, \infty)) \frac{dx}{x} = \int \log_+ \tau \varpi(d\tau),$$

two more instances of the layer cake of Lemma 3.1(2), with the weights x cut off above 1 and x^{-1} cut off below 1. Since $\frac{1}{2}(\tau \wedge 1)^2 + \log_+ \tau \leq \tau^2 \wedge (1 + \log_+ \tau) \leq 2[\frac{1}{2}(\tau \wedge 1)^2 + \log_+ \tau]$ for

$\tau > 0$, the domain condition and the two integrability conditions are the same condition. So the map is onto the cone, and the domain is what the statement says.

That the superposition integral is finite at one $\omega \neq 0$ if and only if at all, and if and only if the domain condition holds, follows. Sufficiency is the paragraph just given together with Lemma 2.10 of [17], which makes an admissible exponent finite at every frequency. For necessity it is enough to dominate $\tau^2 \wedge (1 + \log_+ \tau)$ by a multiple of $\text{Cin}(\tau|\omega|)$ pointwise on $(0, \infty)$, and three regimes do that, using only *lower* bounds on Cin . Below $\tau|\omega| = 1$ the expansion at the origin gives $\text{Cin}(z) \geq \frac{19}{96}z^2$, which dominates τ^2 up to the constant $|\omega|^{-2}$. Past a threshold the expansion at infinity gives $\text{Cin}(z) \geq \frac{1}{2}(1 + \log z)$, which dominates $1 + \log_+ \tau$ up to a constant once τ is large enough that $\log(\tau|\omega|)$ is at least half of $\log \tau$. On the bounded stretch between the two, monotonicity alone suffices: $\text{Cin}(\tau|\omega|) \geq \text{Cin}(1) \geq \frac{19}{96}$ there, while $1 + \log_+ \tau$ is bounded. The map is linear in (a, ϖ) by construction, both clauses being additivity of the integral in the measure and positive homogeneity. It is injective because the pair of F is unique (Proposition 2.3 of [17](3)): the pair determines a and the Lévy measure $k(x)x^{-1}dx$, hence k up to a null set, hence the tails $\varpi((x, \infty))$ at almost every $x > 0$. Two steps carry that to ϖ itself. First, the exceptional set being null, every interval $(x, x + \varepsilon)$ contains a point y at which the two tails agree, and $\varpi_1((x + \varepsilon, \infty)) \leq \varpi_1((y, \infty)) = \varpi_2((y, \infty)) \leq \varpi_2((x, \infty))$; letting ε decrease and using continuity of a measure along the increasing sets $(x + \varepsilon, \infty)$ gives equality of the tails at *every* $x > 0$. Second, the tails at positive points are finite, by the domain condition, so for each $\varepsilon > 0$ the restrictions of ϖ_1 and ϖ_2 to (ε, ∞) are finite measures agreeing on every half-line $(-\infty, t]$, since $\varpi((\varepsilon, t])$ is a difference of two finite tails; two finite measures on $\mathbb{R}$ agreeing on all such half-lines are equal. Letting ε decrease again, and using that ϖ_1 and ϖ_2 give no mass to $(-\infty, 0]$, gives $\varpi_1 = \varpi_2$. The detour through a finite restriction is not avoidable by a direct π -system argument on the line: the Choquet measure of an unbounded profile has infinite mass near the origin, so no exhausting sequence of finite-measure half-lines exists before the restriction is made.

Extremality of the two named families is read at the profile, not through the coordinates. If $\omega^2 = F_1 + F_2$ with both summands admissible, then $F_1 + F_2$ has pair $(1, 0)$ by uniqueness, so $k_1 + k_2$ vanishes almost everywhere on $(0, \infty)$, so each k_i does, and $F_i(\omega) = a_i \omega^2$ with $a_i \geq 0$: the Gaussian ray is extreme. If $\text{Cin}(\tau \cdot) = F_1 + F_2$, then uniqueness gives $a_1 = a_2 = 0$ and $k_1 + k_2 = \mathbf{1}_{(0, \tau)}$ almost everywhere; on the conull set inside $(0, \tau)$ where that holds, each k_i is nonincreasing, being a profile, and nondecreasing, being what is left of a constant after subtracting a nonincreasing function, hence constant there, while above τ both vanish by nonnegativity. So $k_i = c_i \mathbf{1}_{(0, \tau)}$ almost everywhere and $F_i = c_i \text{Cin}(\tau \cdot)$ with $c_i \geq 0$: each Cin ray is extreme.

That there are no other extreme rays is where the coordinates are used, and where the injectivity just proved is spent. Let F be extreme with pair (a, ϖ) . If $\varpi = 0$ then k vanishes almost everywhere and $F(\omega) = a \omega^2$. If $\varpi \neq 0$ then $a = 0$: otherwise splitting F as $a \omega^2$ plus the pure-jump part makes $a \omega^2$ equal to $c_1 F$ for some c_1 , and either $c_1 = 0$, forcing $a = 0$, or F is a multiple of ω^2 , which by uniqueness forces $\varpi = 0$. Now suppose some $s > 0$ has $\varpi((0, s]) > 0$ and $\varpi((s, \infty)) > 0$. The restrictions of ϖ to $(0, s]$ and to (s, ∞) split F into two admissible exponents; extremality makes the first of them $c_1 F$; injectivity turns that into an equality of measures, the restriction of ϖ to $(0, s]$ against $c_1 \varpi$; evaluating both sides on (s, ∞) forces $c_1 = 0$, and then $\varpi((0, s]) = 0$, a contradiction. So every s has mass on at most one side. Put $\tau := \inf\{s > 0 : \varpi((0, s]) > 0\}$, a set that is nonempty because $\varpi \neq 0$. Then $\varpi((x, \infty)) = 0$ for $x > \tau$, by the defining property of the infimum together with the hypothesis just established, hence also at $x = \tau$; and $\varpi((0, x]) = 0$ for $0 < x < \tau$, by minimality. So ϖ is carried by the single point τ , which is positive since otherwise ϖ would be 0, and $F = c \text{Cin}(\tau \cdot)$ with $c = \varpi(\{\tau\})$, finite because $\varpi((\tau/2, \infty))$ is. This is the map's own picture: $(1, 0) \mapsto \omega^2$, the Gaussian ray, and $(0, \delta_\tau) \mapsto \text{Cin}(\tau \cdot)$, whose profile is $k(x) = \delta_\tau((x, \infty)) = \mathbf{1}_{(0, \tau)}(x)$ and whose folded Lévy density is therefore $x^{-1} \mathbf{1}_{(0, \tau)}(x)$. $\square$

The remark that follows compares the cone with that of the causal theory. The causal article's cone is the cone of exponents of the scale spaces on the temporal half-line, its ray off the curve is the *drift*, the pure delay, and its unit-step members are the *Dickman rays*, whose kernels are the laws of the Dickman subordinator ([12]). The Thorin subclass named in it is the subcone of the profiles that are completely monotone, mixtures of exponentials, which §5 treats; the three named families are the subject of that section.

Remark 3.4 (what the cone looks like, and what changed). In the log-displacement coordinate of Remark 2.4 the proposition reads: every nonincreasing profile is a unique positive stack of unit steps, and every admissible family is that stack plus one number a , a point *off* the curve. The three distinguished families of this paper are slices cut by different principles: the symmetric stable family by dilation symmetry (Corollary 2.6), the Matérn family as the single-atom member of the Thorin subclass (Theorem 5.4), and the Student-t family as the image of the causal Bessel corner under the bridge (Proposition 5.10). No member of the three is an extreme ray of the cone, with one exception at an endpoint: the stable family closes at $\alpha = 2$ on the Gaussian ray (Proposition 5.12), which is extreme. All three lie in the Thorin subcone, each by the node that states it: the Matérn family as the one-atom case of Proposition 5.7(2), the stable family by its Thorin density (Proposition 5.12), and the Student-t family by Proposition 5.10(3), whose Thorin half is proved under the hypothesis that the causal delay law is a generalized gamma convolution, a cited fact (Remark 5.11). The Cin rays are the paradigm of what lies outside it: profiles with a hard cutoff, whose kernels carry derivative singularities at the displacements $\pm\tau, \pm2\tau, \dots$, echoes of the step (Lemma 3.5). Two things are new against the causal cone of [18]. The Gaussian ray is an extreme ray of every subcone in the chain $ID_s \supset SD_s \supset GGC_s$ — the symmetric infinitely divisible laws, the symmetric self-decomposable ones, and the symmetric Thorin subclass of §5.3 — being present in each with $k \equiv 0$, whereas the corresponding causal ray, the drift, was the degenerate member. And the Cin rays, unlike their causal counterparts, are not the image of anything under the bridge of §4 (Proposition 4.6). The tail statement of the causal article's cone remark holds here in a different shape but with the same conclusion: by Proposition 5.1(2) a Cin kernel has tails of order $e^{-|x|\log|x|/\tau}$, lighter than any exponential and heavier than any Gaussian; any admissible kernel with a jump catalogue has tails at least that heavy, whatever its Gaussian coefficient; and the only admissible kernels with Gaussian tails are the Gaussians themselves. No truncation of the Gaussian is admissible.

3.3 The delay equation of the Cin kernels

The kernel p_τ of a Cin ray, the density of the law with transform $e^{-\text{Cin}(\tau\omega)}$, has no closed form, and it satisfies a delay equation. A *delay equation* relates the derivative of a function at x to the values of the function at a fixed distance from x ; the classical example is the equation

$$x \rho'(x) = -\rho(x-1)$$

of the Dickman function ρ ([14]; [57], Chapter III.5). It arose in number theory, where $\rho(u)$ is the limiting proportion of the integers up to n with no prime factor above $n^{1/u}$. Up to scale and a normalizing constant it is also the density of the causal unit-step member, the Dickman subordinator of [12]. The lemma of this subsection says that p_τ satisfies the same equation with the average of the two neighbours $p_\tau(x \mp \tau)$ in place of the one delayed value, because displacements come in both signs. The reason to want the equation is what it says about the shape of the kernel. The derivative at x is read off the values at $x \pm \tau$, so a singularity at the origin reappears, smoothed, at $\pm\tau$, then at $\pm2\tau$, and so on down the lattice $\tau\mathbb{Z}$. Figure 1 shows these echoes numerically and Remark 3.8 describes them. The description is not proved here, and no result depends on it. The echoes are the signature of a hard cutoff in the profile, and the smooth members of §5 do not have them.

Lemma 3.5 (the Cin kernels satisfy a delay equation). *Let p_τ be the density of the symmetric law with exponent $\text{Cin}(\tau \cdot)$ — folded profile $\mathbf{1}_{(0,\tau)}$, two-sided Lévy density $\frac{1}{2}|x|^{-1}\mathbf{1}_{(0,\tau)}(|x|)$, no Gaussian part — and let P_τ be its distribution function. Then, in the sense of distributions,*

$$x p_\tau(x) = P_\tau(x) - \frac{1}{2} [P_\tau(x-\tau) + P_\tau(x+\tau)], \quad \text{equivalently} \quad x p'_\tau(x) = -\frac{1}{2} [p_\tau(x-\tau) + p_\tau(x+\tau)].$$

The identity is asserted in the sense of distributions, which is what the transform argument delivers; both sides are locally integrable, so it is also an almost-everywhere identity. Read at the law rather than at the density the two displays are one statement, an identity between two finite signed measures, and that is how the proof is organised; the passage back to the density is a change of variables for the first display and one integration by parts for the second.

Proof. Write μ for the law, $c(\omega) = \int \cos(\omega x) \mu(dx) = e^{-\text{Cin}(\tau\omega)}$ for its transform, and κ_τ for Lebesgue measure on $[0, \tau)$. Everything follows from one identity between finite signed measures on $\mathbb{R}$:

$$2x \mu(dx) = \nu - \nu_\tau, \quad \nu := \mu * \kappa_\tau, \quad \nu_\tau := \text{the image of } \nu \text{ under } x \mapsto x - \tau. \quad (3.2)$$

By Tonelli, ν has Lebesgue density $x \mapsto \mu((x - \tau, x])$ and ν_τ has density $x \mapsto \mu((x, x + \tau])$; both are finite measures of total mass τ . The left-hand side is finite because μ has a finite first absolute moment: $\text{Cin}(z) \leq z^2/4$, so the exponent of μ is quadratically flat, the truncation inequality of Theorem 5.2 of [17] gives $\mu\{|x| > R\} = O(R^{-2})$, and the layer cake integrates that tail.

The transforms. Since μ is symmetric, $\int x e^{-i\omega x} \mu(dx)$ has no real part, the integrand's cosine half being odd, so it equals $-i \int x \sin(\omega x) \mu(dx)$. The first absolute moment lets c be differentiated under the integral sign, giving $c'(\omega) = -\int x \sin(\omega x) \mu(dx)$; and differentiating $c = e^{-\text{Cin}(\tau \cdot)}$ by the chain rule gives $c'(\omega) = -\tau \text{Cin}'(\tau\omega) c(\omega)$ with $\text{Cin}'(v) = (1 - \cos v)/v$. The two readings of c' are the elementary delay ODE

$$\omega c'(\omega) = -c(\omega) (1 - \cos \tau\omega), \quad (3.3)$$

and together they give $\int x e^{-i\omega x} \mu(dx) = -i \tau \text{Cin}'(\tau\omega) c(\omega)$, which for $\omega \neq 0$ is $-i(1 - \cos \tau\omega) c(\omega)/\omega$. On the other side, the transform of a convolution is the product, so

$$\hat{\nu}(\omega) = c(\omega) \int_0^\tau e^{-i\omega u} du = c(\omega) \frac{1 - e^{-i\omega\tau}}{i\omega}, \quad \hat{\nu}_\tau(\omega) = e^{i\omega\tau} \hat{\nu}(\omega),$$

whence, for $\omega \neq 0$,

$$\hat{\nu}(\omega) - \hat{\nu}_\tau(\omega) = c(\omega) \frac{(1 - e^{-i\omega\tau})(1 - e^{i\omega\tau})}{i\omega} = c(\omega) \frac{2 - 2 \cos \omega\tau}{i\omega} = -2i \frac{1 - \cos \omega\tau}{\omega} c(\omega),$$

which is twice the transform of $x \mu(dx)$, that is, the transform of the left-hand side of (3.2); at $\omega = 0$ both sides vanish, the left by symmetry and the right because ν and ν_τ have equal mass.

Uniqueness. Both sides of (3.2) are signed, so Proposition 2.7 of [17] is applied after recombination: writing $x^\pm$ for the positive and negative parts of x , the two *positive* finite measures $2x^+ \mu(dx) + \nu_\tau$ and $2x^- \mu(dx) + \nu$ have the same transform by the display above, hence are equal, which is (3.2).

The two displays. The first is (3.2) read at Lebesgue densities. With $\mu(dx) = p_\tau(x) dx$ the left-hand side has density $2x p_\tau(x)$ and the right-hand side has density

$$\mu((x - \tau, x]) - \mu((x, x + \tau]) = 2 \left(P_\tau(x) - \frac{1}{2} [P_\tau(x - \tau) + P_\tau(x + \tau)] \right),$$

and two integrable functions with the same integral against every bounded measurable test function agree almost everywhere — it is enough to test against the sign of their difference.

The second display is (3.2) paired with φ' for a test function $\varphi \in C_c^\infty$. Since $\int_0^\tau \varphi'(x+u) du = \varphi(x+\tau) - \varphi(x)$, and likewise for ν_τ with x shifted, pairing gives

$$2 \int x \varphi'(x) \mu(dx) = \int [\varphi(x+\tau) + \varphi(x-\tau)] \mu(dx) - 2 \int \varphi d\mu,$$

that is $\int (x\varphi)' d\mu = \frac{1}{2} \int [\varphi(\cdot+\tau) + \varphi(\cdot-\tau)] d\mu$; and one integration by parts against the density turns that into the second display. One route to the same identity convolves μ with the *signed* weight $y \nu_2(dy) = \frac{1}{2} \text{sgn}(y) \mathbf{1}_{|y|<\tau} dy$, which is $\frac{1}{2}$ of κ_τ minus its reflection. The computation above is that one with the two halves kept positive until the subtraction at the end, so that no signed measure is convolved anywhere. $\square$

3.4 The behaviour at the origin

The last question of the section is what an admissible kernel looks like at the origin, the point of zero displacement. A density is *singular* at a point when it is unbounded near it; the kernel of the exponent $\log(1+\omega^2)$, $e^{-|x|}/2$, is bounded with a kink at the origin, the Gaussian is smooth there, and the Cin kernel of Figure 1 is unbounded. The answer, for every admissible kernel without a Gaussian part, rests on two theorems of Sato on self-decomposable laws, and it depends on the profile through one number, its value $c = k(0+)$ at the origin: for $c < 1$ the density is singular of the order $|x|^{c-1}K(|x|)$, for $c > 1$ it is continuous with $\lceil c \rceil - 2$ derivatives, at $c = 1$ it is singular of the order of an integral of K , and when c is infinite the density is of class C^∞ . The factor K is built from the profile, $K(y) = \exp \int_y^1 (c - k(u)) du/u$. It is slowly varying and at least 1, and it measures how fast the profile approaches its value at the origin. When the profile is constant near the origin, as for a Cin ray, K is constant there, the singularity below the threshold is the pure power and the one at the threshold is a logarithm. In general K is unbounded: the profile $\frac{1}{2} - 1/\log(1/x)$ near the origin has $c = \frac{1}{2}$ and a kernel of the order $|x|^{-1/2} \log(1/|x|)$. The intensity of small displacements decides whether the kernel is bounded: a profile that starts above 1 produces enough small displacements to smooth the mass at the origin, and one that starts below 1 does not. The Cin rays start at exactly 1.

The folding. Sato's theorems are stated for a law on the line in terms of a *two-sided* profile k_2 on $\mathbb{R} \setminus \{0\}$ and two constants, $k_2(0+) + k_2(0-)$ and $k_2(0+) - k_2(0-)$, the second of which measures the asymmetry of the law. This paper's profile is folded, as §2.3 said: the Lévy measure $k(x)x^{-1}dx$ lives on $(0, \infty)$ and counts both directions of a displacement together, so a symmetric two-sided profile k_2 and its folding are related by the factor 2, $k = 2k_2$ on $(0, \infty)$, and the sources' first constant is this paper's $k(0+)$ while the second is 0. The lemma states the dictionary, so that the theorem can be applied at the sources' own letter; the factor 2 is the one place where a constant is easily lost in carrying a statement from a source into this paper, and it is exactly what puts the Matérn threshold at $\gamma = \frac{1}{2}$ rather than at $\gamma = 1$ below.

Lemma 3.6 (the folding translation). *Let (a, k) be a pair of the form (2.3) with $a = 0$, and suppose the folded profile has a finite right limit $c := k(0+)$ at the origin. Put $k_2(x) := \frac{1}{2}k(|x|)$ for $x \in \mathbb{R}$. Then k_2 is a two-sided profile in the sense of the probabilistic sources: it is nonnegative, nonincreasing on $(0, \infty)$, nondecreasing on $(-\infty, 0)$, and $\int_{\mathbb{R}} (1 \wedge x^2) k_2(x) |x|^{-1} dx < \infty$. Its Lévy measure $k_2(x)|x|^{-1}dx$ on $\mathbb{R} \setminus \{0\}$ is the unfolding of $k(x)x^{-1}dx$:*

$$\int_{\mathbb{R}} (1 - \cos \omega x) \frac{k_2(x)}{|x|} dx = \int_0^\infty (1 - \cos \omega x) \frac{k(x)}{x} dx \quad (\omega \in \mathbb{R}),$$

so the two forms specify the same symmetric law. Moreover $k_2(x) + k_2(-x) = k(x)$ for every $x > 0$, and both one-sided limits at the origin exist, with

$$k_2(0+) = k_2(0-) = \frac{1}{2}c, \quad \text{so that} \quad k_2(0+) + k_2(0-) = c \quad \text{and} \quad k_2(0+) - k_2(0-) = 0.$$

The hypothesis that c is finite is used only for the two limits; the structural conditions and the exponent identity hold with no hypothesis at the origin.

Two further hypotheses of the sources are met by k_2 and are not visible in the lemma. The singular and threshold regimes are stated for the driftless finite-variation form of the exponent, which asks $\int_{|x|<1} k_2(x) dx < \infty$ and leaves no first-order term: with c finite, $k_2 \leq \frac{1}{2}c$ on each side of the origin by monotonicity, so that integral is at most c ; the drift vanishes because the law is specified through its cosine transform; and there is no Gaussian part because $a = 0$. And the sources normalise a profile to the version left-continuous on $(0, \infty)$ and right-continuous on $(-\infty, 0)$, which k_2 need not be. Passing to that modification changes k_2 on a countable set only, and hence changes neither the Lévy measure $k_2(x)|x|^{-1}dx$, nor the exponent, nor the law, nor the two one-sided limits at the origin, nor the functions K and L below, which are integrals of k_2 . So the theorems apply to the modification and conclude about the same law with the same comparison functions.

Proof. The three structural conditions are the corresponding conditions on k read through $x \mapsto |x|$: nonnegativity is immediate, and for $x \leq y < 0$ one has $|x| \geq |y| > 0$, so $k_2(x) \leq k_2(y)$ because k is nonincreasing on $(0, \infty)$. For the two integrals, note first that the map $x \mapsto -x$ preserves Lebesgue measure and that both integrands below are even, $\cos$ being even and k_2 being a function of $|x|$; so each integral over $\mathbb{R}$ is twice the corresponding integral over $(0, \infty)$, where $k_2 = \frac{1}{2}k$. The two factors 2 cancel, which gives the exponent identity, and the same computation turns $\int_{\mathbb{R}} (1 \wedge x^2) k_2(x) |x|^{-1} dx$ into $\int_0^\infty (1 \wedge x^2) k(x) x^{-1} dx$, which is finite: on $(0, 1)$ it is $\int_0^1 x k(x) dx$ and on $(1, \infty)$ it is $\int_1^\infty k(x) x^{-1} dx$, the two conditions of Lemma 2.3.

The pointwise identity is $k_2(x) + k_2(-x) = \frac{1}{2}k(x) + \frac{1}{2}k(x)$ for $x > 0$. For the limits, k_2 agrees with $\frac{1}{2}k$ on $(0, \infty)$, so its right limit is $\frac{1}{2}c$; and $k_2(x) = \frac{1}{2}k(-x)$ on $(-\infty, 0)$, where $x \mapsto -x$ carries the left neighbourhood filter at the origin to the right one, so its left limit is $\frac{1}{2}c$ as well. The sum and the difference follow. $\square$

The cited fact. The three regimes are two theorems of Sato: the smooth regime is [50], Theorem 28.4, p. 191, and the singular regime and the threshold are Theorem 53.8, pp. 410–411, formulas (53.28) and (53.30) there, all stated for a self-decomposable law on the line with no Gaussian part in terms of its two-sided profile and the two constants above. This is the one fact of this section that is cited rather than proved (§1.1). What the proposition takes from the theorems is the *order* of the singularity, an unbounded representative below and at the threshold and a continuous one above it, and not the constant in Sato’s formula. It claims nothing when $k(0+) = \infty$, which is the case for the stable profiles and for every completely monotone profile with an unbounded Thorin measure; that case is Corollary 3.10 at the end of this subsection, on a further clause of the same theorem.

The order does need the constant in Sato’s formula to be nonzero, since $f \sim Ag$ gives $f \asymp g$ only for $A \neq 0$, and the folding is what makes it so. Below the threshold the constant is $\kappa \sin(k_2(0+)\pi) / (\Gamma(c) \sin(c\pi))$ with $\kappa > 0$ the exponential-integral expression of the source’s (53.27), and $k_2(0+) = c/2$ turns the first factor into $\sin(c\pi/2) / \sin(c\pi) = 1 / (2 \cos(c\pi/2)) > 0$ for $0 < c < 1$; at the threshold it is $(\kappa/\pi) \cos(c'\pi/2)$ with $c' = 0$, hence κ/π . The symmetry is therefore load-bearing twice, once for the antisymmetric constant and once for the nonvanishing.

Proposition 3.7 (the behaviour of an admissible kernel at the origin). *Let F be admissible with Gaussian coefficient 0 and folded profile k , let ϕ_1 be its kernel at canonical scale 1, and suppose the right limit $c := k(0+)$ is positive, so that $c \in (0, \infty]$. The three regimes below are stated for finite c , and nothing is claimed when $c = \infty$. For $0 < y < 1$ set*

$$K(y) := \exp \left[\int_y^1 (c - k(u)) \frac{du}{u} \right], \quad L(y) := \int_y^1 K(z) \frac{dz}{z},$$

both integrals being convergent; $K \geq 1$, since k is nonincreasing with right limit c . Then, near the origin: if $c < 1$, ϕ_1 is unbounded with $\phi_1(x) \asymp |x|^{c-1} K(|x|)$; if $N < c \leq N + 1$ for an

integer $N \geq 1$, ϕ_1 extends continuously and has $N - 1$ derivatives; and if $c = 1$, ϕ_1 is unbounded with $\phi_1(x) \asymp L(|x|)$. Every Cin ray has $c = 1$ with K constant near the origin, so its kernel is logarithmically singular there.

None of the three regimes is empty: the threshold is met by the Cin rays, the singular regime by the profile $c \mathbf{1}_{(0,\tau)}$ with $c < 1$, and the smooth one by the Matérn profile of §5 at $\gamma = 1$, where $c = 2$ and the kernel is the Laplace kernel, continuous with a kink.

Proof. The three regimes are the cited interface, and what has to be said here is how the source's statement becomes this one. The sources state the behaviour at the origin of a self-decomposable law with no Gaussian part in terms of its *two-sided* profile k_2 on $\mathbb{R} \setminus \{0\}$ and the two constants $k_2(0+) + k_2(0-)$ and $k_2(0+) - k_2(0-)$: the first is their c , the second their antisymmetric constant. By Lemma 3.6 the law in question is theirs. Its two-sided datum is $k_2(x) = \frac{1}{2}k(|x|)$, which is a profile of the kind the sources require; its exponent is the folded exponent, so the two descriptions specify the same law; the first constant is $k(0+) = c$; and the second vanishes, the law being symmetric. The three hypotheses of the sources that the dictionary does not display — finite variation near the origin, no drift and no Gaussian part, and the one-sided continuity of the profile — are met for the reasons given after the lemma. Reading the three regimes through that dictionary gives the three statements displayed above, with $k_2(u) + k_2(-u)$ becoming $k(u)$ in the integrand of K , which is the factor both singular clauses carry, and the passage from the source's asymptotic to the two-sided comparison uses the nonvanishing of the source's constant, computed above. The regimes themselves are cited, and the order of the singularity is all that is claimed of them.

The last sentence is this paper's own. The Cin ray of step τ has profile $\mathbf{1}_{(0,\tau)}$ (Lemma 3.1(1)), whose right limit at the origin is 1, so $c = 1$ and the threshold regime applies; and on $(0, \tau)$ the integrand $(1 - k(u))u^{-1}$ of K vanishes, which is the constancy of an indicator on its own interval. So K is constant on $(0, \min(\tau, 1))$, with the value $\max(1, 1/\tau)$: for $\tau \geq 1$ nothing is integrated, and for $\tau < 1$ the window $(\tau, 1)$ contributes $\int_\tau^1 u^{-1} du = \log(1/\tau)$. Hence L is a constant multiple of $\log(1/|x|)$ plus a constant near the origin — $L(x) = \log(1/|x|)$ when $\tau \geq 1$ and $L(x) = \tau^{-1} \log(\tau/|x|) + \tau^{-1} - 1$ when $\tau < 1$ — and in both cases the logarithmic order is all the statement claims. $\square$

Remark 3.8 (the singularity propagates outward). For a Cin ray of step τ the delay equation of Lemma 3.5 carries the singularity outward along the lattice $\tau\mathbb{Z}$, one order of smoothness gained at each step: the derivative of the kernel inherits a logarithmic singularity at $\pm\tau$, so the kernel itself is continuous there with a kink of order $|x \mp \tau| \log |x \mp \tau|$, and so on to $\pm 2\tau$ and beyond. This is a description of the shape of the kernel and not a claim of the proposition: the induction it summarises is not carried out here, and no result depends on it.

The Cin kernels are therefore logarithmically singular at the origin. So is one member of the Matérn family of §5. That family consists of the *variance-gamma* laws, the laws of a Brownian motion run for a Gamma-distributed time ([41]). At the index $\gamma = \frac{1}{2}$ its kernel is a multiple of $K_0(|x|)$, the modified Bessel function of the second kind, which has a logarithmic singularity at the origin and sits at the same threshold. Both are instances of a single boundary in the cone, and where it falls is decided by the profile alone. The corollary says that an admissible kernel with no Gaussian part, and with $k(0+)$ finite and positive, is bounded at the origin exactly when the profile starts above 1. It then reads the condition off for the named members. For the Matérn family it uses the profile $2\gamma e^{-x/\theta}$, which is computed in §5 (Proposition 5.2(1)) by an elementary integral that uses nothing of this section, so the reference ahead is not circular.

Corollary 3.9 (the boundedness threshold). *Let F be admissible with Gaussian coefficient 0 and folded profile k , let ϕ_1 be its kernel at canonical scale 1, and suppose $c := k(0+)$ is finite and positive.*

1. (The threshold.) *If $c > 1$, then ϕ_1 has a representative continuous on the whole line. If $c \leq 1$, then for every representative p of ϕ_1 , every $\delta > 0$ and every M there is an x with $0 < |x| < \delta$ and $p(x) > M$; no representative of ϕ_1 is bounded near the origin. So ϕ_1 is bounded at the origin exactly when $c > 1$.*
2. (The named members.) *Every Cin ray has $c = 1$ and sits at the threshold. The Matérn member of index γ and range θ has $c = 2\gamma$, so its kernel is bounded at the origin exactly for $\gamma > \frac{1}{2}$, and at $\gamma = \frac{1}{2}$ it sits at the threshold.*

Clause (2) is the boundary that the closed form of Proposition 5.3 shows from the other side: the closed form is a Matérn covariance function only for $\gamma > \frac{1}{2}$, and is the Bessel K_0 at $\gamma = \frac{1}{2}$. The corollary claims the threshold and not the rate of the divergence there, which for this member is logarithmic and is the cited Bessel asymptotic rather than anything proved here, and it says nothing in dimension greater than one.

Proof. Both halves of clause (1) come from Proposition 3.7. If $c > 1$, choose the integer $N \geq 1$ with $N < c \leq N + 1$ — which exists for every finite $c > 1$, namely $N = \lceil c \rceil - 1$ — and the smooth regime gives a representative of ϕ_1 that is C^{N-1} on the whole line, in particular continuous. Below the threshold and at it, both comparison functions are handled by one bound on the slowly varying factor: k is nonincreasing on $(0, \infty)$ with right limit c , so $k \leq c$ there, the integrand $(c - k(u))u^{-1}$ of K is nonnegative, and $K \geq 1$ on $(0, 1)$. If $c < 1$ the singular regime gives a representative p_0 and constants $c_1 > 0$, $\delta_0 > 0$ with $p_0(x) \geq c_1|x|^{c-1}K(|x|) \geq c_1|x|^{c-1}$ for $0 < |x| < \delta_0$; if $c = 1$ the threshold regime gives the same with $c_1 L(|x|)$ in place of $c_1|x|^{c-1}K(|x|)$, and $L(x) \geq \int_{|x|}^1 z^{-1}dz = \log(1/|x|)$ by the same bound. In both cases the lower bound tends to $+\infty$ as $x \rightarrow 0$.

That carries to every representative, and not only to p_0 , by comparing measures rather than functions. Suppose some representative p satisfied $p \leq M$ on the punctured δ -neighbourhood of the origin, and let $0 < \varepsilon \leq \min(\delta, \delta_0)$. On $(0, \varepsilon)$ the regime's lower bound is at least its value at ε , both $|x|^{c-1}$ and L being nonincreasing in $|x|$: below the threshold $p_0 \geq c_1\varepsilon^{c-1}$ there, and at it $p_0 \geq c_1 \log(1/\varepsilon)$. So $\phi_1((0, \varepsilon)) \geq c_1\varepsilon^{c-1} \cdot \varepsilon$ in the first case and $\geq c_1 \log(1/\varepsilon) \cdot \varepsilon$ in the second, while the assumed bound on p gives $\phi_1((0, \varepsilon)) \leq M\varepsilon$. Dividing by ε leaves $c_1\varepsilon^{c-1} \leq M$, respectively $c_1 \log(1/\varepsilon) \leq M$, and both left-hand sides tend to $+\infty$ as ε decreases. Choosing ε small enough contradicts the assumption. Neither step needs p to be measurable: a density integrates over a Borel set whatever representative names it, and the two bounds are bounds on that integral.

Clause (2) is the profile computation in each case. The Cin ray of step τ has profile $\mathbf{1}_{(0, \tau)}$, whose right limit at the origin is 1 (Lemma 3.1(1)); the Matérn profile at range θ is $2\gamma e^{-x/\theta}$ (Proposition 5.2(1)), whose right limit is 2γ . Reading $c = 2\gamma$ through clause (1) puts the threshold at $\gamma = \frac{1}{2}$. $\square$

The corollary leaves one case open, and it is the case of two of the three families of §5: a profile that is unbounded at the origin, $k(0+) = \infty$. There the same theorem of Sato on the smooth regime has a clause of its own — if the two-sided constant is infinite the law has a C^∞ density ([50], Theorem 28.4, p. 191, clause (ii)) — and it settles the case in the other direction: an unbounded profile produces so many small displacements that the kernel is not merely bounded at the origin but smooth there.

Corollary 3.10 (an unbounded profile makes the kernel smooth). *Let F be admissible with Gaussian coefficient 0 and folded profile k , and let ϕ_1 be its kernel at canonical scale 1. If $k(0+) = \infty$, then ϕ_1 has a representative of class C^∞ on $\mathbb{R}$; in particular that representative is bounded on every bounded set, so ϕ_1 is bounded at the origin. The symmetric stable members of index $0 < \alpha < 2$ are of this kind: their folded profile is a positive multiple of $x^{-\alpha}$ (Proposition 5.12), whose right limit at the origin is $+\infty$.*

With the Gaussian coefficient 0 and $F \neq 0$ the value $c = k(0+)$ lies in $(0, \infty]$, a nonincreasing profile with $k(0+) = 0$ vanishing identically. The two corollaries therefore cover the whole range, and together they say one thing: *an admissible kernel with no Gaussian part is bounded at the origin exactly when its profile starts above 1*. What is claimed of the smooth case is the smoothness and nothing else — no rate, no bound on the density or its derivatives. The Student-t members have an unbounded profile as well, but this paper does not prove it: what it has of that family's profile is the Gaussian mixture (4.2) of a causal catalogue it does not compute. Nothing is lost, since the Student-t kernel is bounded at the origin by its own explicit density (Proposition 5.10(1)).

Proof. The cited theorem is stated for a self-decomposable law on $\mathbb{R}$ with no Gaussian part, presented through a two-sided profile k_2 — nonnegative, nonincreasing on $(0, \infty)$ and nondecreasing on $(-\infty, 0)$ — and reads its conclusion off the constant $c = k_2(0+) + k_2(0-)$; at $c = \infty$ that conclusion is a C^∞ density on the line. That presentation is itself the hypothesis: the source says of it that it is the general form of a purely non-Gaussian self-decomposable law on $\mathbb{R}$, so exhibiting a law in it is exhibiting it as one. Put $k_2(x) := \frac{1}{2}k(|x|)$. The structural conditions and the exponent identity are the first two paragraphs of the proof of Lemma 3.6, which read only the monotonicity of k and the two integrability conditions of Lemma 2.3: that lemma's hypothesis that $k(0+)$ is finite is used for its two one-sided limits alone, and is not available here. So k_2 is a two-sided profile of the required kind, and its Lévy measure $k_2(x)|x|^{-1}dx$ specifies the same symmetric law as the pair $(0, k)$, whose kernel at canonical scale 1 is ϕ_1 . Both one-sided limits of k_2 at the origin are $+\infty$: on $(0, \infty)$ it is $\frac{1}{2}k$, whose right limit is $+\infty$ by hypothesis, and on $(-\infty, 0)$ it is $x \mapsto \frac{1}{2}k(-x)$, where $x \mapsto -x$ carries the left neighbourhood filter at the origin to the right one. So the source's constant is $c = \infty$ and its clause applies. The one hypothesis the dictionary does not display is the source's normalisation of a k -function to its one-sided-continuous version, discussed after Lemma 3.6: passing to that modification changes k_2 on a countable set, hence changes neither the Lévy measure nor the exponent nor the law, and leaves both one-sided limits infinite.

The two consequences are elementary. A function of class C^∞ on $\mathbb{R}$ is continuous, hence bounded on every compact set, so the representative the theorem produces is bounded on a neighbourhood of the origin. And the symmetric stable member of index α has folded profile $C_\alpha^{-1}x^{-\alpha}$ with $C_\alpha \in (0, \infty)$ (Proposition 5.12), which tends to $+\infty$ as x decreases to 0. $\square$

4 The subordination bridge

The causal theory of [18] classifies the scale spaces of a signal in time, where a measurement may use the past and not the future, under the same weakening of the semigroup to a hemigroup that the line paper applied in space. The two theories have the same shape, and a classical construction connects them, Bochner's subordination ([50], Theorem 30.1, p. 197). Run a Brownian motion for a random time drawn from a causal kernel, and the displacement reached has a symmetric law. This section makes that construction a map from the causal cone into the admissible cone, family to family, and asks what it reaches.

Why the bridge. Two later results need it, and one question is answered by it. The Student-t family of §5 is defined as the image of the causal Bessel corner, and its properties are read off through the map; the exact implementation of the Matérn family in §7 is the causal Gamma cascade carried across, its rational transform becoming a rational spectrum. The question is which of the two theories is larger. The bridge is injective, since a subordinated exponent determines its causal ancestor (Proposition 4.6(1)), so the spatial theory contains a copy of the causal one. It is not onto. The Cin rays of §3, and every member with no Gaussian part whose profile is a lattice step, have no causal ancestor. So the symmetric theory is strictly larger than the causal theory in Brownian disguise. The section then places the image inside a chain of

classes indexed by dimension, between the image and the cone, and shows that the chain is strict at its bottom.

The map of the section. §4.1 recalls the causal theory in the form used here, one definition. §4.2 is the construction: on exponents the bridge is a substitution, $F(\omega) = F_1(\omega^2/2)$, and Lemma 4.2 says that it lands in the admissible cone and computes the profile of the image. §4.3 lifts the map to families and follows the three causal corners across (Proposition 4.5, Figure 3). §4.4 proves that the map is not onto and says which members of the image are the named families (Proposition 4.6 and two lemmas). §4.5 measures the gap between the image and the cone by a chain of classes indexed by dimension (Proposition 4.9), §4.6 gives the one explicit family that shows the chain is strict at the bottom (Lemma 4.12), and §4.7 lists what else of the causal theory crosses. Throughout, W is a standard Brownian motion, so $\mathbb{E} \cos(\omega W_u) = e^{-u\omega^2/2}$, and g_u is its law at time u , with $g_0 = \delta_0$; a different diffusion constant is a change of spatial gauge and is not tracked.

4.1 The causal theory, in brief

The causal article axiomatizes the measurement of a temporal signal at every scale at once, on the same pattern as §2.2. The operators are linear and positive, they form a two-parameter hemigroup in scale, and they are continuous and scale covariant. In place of translation and reflection covariance in space they have translation covariance in time together with causality: the output at a time depends on the input up to that time only.

Its characterization theorem ([18], Theorem 7.3) is the twin of Theorem 2.5. An admissible causal family is, in its canonical gauge, a family of convolutions by laws on the half-line. The kernel at scale x is the law of xT_1 for one random delay $T_1 \geq 0$, and the delays that occur are exactly the self-decomposable laws on the half-line.

The dictionary between the two theories is short. Where a law on the line is described by its cosine transform and an exponent F , a law on the half-line is described by its Laplace transform and a Laplace exponent, $\mathbb{E} e^{-\sigma T_1} = e^{-F_1(\sigma)}$. The self-decomposable delays are those whose exponent is of the form below: a drift b_0 , the deterministic part of the delay, plus a catalogue of random delays of every size u arriving at rate $k_1(u) du/u$, with k_1 nonincreasing. This is the form of §2.3 with $1 - e^{-\sigma u}$ in place of $1 - \cos \omega x$. The Gaussian coefficient a of the line has the drift b_0 as its counterpart, and the folded profile k has the delay profile k_1 . The causal theory has three named corners. The Gamma family has exponent $\gamma \log(1 + \sigma)$ and the Gamma laws as kernels. The Bessel family has inverse-gamma laws as kernels, and its scale evolution is a local equation. The stable family is the semigroup case.

One word of the definition needs a gloss. A *Bernstein function* is a nonnegative smooth function on $(0, \infty)$ whose derivative is completely monotone, that is whose derivatives alternate in sign ([52], Def. 3.1 and Thm. 3.2, p. 21). Except for Proposition 4.5, which starts from a causal family, the statements of this section read nothing of the causal axioms. What they read is the class of exponents, recorded as a definition.

Definition 4.1 (the causal admissible cone). A function $F_1 : [0, \infty) \rightarrow [0, \infty)$ is *causally admissible* if

$$F_1(\sigma) = b_0 \sigma + \int_0^\infty (1 - e^{-\sigma u}) k_1(u) \frac{du}{u}, \quad b_0 \geq 0, \quad k_1 : (0, \infty) \rightarrow [0, \infty) \text{ nonincreasing},$$

with $\int_0^1 k_1(u) du < \infty$ and $\int_1^\infty k_1(u) u^{-1} du < \infty$. These are exactly the exponents the time-causal article's characterization theorem attaches to its axioms, in its canonical gauge; (b_0, k_1) is its data, its drift and its delay profile. Such an F_1 is a Bernstein function vanishing at the origin, so e^{-F_1} is completely monotone with value 1 there; that it is therefore the Laplace transform of the law of a random delay $T_1 \geq 0$ — the subordination reading of this chapter — is cited at Remark 4.3 and is not asserted here.

The functions of the displayed form are Bernstein functions. Conversely every Bernstein function vanishing at the origin has such a representation with a measure in place of $k_I(u) du/u$ ([52], Thm. 3.2, p. 21). So what the definition adds to being a Bernstein function is that the catalogue has a density and that the density is nonincreasing, which is the self-decomposability of the delay. The two integrability conditions are the causal counterpart of Lemma 2.3.

4.2 The bridge on exponents

The construction of this section, in its simplest form, is a substitution. Take a causally admissible exponent F_I and put

$$\Psi(\omega) := F_I(\tfrac{1}{2}\omega^2).$$

The reason this is the right substitution is the Brownian motion: its law at time u has cosine transform $e^{-u\omega^2/2}$, so replacing the Laplace variable σ by $\omega^2/2$ is what one gets by running W for a random time T_I and asking for the law of the displacement, $\mathbb{E} \cos(\omega W_{T_I}) = \mathbb{E} e^{-T_I \omega^2/2} = e^{-F_I(\omega^2/2)}$. The lemma shows three things about Ψ . It is admissible, that is of the form (2.3): its Gaussian coefficient is half the causal drift, and its folded profile is the delay catalogue seen through the Gaussian,

$$k(x) = 2x \int_0^\infty g_u(x) k_I(u) \frac{du}{u} :$$

a delay of size u produces a Gaussian displacement of variance u , and the profile is the mixture of these over the catalogue, folded onto the half-line. The profile is nonincreasing, which is the one property not visible in that formula and is what makes the image self-decomposable. And the law of W_{T_I} is the law with exponent Ψ , which is what justifies the name. The lemma states the three, with the delay entering the third as a hypothesis (Remark 4.3).

Lemma 4.2 (the bridge on exponents). *Let F_I be causally admissible with data (b_0, k_I) and put*

$$\Psi(\omega) := F_I(\tfrac{1}{2}\omega^2), \quad \omega \in \mathbb{R}. \quad (4.1)$$

Then Ψ is admissible in the sense of Theorem 2.5, that is of the form (2.3), with Gaussian coefficient $a = \frac{1}{2}b_0$ and folded profile

$$k(x) = 2x \int_0^\infty g_u(x) k_I(u) \frac{du}{u}, \quad x > 0, \quad (4.2)$$

the Gaussian mixture of the delay catalogue. Moreover, if $T_I \geq 0$ is a random delay independent of W with $\mathbb{E} e^{-\sigma T_I} = e^{-F_I(\sigma)}$ for every $\sigma \geq 0$, then the law of W_{T_I} — the mixture of the Gaussian laws against the law of T_I — is a symmetric probability law with cosine transform $e^{-\Psi}$.

The construction is classical, and its two halves have different sources. The profile (4.2) is [51]’s (3.3), p. 8, in the two-sided convention: his $k^\sharp(x) = x \int_0^\infty (2\pi s)^{-1/2} e^{-x^2/2s} k(s) ds/s$ is half the folded profile above, and his (4.2)–(4.3), p. 9, record the same folding at the origin, $k^\sharp(0+) = k(0+)$ and $c_Y = 2c_Z$. That the image of a self-decomposable delay is self-decomposable is [25], §3, p. 15, for a pure variance mixture, and [51], Theorem 1.1, for a general self-decomposable subordinator. What this lemma adds is the form: it *prescribes* the pair (a, k) and checks it against the requirements of (2.3) one by one, so that no representation theorem is spent and the delay of the last clause is a hypothesis rather than something to be produced. The monotonicity of the profile becomes visible after one substitution.

Proof. Write $\nu_2(x) = \int_0^\infty g_u(x) k_I(u) du/u$ and $k(x) = 2x \nu_2(x)$, and put $a = \frac{1}{2}b_0$. We verify that (a, k) is a pair of the kind (2.3) requires and that its exponent is Ψ ; admissibility is then Theorem 2.5.

ν_2 is finite off the origin. Fix $x \neq 0$. Below $u = 1$ the mixture integrand is dominated by a multiple of k_I : from $y^2 \leq 4e^y$ at $y = x^2/2u$ we get $e^{-x^2/2u} \leq 16u^2/x^4$, and $u \leq \sqrt{2\pi u}$ for

$u \leq 1$, so $g_u(x)/u \leq 16/x^4$ and the integrand is at most $(16/x^4) k_I(u)$, integrable by $\int_0^1 k_I < \infty$. Above $u = 1$, $g_u \leq 1$ and the integrand is at most $k_I(u)u^{-1}$, integrable by the second condition of Definition 4.1. The two conditions pay in different places, and the first pays only because of the Gaussian: the mixture carries du/u where that condition carries du , and the missing power of u is supplied by the flatness of $g_u(x)$ as $u \downarrow 0$ at a fixed $x \neq 0$.

k is nonincreasing. This is the clause that is not visible term by term. It becomes visible in the right variable. Substituting $u = x^2v$, which leaves du/u invariant,

$$x \nu_2(x) = \int_0^\infty \frac{e^{-1/2v}}{v\sqrt{2\pi v}} k_I(x^2v) dv, \quad x > 0. \quad (4.3)$$

The weight no longer involves x ; the whole dependence sits in $k_I(x^2v)$, which is nonincreasing in $x > 0$ for each fixed $v > 0$ because k_I is nonincreasing. Hence $k = 2x\nu_2(x)$ is nonincreasing on $(0, \infty)$.

The Lévy condition. Since $\int_{\mathbb{R}} (1 \wedge x^2) g_u(x) dx$ is at most the total mass 1 and at most the variance u , Tonelli gives

$$\int_0^\infty (1 \wedge x^2) k(x) \frac{dx}{x} = \int_0^\infty \left(\int_{\mathbb{R}} (1 \wedge x^2) g_u(x) dx \right) k_I(u) \frac{du}{u} \leq \int_0^\infty (1 \wedge u) k_I(u) \frac{du}{u} < \infty,$$

the last integral being the single-integral reading of the two conditions of Definition 4.1. By Lemma 2.3 this is the pair of conditions (2.3) asks of k . So the spatial Lévy condition *is* the causal integrability condition, with the Gaussian performing the truncation.

The exponent. Since $e^{-u\omega^2/2} = \int \cos(\omega x) g_u(x) dx$ and $\int g_u = 1$, we have $1 - e^{-u\omega^2/2} = \int_{\mathbb{R}} (1 - \cos \omega x) g_u(x) dx$, and Tonelli gives

$$a\omega^2 + \int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x} = \frac{1}{2}b_0\omega^2 + \int_0^\infty (1 - e^{-u\omega^2/2}) k_I(u) \frac{du}{u} = F_I(\frac{1}{2}\omega^2) = \Psi(\omega),$$

the folding factor 2 in $k = 2x\nu_2$ being exactly the factor the half-line reading of an even integral needs.

The mixture reading. Let $T_1 \geq 0$ be as in the statement. The mixture of the Gaussian laws against the law of T_1 is a probability measure, each g_u being one, and it is symmetric, each g_u being symmetric and symmetry passing through the mixture set by set. Conditioning on T_1 and using $\mathbb{E} \cos(\omega W_u) = e^{-u\omega^2/2}$,

$$\mathbb{E} \cos(\omega W_{T_1}) = \mathbb{E} e^{-T_1\omega^2/2} = e^{-F_I(\omega^2/2)} = e^{-\Psi(\omega)},$$

the middle equality being the hypothesis read at $\sigma = \omega^2/2$. So the law of W_{T_1} is a symmetric probability law with cosine transform $e^{-\Psi}$. Nothing here reads where the delay law comes from: only its Laplace transform is used. $\square$

Remark 4.3 (the bridge is subordination). A delay of the kind the lemma's second clause hypothesises always exists, and that is what makes the bridge an instance of Bochner's subordination rather than a formal identity between exponents. A causally admissible F_I is a Bernstein function vanishing at the origin, so e^{-F_I} is completely monotone — [52], Thm. 3.7(iii) — with value 1 at the origin, and a completely monotone function with value 1 at the origin is the Laplace transform of a probability law on $[0, \infty)$ — [52], Thm. 1.4, p. 3. That law is the law of a random delay $T_1 \geq 0$, unique because a law on the half-line is determined by its Laplace transform, and the lemma then says that the symmetric family attached to Ψ is the family of laws of W_{T_1} : Brownian motion run for the causal delay. The catalogue formula (4.2) is the same statement on profiles. Nothing in the paper depends on the two cited facts: every consumer of the bridge quantifies over the delay law instead.

Remark 4.4 (why the image is self-decomposable, in one line). There is a second route to the admissibility clause, which produces the profile instead of checking a prescribed one and shows in one line *why* the image is self-decomposable. For $c \in (0, 1)$ the function $g = F_1 - F_1(c^2 \cdot)$ is again a Bernstein function vanishing at the origin, so $e^{-\tau g}$ is completely monotone for every $\tau > 0$ and hence the Laplace transform of a probability law ρ_τ on $[0, \infty)$; then $e^{-\tau(\Psi(\omega) - \Psi(c\omega))} = \int_{[0, \infty)} e^{-u\omega^2/2} \rho_\tau(du)$ is a positive mixture of positive definite functions, so $\Psi - \Psi(c \cdot) \in \text{ND}_s$ for every c , which is the self-decomposability condition of the line paper's Lemma 7.1 and produces (2.3). The route rests on the two cited facts of the previous remark; the proof above rests on none, which is why it is the proof of record.

4.3 The bridge on families, and the corners

The map on exponents lifts to a map on families. Given a causal family in its canonical gauge, mix its kernel from causal scale s^2 to causal scale t^2 against the Gaussian laws and call the result the spatial kernel from s to t . The proposition says that this is an admissible family on the line, in canonical gauge, with exponent $F_1(\omega^2/2)$, and it follows the named members across:

- the causal pure delay becomes the Gaussian;
- the Gamma family becomes the Matérn family;
- the Bessel family becomes the Student-t family;
- the stable family of index α becomes the symmetric stable family of index 2α ;
- a causal scale x becomes the spatial scale $\sqrt{x}$.

Figure 3 shows two of the corners crossing.

Proposition 4.5 (the bridge on families; the corners; the gauges). *Let $(\Phi_{x,y}^I)$ be a time-causal family satisfying the causal axioms, with kernels $\mu_{x,y}^I$ on $[0, \infty)$ written in its canonical gauge and exponent F_1 . Define, for $0 \leq s \leq t$,*

$$\mu_{s,t} := \int_{[0, \infty)} g_u \mu_{s^2, t^2}^I(du), \quad \Phi_{s,t} f := \mu_{s,t} * f. \quad (4.4)$$

Then $(\Phi_{s,t})$ satisfies (A1)–(A8) and (ND) with $S_\lambda t = \lambda t$; the coordinate t is its canonical gauge, and its exponent is $F = F_1(\cdot^2/2)$. In particular:

1. *the causal pure delay, $\mu_{x,y}^I = \delta_{y-x}$, maps to the Gaussian family of Proposition 3.3 of [17], $\mu_{s,t} = g_{t^2-s^2}$; the causal drift ray $b_0\sigma$ maps to the Gaussian ray $\frac{1}{2}b_0\omega^2$;*
2. *the causal Gamma family, $F_1(\sigma) = \gamma \log(1+\sigma)$, maps to the Matérn family at range $1/\sqrt{2}$: $F(\omega) = \gamma \log(1 + \omega^2/2)$, folded profile $k(x) = 2\gamma e^{-\sqrt{2}x}$;*
3. *if the inverse-gamma laws of shape a are the delay kernels of such a family — the causal Bessel family, whose existence is cited at Remark 5.11 and asserted nowhere here — then it maps to the Student-t family with $2a$ degrees of freedom. The mixture identity behind that clause is unconditional: mixing the Gaussian kernels against the inverse-gamma law of shape a gives the Student-t law with $2a$ degrees of freedom, which is Proposition 5.10(1);*
4. *the causal stable family, $F_1(\sigma) = \sigma^\alpha$ with $0 < \alpha \leq 1$, maps to the symmetric stable family $F(\omega) = 2^{-\alpha}|\omega|^{2\alpha}$ of index $2\alpha \in (0, 2]$; the boundary $\alpha = 1$ is item (1);*
5. *the spatial canonical gauge is the square root of the causal one: a causal scale x becomes the spatial scale $t = \sqrt{x}$, which is the parabolic gauge of the causal article.*

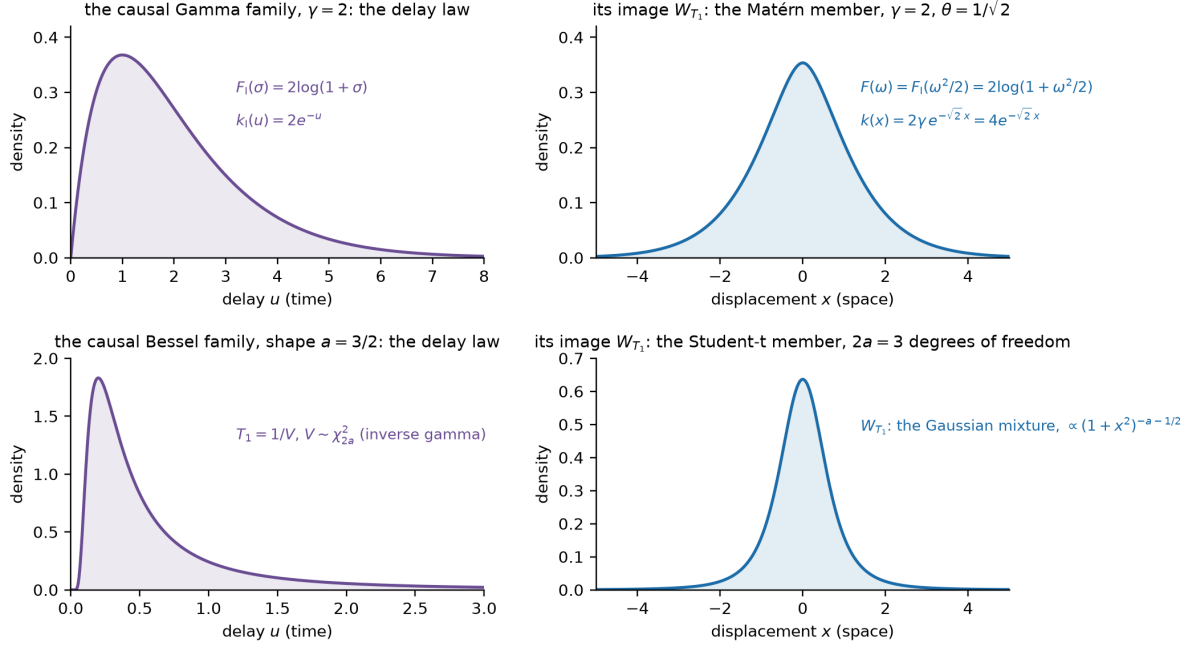


Figure 3: The bridge on two corners, at canonical scale 1. Left, the causal kernel: the law of the delay T_1 on the half-line. Right, its image: the law of W_{T_1} , Brownian motion run for that delay, on the line. Top row, the causal Gamma family at $\gamma = 2$, delay density ue^{-u} , whose image is the Matérn member of index 2 at range $1/\sqrt{2}$ (Proposition 4.5(2)); the exponents are related by the substitution $\sigma = \omega^2/2$ and the profiles by the Gaussian mixture (4.2). Bottom row, the causal Bessel family at shape $a = \frac{3}{2}$, the inverse-gamma delay $T_1 = 1/V$ with V chi-square on 3 degrees of freedom, whose image is the Student-t member with 3 degrees of freedom in the scale of Proposition 5.10, with density $(2/\pi)(1 + x^2)^{-2}$ (clause (3)). The heavy tail of the delay becomes the heavy tail of the displacement.

The proof has one step beyond Lemma 4.2, the identification of the mixtures (4.4) with the kernels the characterization theorem attaches to F , and the corners are then computations on profiles and exponents; item (2) is computed from the causal *profile*, so that the Gamma exponent comes out rather than going in.

Proof. The family. $\hat{\mu}_{s,t}(\omega) = \int e^{-u\omega^2/2} \mu_{s^2,t^2}^I(du) = \widehat{\mu_{s^2,t^2}^I}(\omega^2/2) = \exp[-(F_1(t^2\omega^2/2) - F_1(s^2\omega^2/2))]$, using the causal similarity form in the canonical gauge. So $g_{s,t}(\omega) = F(t\omega) - F(s\omega)$ with $F = F_1(\cdot^2/2)$, admissible by Lemma 4.2; the family is therefore the one Theorem 2.5 attaches to F in the canonical gauge, and that construction verified (A1)–(A8) and (ND), with $S_\lambda t = \lambda t$ because $F(\lambda t\omega) - F(\lambda s\omega) = g_{s,t}(\lambda\omega)$. Directly: $D_\lambda g_u = g_{\lambda^2 u}$, so a spatial dilation by λ is a causal dilation by λ^2 , and the causal covariance $D_{\lambda^2} \mu_{x,y}^I = \mu_{\lambda^2 x, \lambda^2 y}^I$ becomes $D_\lambda \mu_{s,t} = \mu_{\lambda s, \lambda t}$; the cascade is inherited from the independent increments of both $\tilde{W}$ and the delay.

Two identifications in that paragraph need justification, and they are where the work is once Lemma 4.2 is in hand. First, the kernels Theorem 2.5 attaches to F and the mixtures (4.4) are *a priori* two different families; they agree because both are symmetric probability measures and their cosine transforms agree, so Proposition 2.7 of [17] identifies them. Second, the scaling action is $S_\lambda t = \lambda t$ on the nose, not merely up to the gauge: the construction of Theorem 2.5 produces the gauge conjugate $\chi^{-1}(\lambda\chi(t))$, which in the canonical gauge $\chi = \text{id}$ is λt for every $t \geq 0$, and every clause of (A8) reads S_λ on $[0, \infty)$ only.

(1) $\mu_{x,y}^I = \delta_{y-x}$ gives $\mu_{s,t} = g_{t^2-s^2}$, and $F_1(\sigma) = b_0\sigma$ gives $F(\omega) = \frac{1}{2}b_0\omega^2$.

(2) Take the causal Gamma family by its *profile*, $k_1(u) = \gamma e^{-u}$ with no drift; the exponent then comes out of the computation rather than into it. By (4.2) the image has folded profile

$$k(x) = 2\gamma x \int_0^\infty g_u(x) e^{-u} \frac{du}{u} = \frac{2\gamma x}{\sqrt{2\pi}} \int_0^\infty u^{-3/2} e^{-x^2/2u} e^{-u} du = 2\gamma e^{-\sqrt{2}x},$$

by the identity $\int_0^\infty u^{-3/2} e^{-A/u-Bu} du = \sqrt{\pi/A} e^{-2\sqrt{AB}}$ at $A = x^2/2$, $B = 1$ — equivalently, the Laplace transform $\int_0^\infty \frac{x}{\sqrt{2\pi u^3}} e^{-x^2/2u} e^{-\sigma u} du = e^{-x\sqrt{2\sigma}}$ of the first-passage law of Brownian motion at level x , at $\sigma = 1$. That is the folded profile $2\gamma e^{-x/t}$ of Proposition 5.2 at $t = 1/\sqrt{2}$. The exponent now follows without a second integral: (4.1) says that $F_1(\omega^2/2)$ is the exponent of the admissible datum whose folded profile is (4.2), and that profile has just been identified, so Proposition 5.2(1) evaluates it and $F(\omega) = \gamma \log(1 + \omega^2/2) = \gamma \log(1 + t^2\omega^2)$ at $t = 1/\sqrt{2}$. In particular the causal exponent $\gamma \log(1 + \sigma)$ is a consequence of the causal profile and need not be assumed.

(3) is the classical normal variance-mixture representation, computed in Proposition 5.10(1); that identity needs no hypothesis on the causal side. Under the clause's hypothesis — that the inverse-gamma law of shape a is the delay law of a causally admissible F_1 — Lemma 4.2 makes the image admissible and self-decomposable, and Lemma 4.8 is the same statement read as membership in $\mathcal{S}$.

(4) $(\omega^2/2)^\alpha = 2^{-\alpha}|\omega|^{2\alpha}$, and $\alpha \leq 1$ maps onto index at most 2.

(5) The causal accumulated exponent in the canonical gauge is $F_1(x\sigma)$, and the image has $\hat{\mu}_{0,t}(\omega) = e^{-F_1(t^2\omega^2/2)} = e^{-F(t\omega)}$, so the spatial gauge t satisfies $t^2 = x$. $\square$

The causal article names its parabolic gauge because in it the memory-line evolution of its Bessel family is a Bessel operator, a heat-type generator; the reason is now visible from the other side, that parameter being the spatial scale of the subordinated family, with the factor $\frac{1}{2}$ from Brownian scaling. The two articles' natural coordinates agree through $x = t^2$, causal scale being a variance and spatial scale a standard deviation — which is also why the scale-space literature's diffusion variable is the square of this paper's t (Remark 6.7 of [17]).

4.4 Strictness: the bridge is not onto

The image of the bridge is a subcone of the admissible cone, and the proposition says what separates it from the whole: a subordinated exponent is strictly increasing on the half-line, because it is a Bernstein function of ω^2 , whereas an admissible exponent may have stationary points, and exactly the ones with no Gaussian part whose profile is a step function with all its jumps on a lattice do. The Cin rays are the simplest case, one jump. So the extreme rays of the cone, other than the Gaussian, have no causal ancestor, and the image is strictly smaller than the cone. The two lemmas that follow say which named families do lie in the image: the Gaussian ray and the Matérn and stable families unconditionally, and the Student-t family under a hypothesis on its delay law that the literature supplies.

Proposition 4.6 (strictness: the bridge is not onto). *Call an admissible F subordinated if $F(\omega) = F_1(\omega^2/2)$ for some causally admissible F_1 , and write $\mathcal{S}$ for the set of subordinated exponents.*

1. F is subordinated if and only if $\sigma \mapsto F(\sqrt{2\sigma})$ is causally admissible.
2. Every subordinated F other than 0 is strictly increasing on $(0, \infty)$, with $F' > 0$ there.
3. No Cin ray is subordinated, since $\text{Cin}'(z) = (1 - \cos z)/z$ vanishes at every $z \in 2\pi\mathbb{Z} \setminus \{0\}$. More generally, an admissible F has a stationary point $\omega_0 \neq 0$ if and only if $a = 0$ and the Choquet measure ϖ of Proposition 3.3 is carried by the lattice $\frac{2\pi}{|\omega_0|}\mathbb{Z}$, that is, if and only if $a = 0$ and the profile k is a step function with all its jumps on that lattice; and no such F with $F \not\equiv 0$ is subordinated.

4. $\mathcal{S}$ is a convex subcone of the admissible cone, closed under the dilations $F \mapsto F(\lambda \cdot)$; it contains none of the extreme rays $\text{Cin}(\tau \cdot)$.

Proof. (1) is Definition 4.1 read on $F_I(\sigma) := F(\sqrt{2\sigma})$.

(2) Let $F = F_I(\cdot^2/2)$ with F_I causally admissible, and suppose $F \neq 0$. Fix $\sigma > 0$. On $(1, \infty)$ the elementary bound $ue^{-\sigma u/2} \leq 2/\sigma$ turns $e^{-\sigma u/2}k_I(u)$ into $(2/\sigma)k_I(u)u^{-1}$, integrable by the second condition of Definition 4.1, and on $(0, 1)$ it is at most k_I , integrable by the first; so $e^{-\sigma u/2}k_I(u)$ dominates the σ -derivative $e^{-\sigma u}k_I(u)$ of the integrand uniformly for σ in a neighbourhood of the given one, and differentiation under the integral sign gives

$$F'_I(\sigma) = b_0 + \int_0^\infty e^{-\sigma u} k_I(u) du, \quad \sigma > 0.$$

Both terms are nonnegative. If the sum vanishes at one $\sigma > 0$ then $b_0 = 0$ and, the exponential being strictly positive, $k_I = 0$ almost everywhere on $(0, \infty)$; but then the representation makes F_I , and so F , identically zero. Hence $F'_I > 0$ on $(0, \infty)$, and the chain rule gives $F'(\omega) = \omega F'_I(\omega^2/2) > 0$ for $\omega > 0$. A function continuous on $(0, \infty)$ with positive derivative there is strictly increasing there, and F is continuous, its representation (2.3) being so by dominated convergence.

The derivative formula is the only place the causal representation is differentiated, and one classical step does *not* enter it. Read as a function of σ , F'_I is the Laplace transform of the positive measure $b_0\delta_0 + k_I(u) du$, so it is completely monotone and one may argue, as the causal article does, that a completely monotone function vanishing at an interior point vanishes identically. That is Bernstein's theorem, and it is not needed: it *produces* a representing measure, and Definition 4.1 has already given one. What is left is the positivity of an integral, which is the display above.

(3) $\text{Cin}'(2\pi n) = 0$ for every nonzero integer n , which contradicts (2); so no Cin ray is subordinated. For the general statement, (2.4) gives $\omega F'(\omega) = B(\omega) = 2a\omega^2 + \int (1 - \cos \omega u) \varpi(du)$, a sum of nonnegative terms, which vanishes at $\omega_0 \neq 0$ if and only if $a = 0$ and $\cos(\omega_0 u) = 1$ for ϖ -almost every u ; the latter says, by the argument of Lemma 2.11 of [17] applied to ϖ , that ϖ is carried by $\frac{2\pi}{|\omega_0|}\mathbb{Z}$, which is the profile $k = \varpi((\cdot, \infty))$ being a step function with jumps on that lattice. Such an F has a stationary point off the origin, so it is not subordinated, by (2).

(4) $\mathcal{S}$ is the image of the causal cone under $F_I \mapsto F_I(\cdot^2/2)$, so the three closure properties are closure properties of the causal cone, and the map carries them across because it is linear and because $F_I(\lambda^2 \cdot)$ maps to $F(\lambda \cdot)$ — a spatial dilation by λ is a causal dilation by λ^2 . On the causal side, sums and nonnegative multiples act on the data (b_0, k_I) termwise and the two conditions of Definition 4.1 are preserved term by term. The dilation $F_I \mapsto F_I(c \cdot)$, $c > 0$, has data $(cb_0, k_I(\cdot/c))$, and for it the two conditions are better read as the single one

$$\int_0^\infty (1 \wedge u) k_I(u) \frac{du}{u} < \infty,$$

which they are equivalent to, $(1 \wedge u)/u$ being 1 below 1 and u^{-1} above it. The change of variables $u = cx$ leaves du/u invariant, so it transforms that integral into $\int_0^\infty (1 \wedge cx) k_I(x) dx/x$, and $1 \wedge cx \leq \max(1, c)(1 \wedge x)$ bounds it by $\max(1, c)$ times the undilated one. Splitting the two windows first would instead require the change of variables on two moving intervals. The exclusion of the Cin rays is (3), whose Choquet coordinates are Proposition 3.3. $\square$

The next lemma exhibits a causal ancestor for each of three named families.

Lemma 4.7 (the subordinated slice's members). *$\mathcal{S}$ contains the Gaussian ray and the Matérn and symmetric stable families.*

Proof. Membership in $\mathcal{S}$ asks for a causally admissible datum (b_0, k_1) whose exponent, read at $\omega^2/2$, is the given exponent. One datum is exhibited per family, and the named families at their general parameters are then reached by the closure clauses of Proposition 4.6(4).

The Gaussian ray. For $b_0 \geq 0$ the datum $(b_0, 0)$ satisfies Definition 4.1 with nothing to check, and its exponent is $F_1(\sigma) = b_0\sigma$ for $\sigma \geq 0$. With $b_0 = 2a$ it gives $F_1(\omega^2/2) = a\omega^2$.

The Matérn family. Let $b_0 = 0$ and $k_1(u) = \gamma e^{-u}$ for $u > 0$. The profile is nonincreasing, bounded by γ on $(0, 1)$, and $k_1(u)/u \leq \gamma e^{-u}$ on $(1, \infty)$, so both conditions of Definition 4.1 hold. By Proposition 4.5(2), $F_1(\omega^2/2) = \gamma \log(1 + \omega^2/2)$, the Matérn exponent at range $1/\sqrt{2}$. For $\theta > 0$,

$$\gamma \log(1 + \theta^2 \omega^2) = \gamma \log(1 + \tfrac{1}{2}(\sqrt{2}\theta\omega)^2),$$

so the Matérn exponent at range θ is the dilation by $\lambda = \sqrt{2}\theta$ of the one at range $1/\sqrt{2}$, and it lies in $\mathcal{S}$ because $\mathcal{S}$ is closed under dilations.

The symmetric stable family. Let $0 < \beta < 1$ and $c_\beta = \beta/\Gamma(1 - \beta)$, and take $b_0 = 0$ and $k_1(u) = c_\beta u^{-\beta}$ for $u > 0$. The profile is nonincreasing, and $\int_0^1 u^{-\beta} du = (1 - \beta)^{-1}$ and $\int_1^\infty u^{-\beta-1} du = \beta^{-1}$ are finite, the first because $\beta < 1$ and the second because $\beta > 0$. Its exponent is

$$F_1(\sigma) = \sigma^\beta, \quad \sigma \geq 0.$$

The profile is homogeneous: $k_1(u/c) = c^\beta k_1(u)$ for $c > 0$. So the dilated datum $(cb_0, k_1(\cdot/c))$ of the proof of Proposition 4.6(4), whose exponent is $F_1(c \cdot)$, is the datum $(c^\beta b_0, c^\beta k_1)$, whose exponent is $c^\beta F_1$; hence $F_1(\sigma) = \sigma^\beta F_1(1)$ for every $\sigma > 0$. The two sides agree on a neighbourhood of $\sigma = 1$, so their derivatives there agree: the right side has derivative $\beta F_1(1)$, and by the derivative formula in the proof of Proposition 4.6(2)

$$F_1'(1) = \int_0^\infty e^{-u} c_\beta u^{-\beta} du = c_\beta \Gamma(1 - \beta) = \beta.$$

Hence $F_1(1) = 1$, and $F_1(0) = 0$ by the representation. By Proposition 4.5(4), $F_1(\omega^2/2) = 2^{-\beta}|\omega|^{2\beta}$. Given $0 < \alpha < 2$, take $\beta = \alpha/2$: then $|\omega|^\alpha = 2^{\alpha/2}F_1(\omega^2/2)$ is a positive multiple of a member of $\mathcal{S}$, and lies in $\mathcal{S}$, which is a cone. $\square$

For the Student-t family the delay is the inverse-gamma law, and what has to be known is that this law is the delay law of a causally admissible exponent, that is that it is self-decomposable on the half-line. That is a theorem of Halgreen's ([25], §2, pp. 14–15, recorded as Remark 5.11). The lemma takes it as a hypothesis and proves the membership from it.

Lemma 4.8 (the Student-t family is subordinated). *Let $a > 0$, let T_1 be the inverse-gamma law of shape a of Proposition 5.10, and suppose T_1 is the delay law of a causally admissible exponent: that for some F_1 as in Definition 4.1,*

$$\mathbb{E}e^{-\sigma T_1} = e^{-F_1(\sigma)}, \quad \sigma \geq 0.$$

Then $\mathcal{S}$ contains the Student-t family of shape a , with F_1 as its causal ancestor. That T_1 is such a delay law is cited at Remark 5.11 and is not asserted here.

Proof. Conditioning the Brownian kernel on T_1 , the Student-t law of shape a is the mixture of the Gaussian laws against the law of T_1 — Proposition 4.5(3) — so its cosine transform at ω is $\mathbb{E}e^{-T_1\omega^2/2}$, which is the mixture identity of Lemma 4.2. The hypothesis read at $\sigma = \omega^2/2$ turns that into $e^{-F_1(\omega^2/2)}$. So an exponent F of the Student-t family — one with $e^{-F(\omega)} = \hat{\phi}_1(\omega)$ at every frequency — satisfies $e^{-F(\omega)} = e^{-F_1(\omega^2/2)}$ there, hence $F(\omega) = F_1(\omega^2/2)$ pointwise, the exponential being injective. That is membership in $\mathcal{S}$, with F_1 as the ancestor. $\square$

The criterion in clause (3) of Proposition 4.6 is the second appearance of one phenomenon. The first is in the line paper. A zero of an *exponent* at $\omega_0 \neq 0$ means that the law is carried by the lattice $\frac{2\pi}{\omega_0}\mathbb{Z}$ (Lemma 2.11 of [17]), and scale covariance excludes such laws (Lemma 6.2 of [17]). Here a zero of the *symbol* $\omega F'$ at $\omega_0 \neq 0$ means a lattice *profile*, a step function whose jumps lie on a lattice, that is a tail measure ϖ carried by the lattice, and no axiom excludes it. The Cin rays are the case of a single step. What excludes it is subordination, because a Gaussian mixture spreads every delay over the whole line.

Symmetric self-decomposable laws with a lattice-step profile are therefore members of the spatial theory with no causal ancestor. The converse — that a member with no causal ancestor must have a step profile — is not asserted, and it fails: an admissible F whose *tail measure* $\varpi = -dk$ is carried by a compact subset of $(0, \infty)$ and has no atom there, equivalently whose profile is continuous and of bounded support, has no ancestor either, because a subordinated profile is a Gaussian mixture of the delay catalogue ((4.2)) and is therefore positive at every $x > 0$. (The catalogue itself is carried by no compact subset of $(0, \infty)$: a nonincreasing $k \neq 0$ is positive on a neighbourhood of the origin.) The next two subsections place $\mathcal{S}$ inside a chain of larger classes.

4.5 The dimension filtration

Clause (1) of Proposition 4.6 turns membership of $\mathcal{S}$ into a condition on one function of one variable: F is subordinated exactly when $\sigma \mapsto F(\sqrt{2\sigma})$ is causally admissible. Definition 4.1 asks two things of that function, that it be a Bernstein function vanishing at the origin and that its delay profile be nonincreasing, and dropping the second leaves a condition with a geometric meaning. A theorem of Schoenberg's, the *radial theorem*, says, up to a dilation of ξ that changes nothing, that $f(\sigma) = F(\sqrt{2\sigma})$ is a Bernstein function exactly when the radial extension $\xi \mapsto F(|\xi|)$ is a Lévy exponent on $\mathbb{R}^d$ for every dimension d at once. So the larger class is the set of line exponents that are radial profiles of isotropic infinitely divisible laws in every dimension, and the question how far $\mathcal{S}$ is from the admissible cone can be asked dimension by dimension: for which d is $F(|\xi|)$ the exponent of a self-decomposable law on $\mathbb{R}^d$? This is the natural question for a scale-space theory, since the isotropic scale spaces of the plane and of space have kernels of exactly that form, and their restrictions to a line are members of the line's theory: the classes below are, heuristically, the exponents that come from an isotropic admissible family in dimension d . The proposition says that the classes form a chain, that the image of the bridge sits at the bottom of every one of them, that the lattice-step members with no Gaussian part lie outside their intersection, and that the intersection is exactly the image of the bridge. Figure 4 draws the chain with what is known about it and what is not.

The cited facts. The radial theorem is [52], Thm. 13.14, p. 212: a function $f : (0, \infty) \rightarrow [0, \infty)$ is a Bernstein function if and only if, for all $d \in \mathbb{N}$, the function $\xi \mapsto f(|\xi|^2)$ on $\mathbb{R}^d$ is continuous and negative definite; the direction from every dimension to Bernstein, which is the one used, is proved by a limit in the dimension. With it comes the representation of a Bernstein function with a unique triplet, Thm. 3.2, p. 21, of the same source. The other facts are the ones the line paper already cites, read in dimension d where their sources state them: the Lévy–Khintchine representation on $\mathbb{R}^d$ with its converse and uniqueness ([50], Thm. 8.1, pp. 37–38), the definition of a self-decomposable law on $\mathbb{R}^d$, that such a law is infinitely divisible, and the polar criterion, that an infinitely divisible law on $\mathbb{R}^d$ is self-decomposable if and only if its Lévy measure has, along every ray, a nonincreasing profile in the polar coordinate ([50], Def. 15.1, p. 90, Prop. 15.5, p. 93, Thm. 15.10, p. 95), and that the exponent of an infinitely divisible law is negative definite in the kernel sense ([52], Prop. 4.4, p. 36). Clause (4) uses three more. The remainder ρ_b in the definition of a self-decomposable law on $\mathbb{R}^d$ is itself infinitely divisible ([50], Prop. 15.5, p. 93). The exponential of the negative of a Bernstein function is completely monotone, and is therefore the Laplace transform of a positive measure ([52], Thm. 3.7, p. 27, and Thm. 1.4, p. 3). And an infinitely divisible law carried by the half-line has

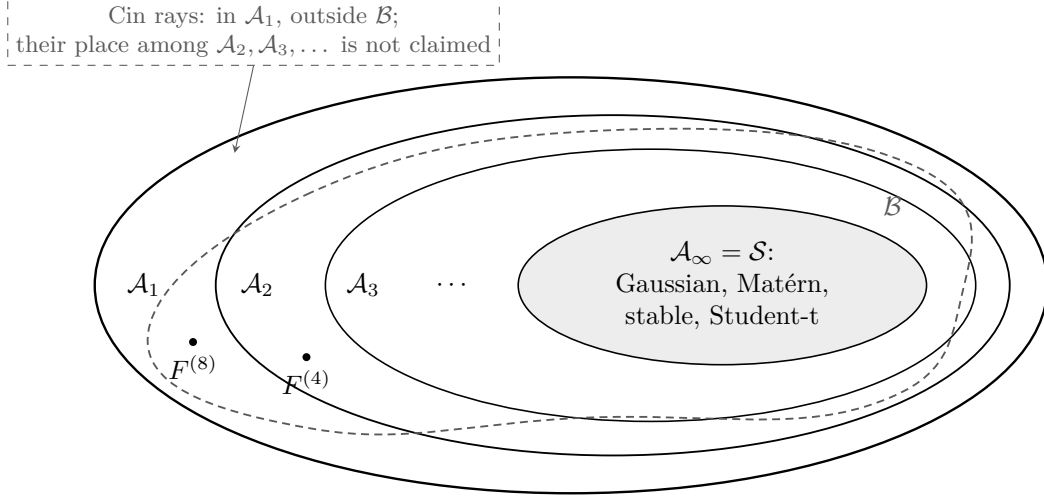


Figure 4: The dimension filtration, schematically. The class $\mathcal{A}_d$ holds the admissible F whose radial extension $F(|\xi|)$ is the exponent of a self-decomposable law on $\mathbb{R}^d$. The classes are nested, $\mathcal{A}_1$ is the whole admissible cone, and their intersection is the image $\mathcal{S}$ of the bridge, which holds the named families, the Student-t family under the hypothesis of Remark 5.11 (Proposition 4.9). The members $F^{(8)}$ and $F^{(4)}$ of Lemma 4.12 show that the first two inclusions are strict, and both lie in the class $\mathcal{B}$ of the Bernstein condition, which contains $\mathcal{A}_\infty$. Nothing is claimed about $\mathcal{B}$ against $\mathcal{A}_d$ for finite $d > 1$, so $\mathcal{B}$ is a dashed contour and not a region of the chain. The Cin rays lie outside $\mathcal{B}$, and the figure places them no more precisely than that.

no Gaussian part, a Lévy measure on $(0, \infty)$ and a nonnegative drift, with the Laplace exponent $\gamma_0 u + \int (1 - e^{-ux}) \nu(dx)$ ([50], Thm. 24.11, p. 153, and Remark 21.6, p. 138). The proposition is proved in prose from them and is not machine-checked; this paper's development has no laws on $\mathbb{R}^d$.

Proposition 4.9 (the dimension filtration). *For an integer $d \geq 1$ let $\mathcal{A}_d$ be the set of admissible F for which $\xi \mapsto F(|\xi|)$, $\xi \in \mathbb{R}^d$, is the exponent of a self-decomposable law on $\mathbb{R}^d$; let $\mathcal{A}_\infty$ be the intersection of all the $\mathcal{A}_d$; and let $\mathcal{B}$ be the set of admissible F for which $\sigma \mapsto F(\sqrt{2}\sigma)$ is a Bernstein function on $(0, \infty)$.*

1. (The bottom, and the nesting.) $\mathcal{A}_1$ is the whole admissible cone, and $\mathcal{A}_{d+1} \subseteq \mathcal{A}_d$ for every $d \geq 1$.
2. (The slice sits in every dimension.) Let (b_0, k_1) be causally admissible with exponent F_1 . For every d the function $\xi \mapsto F_1(|\xi|^2/2)$ on $\mathbb{R}^d$ is the exponent of an isotropic infinitely divisible law, with Gaussian covariance $b_0 I$ and with a Lévy measure of radial density $f_d(r) = \int_0^\infty (2\pi u)^{-d/2} e^{-r^2/2u} k_1(u) du/u$, whose polar profile is

$$|S^{d-1}| r^d f_d(r) = \frac{2}{\Gamma(d/2)} \int_0^\infty v^{d/2-1} e^{-v} k_1\left(\frac{r^2}{2v}\right) dv, \quad r > 0, \quad (4.5)$$

nonincreasing in r ; so that law is self-decomposable, and $\mathcal{S} \subseteq \mathcal{A}_d$. At $d = 1$ the polar profile is the folded profile (4.2).

3. (The top.) $\mathcal{A}_\infty \subseteq \mathcal{B}$. Consequently every $F \in \mathcal{A}_\infty$ with $F \not\equiv 0$ has $F' > 0$ on $(0, \infty)$, and no admissible $F \not\equiv 0$ with Gaussian coefficient 0 whose profile is a step function with its jumps on a lattice — in particular no Cin ray — lies in $\mathcal{A}_\infty$.
4. (The intersection is the slice.) $\mathcal{A}_\infty = \mathcal{S}$: an admissible F whose radial extension is the exponent of a self-decomposable law on $\mathbb{R}^d$ for every d is subordinated.

So $\mathcal{S} = \mathcal{A}_\infty \subseteq \dots \subseteq \mathcal{A}_2 \subseteq \mathcal{A}_1$, and the lattice-step members of the line's theory with no Gaussian part lie outside $\mathcal{A}_\infty$.

Proof. (1) An admissible F is of the form (2.3), which is the symmetric Lévy–Khintchine form of a self-decomposable law on the line (Proposition 2.2), so $F \in \mathcal{A}_1$; and $\mathcal{A}_1$ consists of admissible F by definition. For the nesting let $F \in \mathcal{A}_{d+1}$, let μ be a self-decomposable law on $\mathbb{R}^{d+1}$ with exponent $\xi \mapsto F(|\xi|)$, and let $\Pi : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^d$ be the projection that drops the last coordinate, so that $\Pi^\top$ is an isometric embedding. The image law $\Pi_*\mu$ has transform $\eta \mapsto \hat{\mu}(\Pi^\top \eta) = e^{-F(|\eta|)}$, so its exponent is $\eta \mapsto F(|\eta|)$ on $\mathbb{R}^d$. It is self-decomposable: for $b \in (0, 1)$ the definition gives a law ρ_b on $\mathbb{R}^{d+1}$ with $\hat{\mu}(\xi) = \hat{\mu}(b\xi) \hat{\rho}_b(\xi)$ — the source states it at the reciprocal parameter, quantifying over $1/b > 1$ — for every ξ , and reading this at $\xi = \Pi^\top \eta$ gives $\widehat{\Pi_*\mu}(\eta) = \widehat{\Pi_*\mu}(b\eta) \widehat{\Pi_*\rho_b}(\eta)$, which is the definition on $\mathbb{R}^d$ with $\Pi_*\rho_b$ as the residual. So $F \in \mathcal{A}_d$.

(2) Write $g_u^{(d)}$ for the centred Gaussian density on $\mathbb{R}^d$ of covariance uI , and put $\nu_d(dx) := (\int_0^\infty g_u^{(d)}(x) k_1(u) du/u) dx$ on $\mathbb{R}^d \setminus \{0\}$, a positive measure with radial density f_d . It is a Lévy measure: since $\int_{\mathbb{R}^d} (1 \wedge |x|^2) g_u^{(d)}(x) dx$ is at most the total mass 1 and at most the trace du of the covariance, Tonelli gives $\int (1 \wedge |x|^2) \nu_d(dx) \leq \int_0^\infty (1 \wedge du) k_1(u) du/u \leq d \int_0^\infty (1 \wedge u) k_1(u) du/u < \infty$, the last integral being the single-integral reading of the two conditions of Definition 4.1. Since $e^{-u|\xi|^2/2} = \int_{\mathbb{R}^d} \cos(\xi \cdot x) g_u^{(d)}(x) dx$ and $\int g_u^{(d)} = 1$, a second Tonelli exchange gives

$$F_1(\tfrac{1}{2}|\xi|^2) = \tfrac{1}{2} \xi \cdot (b_0 I) \xi + \int_{\mathbb{R}^d} (1 - \cos \xi \cdot x) \nu_d(dx), \quad \xi \in \mathbb{R}^d,$$

the symmetric Lévy–Khintchine form on $\mathbb{R}^d$ with Gaussian covariance $b_0 I$ and Lévy measure ν_d ; both are invariant under the orthogonal group. By the converse clause of the representation theorem in dimension d this is the exponent of an infinitely divisible law, isotropic because its transform is, and by the uniqueness clause that law's Lévy measure is ν_d . Its polar form is read off the radial density: in polar coordinates $\nu_d(dx) = f_d(r) r^{d-1} dr d\sigma(\theta)$, so the profile along every ray, in the sense of the polar criterion, is $r \mapsto r^d f_d(r)$ up to the constant $|S^{d-1}|$. The substitution $u = r^2/2v$, which leaves du/u invariant, turns $|S^{d-1}| r^d \int_0^\infty (2\pi u)^{-d/2} e^{-r^2/2u} k_1(u) du/u$ into (4.5), using $|S^{d-1}| = 2\pi^{d/2}/\Gamma(d/2)$ and $(2\pi u)^{-d/2} r^d = \pi^{-d/2} v^{d/2}$. The weight $v^{d/2-1} e^{-v}$ no longer involves r , and $k_1(r^2/2v)$ is nonincreasing in r for each fixed v because k_1 is nonincreasing; so the polar profile is nonincreasing, and by the polar criterion the law is self-decomposable. Its exponent restricted to a line is $\omega \mapsto F_1(\omega^2/2)$, which is admissible by Lemma 4.2, so every subordinated F lies in $\mathcal{A}_d$. At $d = 1$, $|S^0| = 2$ and (4.5) is (4.3) with the folding factor in place, that is the folded profile (4.2).

(3) Let $F \in \mathcal{A}_\infty$ and fix d . The law on $\mathbb{R}^d$ with exponent $\xi \mapsto F(|\xi|)$ is self-decomposable, hence infinitely divisible, so $e^{-tF(|\xi|)}$ is a characteristic function for every $t > 0$, hence positive definite, and $\xi \mapsto F(|\xi|)$ is continuous and negative definite in the kernel sense. That holds for all d , and negative definiteness is unchanged by the dilation $\xi \mapsto \sqrt{2}\xi$, so with $f(\sigma) := F(\sqrt{2}\sigma)$ the function $\xi \mapsto f(|\xi|^2) = F(\sqrt{2}|\xi|)$ is continuous and negative definite on every $\mathbb{R}^d$. By the radial theorem f is a Bernstein function on $(0, \infty)$, and $F \in \mathcal{B}$. For the derivative, the representation gives $f(\sigma) = a + b\sigma + \int_{(0, \infty)} (1 - e^{-\sigma u}) m(du)$ with $a, b \geq 0$ and $\int (1 \wedge u) m(du) < \infty$; $a = \lim_{\sigma \downarrow 0} f(\sigma) = F(0) = 0$, F being continuous with $F(0) = 0$. On $[\sigma_0, \infty)$ for any $\sigma_0 > 0$ the σ -derivative $u e^{-\sigma u}$ of the integrand is dominated by u on $(0, 1]$ and by $1/(e\sigma_0)$ on $(1, \infty)$, both m -integrable, so $f'(\sigma) = b + \int_{(0, \infty)} u e^{-\sigma u} m(du)$ for $\sigma > 0$, a sum of nonnegative terms that vanishes only if $b = 0$ and $m = 0$, that is only if $f \equiv 0$, that is only if $F \equiv 0$. So $F \not\equiv 0$ gives $f' > 0$, and the chain rule gives $F'(\omega) = \omega f'(\omega^2/2) > 0$ for $\omega > 0$. Finally, an admissible $F \not\equiv 0$ with Gaussian coefficient 0 whose profile is a step function with its jumps on a lattice has, by Proposition 4.6(3), a stationary point $\omega_0 \neq 0$, and by evenness one with $\omega_0 > 0$; that contradicts $F' > 0$, so no such F lies in $\mathcal{A}_\infty$. The Cin rays are the case of a single jump.

(4) $\mathcal{S} \subseteq \mathcal{A}_\infty$ is clause (2). Conversely let $F \in \mathcal{A}_\infty$ and put $f(\sigma) := F(\sqrt{2}\sigma)$, a Bernstein function with $f(0+) = 0$ by clause (3) and the continuity of F . Fix $0 < c < 1$. For each d let

μ_d be the self-decomposable law on $\mathbb{R}^d$ with exponent $\xi \mapsto F(|\xi|)$. By the definition there is a law $\rho_{c,d}$ with $\hat{\mu}_d(\xi) = \hat{\mu}_d(c\xi) \hat{\rho}_{c,d}(\xi)$, and by the cited proposition $\rho_{c,d}$ is infinitely divisible. Its transform is $e^{-(F(|\xi|)-F(c|\xi|))}$, the transform of μ_d having no zero, so its exponent, being continuous with value 0 at the origin, is $\xi \mapsto F(|\xi|) - F(c|\xi|)$. That function is therefore continuous and negative definite on $\mathbb{R}^d$, for every d , and so is its dilate $\xi \mapsto F(\sqrt{2}|\xi|) - F(c\sqrt{2}|\xi|) = g_c(|\xi|^2)$ with $g_c(\sigma) := f(\sigma) - f(c^2\sigma)$. That g_c takes values in $[0, \infty)$, which the radial theorem asks of its argument, is the monotonicity of the Bernstein function f at $c^2\sigma \leq \sigma$. By the radial theorem, applied as in clause (3), g_c is a Bernstein function, and $g_c(0+) = 0$.

The exponential of the negative of a Bernstein function is completely monotone, and a completely monotone function is the Laplace transform of a positive measure on $[0, \infty)$, whose mass is the value at $0+$. So there are probability laws T and R_c on $[0, \infty)$ with Laplace transforms e^{-f} and e^{-g_c} , and $e^{-f(\sigma)} = e^{-f(c^2\sigma)} e^{-g_c(\sigma)}$ says that T has the law of $c^2T' + R$ with T' distributed as T , R as R_c , and the two independent, a Laplace transform determining a law on the half-line. As c ranges over $(0, 1)$ so does c^2 , so T is self-decomposable. By the one-dimensional characterization of self-decomposable Lévy measures — [50], Corollary 15.11, p. 95, the two-sided form for a law on the line, and not Proposition 2.2, which is stated for symmetric laws while this T is one-sided — its Lévy measure is $k_I(u) du/u$ on $(0, \infty)$ with k_I nonincreasing, the law being carried by the half-line, and its Laplace exponent is $f(\sigma) = b_0\sigma + \int_0^\infty (1 - e^{-\sigma u}) k_I(u) du/u$ with $b_0 \geq 0$ and $\int_0^\infty (1 \wedge u) k_I(u) du/u < \infty$, which is the single-integral reading of the two conditions of Definition 4.1. So f is causally admissible, and $F \in \mathcal{S}$ by Proposition 4.6(1). $\square$

The remark records the readings that go with the proposition, the two questions the filtration raises and what the literature says about them, and the family behind the lemma of the next subsection.

Remark 4.10 (the dimension filtration: readings, questions, numerics). *The geometric reading.* By the radial theorem, $\mathcal{B}$ is exactly the set of admissible F whose radial extension $\xi \mapsto F(|\xi|)$ is a Lévy exponent on $\mathbb{R}^d$ for every d at once, and the laws so obtained are the variance mixtures of the centred normal law with an infinitely divisible mixing law, the type G laws, the mixing law being the delay of this section ([42], (1.1), p. 323, and for $\mathbb{R}^d$ p. 325, who attribute the class to Marcus and to Rosiński; [2], Definition 1.1, p. 148; the one-dimensional mixture form is also [56], (2.7), p. 345); the criterion is about all dimensions together, the hard direction of the theorem taking a limit in the dimension. The classes $\mathcal{A}_d$ read, heuristically, as the radial restrictions of the isotropic admissible families of $\mathbb{R}^d$. Take a family of operators on $L^1(\mathbb{R}^d)$ satisfying the evident analogue of (A1)–(A8) and (ND), with kernels invariant under the orthogonal group, and push the kernels forward under an orthogonal projection onto a line. What comes out satisfies (A1)–(A8) and (ND) on the line, with the restriction of $\xi \mapsto F(|\xi|)$ as its exponent: pushforward under a linear map commutes with convolution and carries probability measures to probability measures and dilations to dilations; the reflection of the line is induced by an orthogonal map; and a rotation invariant measure carried by a hyperplane is δ_0 . A linear image of a self-decomposable law is self-decomposable, which is the nesting argument of Proposition 4.9(1). The converse would need the analogue of Theorem 2.5 in dimension d , which this paper does not prove. So the classes are defined by the exponent condition and not by the families, and only the exponent condition is used. In these terms Proposition 4.9(3) says that the lattice-step members of the line’s theory, the Cin rays among them, are not radial restrictions of isotropic admissible families valid in every dimension, and are outside the type G laws, not merely outside $\mathcal{S}$.

Two questions, and what the lemma below settles. The first is settled in the literature, by a theorem of Sato’s; the second, to the author’s knowledge, is not. The first is whether $\mathcal{S}$ is all of the type G slice, that is whether the mixing subordinator of a self-decomposable Gaussian variance mixture is itself self-decomposable. The held sources prove the implication

in the direction this section uses and not its converse. [25] announces on p. 14 and proves on pp. 15–16 that for a pure variance mixture self-decomposability of the mixing law implies self-decomposability of the mixture, and the question left open there is a different one, the mean-variance case. [3] gives infinite divisibility of the mixture from infinite divisibility of the mixing law, in $\mathbb{R}^r$ for every r (pp. 309–311). One level up the corresponding statement is an equivalence: $\varphi(s^2/2)$ is the transform of a symmetric extended generalized gamma convolution if and only if φ is the transform of a generalized gamma convolution ([8], Thm. 7.3.1, pp. 115–116, proved in both directions), which is the Thorin-level twin of Proposition 5.7(3). At the level of class L the converse is false, and this is known: [51], Theorem 1.2, constructs a self-decomposable law of type G that is the Gaussian mixture of no self-decomposable law on the half-line, the mixing law being determined by the mixture (his Lemma 3.1). His witness is explicit as well: the delay profile of his (3.7), p. 8, is $2s^{-1/2}$ on $(0, 1/(a+b)) \cup [1/a, \infty)$ and $-2s^{-1/2} + c$ on $[1/(a+b), 1/a]$, for fixed $a > b > 0$ and a constant c chosen to keep it nonnegative — a power with a rising piece inserted, which is the design of the family below with a slope where that family has a jump. What is new here is therefore not the explicitness. It is the crossing of the dimensions one at a time, Lemma 4.12(3), and the identification at the top, Proposition 4.9(4): Lemma 4.12(1) exhibits an admissible exponent that is of the type G form of Definition 4.11 and is not subordinated. Reading that as an answer to the question as posed uses the radial theorem, which makes the type G form the class of variance mixtures, and the existence of the delay law at Remark 4.3, which makes the mixing law a law at all. The second question is whether the filtration is strict inside the admissible cone, and in particular whether $\mathcal{A}_2 = \mathcal{A}_\infty$. It is strict at the bottom, which is Lemma 4.12(3): one admissible exponent lies in $\mathcal{A}_2$ and not in $\mathcal{A}_3$, another in $\mathcal{A}_1$ and not in $\mathcal{A}_2$, and the first separates $\mathcal{A}_2$ from $\mathcal{A}_\infty$ as well. Admissibility in the plane therefore does not already force the subordination picture. The top of the filtration is settled by Proposition 4.9(4): $\mathcal{A}_\infty = \mathcal{S}$, by the radial theorem applied to the remainders of self-decomposability. *We have not found this equivalence stated.* The sufficiency half is classical and is in print more than once: in dimension one ([25]; [51], Theorem 1.1); for normal variance mixtures in a dimension r that does not enter the hypothesis on the mixing law, which are self-decomposable when the mixing law is, and “self-decomposable in any dimension” for the generalized hyperbolic family at zero drift ([4], pp. 147 and 149); and in a fixed dimension for an operator-stable subordinand ([6], Theorem 6.1, whose Example 6.2 shows that self-decomposability of the subordinand alone does not suffice). None of these states a converse. The literature on type G laws on $\mathbb{R}^d$ ([42]; [2]; [5]), read in full, takes an infinitely divisible mixing law, fixes the dimension, and does not treat the question. So the direction from self-decomposability in every dimension to a self-decomposable mixing law is, as far as these sources go, stated here for the first time; the sentence records what was read, on 2026-09-19 and 2026-09-20, and not an exhaustive search. In the coordinate $x = \log r^2$ the identification has a second reading: (4.5) says that the polar profile is the convolution of the delay measure with the law of $\log w$, w a chi-square variable with d degrees of freedom, and those laws recentred at $\log d$ are an approximate identity, which is a second way to see why monotonicity in every dimension forces monotonicity of k_1 . That route is not carried out here, and clause (4) does not need it.

The family behind the lemma, and the numerics. Writing the reading above produced an answer to the first question and a witness for the second, and both are Lemma 4.12, stated at one parameter point of the following family. Take $0 < \beta < 1$, $0 < a < b$, $c > 0$ and the delay profile $k_1(u) = u^{-\beta} + c \mathbf{1}_{[a,b]}(u)$. It satisfies the two integrability conditions of Definition 4.1 and has no nonincreasing version, so its F_1 is a Bernstein function and is not causally admissible. By (4.5) the polar profiles of the associated isotropic laws are

$$2^{1+\beta} \frac{\Gamma(d/2 + \beta)}{\Gamma(d/2)} r^{-2\beta} + 2c \left[P\left(\frac{d}{2}, \frac{r^2}{2a}\right) - P\left(\frac{d}{2}, \frac{r^2}{2b}\right) \right],$$

P here being the regularized lower incomplete gamma function. The lemma takes $\beta = \frac{1}{2}$, $a = 1$, $b = 2$, where the bracket is $\text{erf}(r/\sqrt{2}) - \text{erf}(r/2)$ at $d = 1$ and $e^{-r^2/4} - e^{-r^2/2}$ at $d = 2$.

What stays here rather than in the lemma is the numerics. At those parameters the polar profile is nonincreasing in dimension d exactly for $c \leq c_d$, with $c_1 \approx 12.61$, $c_2 \approx 5.24$, $c_3 \approx 3.21$ and $c_d \rightarrow 0$; the thresholds were computed numerically from those one-variable inequalities, and the lemma asserts memberships only at $c = 4$ and $c = 8$, where each inequality is elementary and is proved. Two sufficient conditions at $d = 1$ are worth comparing, since the second is what lets one hypothesis carry both values of c . Dropping the negative half of the derivative of the bump term leaves $c r^2 e^{-r^2/2} \leq 1$ for every r , that is $c \leq e/2$; keeping it and bounding $\log(1/p) \leq (1-p)/\sqrt{p}$ leaves $c(\frac{3}{2} - \sqrt{2}) \leq 1$, that is $c \leq 6 + 4\sqrt{2} \approx 11.66$, which is within eight per cent of c_1 and is the hypothesis of Lemma 4.12(1). Because $c_d \rightarrow 0$ the family leaves every $\mathcal{A}_d$ eventually while staying of type G for every $c > 0$, which is why $\mathcal{A}_\infty$ is not the type G slice.

4.6 A witness: the slice is strictly inside the type G exponents

The proposition identifies the top of the chain with $\mathcal{S}$ and places it inside the class $\mathcal{B}$, and says nothing about whether that inclusion, or any step of the chain, is strict. One explicit family of delay profiles settles both: $\mathcal{S}$ is strictly inside the type G exponents, and the chain is strict at its bottom. The family is a profile that is a power law with a bump added on an interval, so that it fails to be nonincreasing and yet, for a small enough bump, its Gaussian mixture on the line is nonincreasing. The resulting exponent is admissible on the line and of the Bernstein form, and it has no causal ancestor, because the datum of a causally admissible exponent is unique and this one is not nonincreasing. Raising the bump, the same family produces polar profiles that are nonincreasing in dimension 2 and not in dimension 3, and then in dimension 1 and not in dimension 2, so the chain of the proposition is strict at its bottom. The vocabulary the lemma needs is the causal definition with the monotonicity of the delay profile dropped, and the polar profile of the proposition restated so that the lemma stands on its own.

Definition 4.11 (delay data, the type G exponents, and the polar profiles). A *delay datum* is a pair (b_0, k_I) with $b_0 \geq 0$ and $k_I : (0, \infty) \rightarrow [0, \infty)$ measurable, subject to $\int_0^1 k_I(u) du < \infty$ and $\int_1^\infty k_I(u) u^{-1} du < \infty$. Its exponent is

$$F_I(\sigma) = b_0 \sigma + \int_0^\infty (1 - e^{-\sigma u}) k_I(u) \frac{du}{u}, \quad \sigma \geq 0,$$

and its *polar profile in dimension d* , for an integer $d \geq 1$, is

$$k_d(r) := \frac{2}{\Gamma(d/2)} \int_0^\infty v^{d/2-1} e^{-v} k_I\left(\frac{r^2}{2v}\right) dv, \quad r > 0. \quad (4.6)$$

Write $\mathcal{G}$ for the set of functions of the form $F(\omega) = F_I(\omega^2/2)$, $\omega \in \mathbb{R}$, for a delay datum. These are type G exponents, the exponents of Gaussian variance mixtures with an infinitely divisible mixing law, and they are those whose mixing law has a Lévy measure of the form $k_I(u) du/u$; a mixing law with an atomic Lévy measure gives a type G exponent that is not in $\mathcal{G}$.

The lemma is the one statement of this paper that is stated and not proved in the machine-checked development; the proof below is the proof of record, and no statement of the paper reads any of its clauses.

Lemma 4.12 (the slice inside the type G exponents, and the polar profiles across dimensions). *For $c > 0$ let $D^{(c)}$ be the delay datum with drift 0 and profile*

$$k_I^{(c)}(u) = u^{-1/2} + c \mathbf{1}_{[1,2]}(u), \quad u > 0,$$

let $F_I^{(c)}$ be its exponent, put $F^{(c)}(\omega) := F_I^{(c)}(\omega^2/2)$, and write $k_d^{(c)}$ for its polar profile (4.6).

1. The subordinated slice is strictly smaller than the type G exponents. $\mathcal{S} \subseteq \mathcal{G}$; and for $0 < c \leq 6 + 4\sqrt{2}$ the function $F^{(c)}$ is admissible in the sense of Theorem 2.5, with Gaussian coefficient 0 and folded profile $k_1^{(c)}$, and lies in $\mathcal{G} \setminus \mathcal{S}$. Hence $\mathcal{S} \subsetneq \mathcal{G}$.
2. The polar profiles cross the dimensions one at a time. For $c = 4$ the profiles $k_1^{(4)}$ and $k_2^{(4)}$ are nonincreasing on $(0, \infty)$ and $k_3^{(4)}$ is not; for $c = 8$ the profile $k_1^{(8)}$ is nonincreasing on $(0, \infty)$ and $k_2^{(8)}$ is not.
3. The filtration is strict at the bottom. In the notation of Proposition 4.9, $F^{(8)} \in \mathcal{A}_1 \setminus \mathcal{A}_2$ and $F^{(4)} \in \mathcal{A}_2 \setminus \mathcal{A}_3$. Hence $\mathcal{A}_1 \supsetneq \mathcal{A}_2 \supsetneq \mathcal{A}_3$, and $F^{(4)}$ separates $\mathcal{A}_2$ from $\mathcal{A}_\infty$ as well, that class lying inside $\mathcal{A}_3$.

Clauses (1) and (2) use no cited fact. Clause (3) reads clause (2) through two of the facts Proposition 4.9 uses in dimension d : the polar criterion, at both of its directions, and the Lévy–Khintchine representation there, at its converse and its uniqueness. Its proof writes out two steps that the earlier clauses leave implicit — that $k_d^{(c)}$ is the polar profile of the law with radial exponent $F^{(c)}(|\xi|)$, and that a rotation-invariant Lévy measure of a self-decomposable law has a nonincreasing radial profile.

The proof, in outline. The proof is in Appendix A. It has three parts. The profile of the datum jumps upward at $u = 1$, so it has no nonincreasing version, and a causally admissible ancestor would have the same profile, by the uniqueness of a Laplace representation (Proposition 2.9), which keeps $F^{(c)}$ out of the slice. The polar profile $k_d^{(c)}$ is computed in closed form with the incomplete gamma function, and its monotonicity is decided by explicit numerical inequalities: it is nonincreasing in dimension d when c is small and increases somewhere when c is large, with thresholds that fall with the dimension. Clause (3) reads those computations through the polar criterion in dimension d .

4.7 What else crosses the bridge

The bridge carries more than exponents, and the remark collects what else it carries. It is this section’s relation to the second half of the causal article, and it is kept to what can be said without that article’s vocabulary.

Remark 4.13 (what crosses the bridge). Three of the causal article’s later results have images on the line. The *exact cascade implementation* crosses: the rational Laplace transform of the causal Gamma increments becomes the rational spectrum of the Matérn increments, and the causal pole–zero sections become forward–backward first-order sections in x (Proposition 7.2). The *cone geometry* crosses as a map of cones but not as a map of extreme rays, by Proposition 4.6: the image of a causal unit-step ray has the smooth positive profile of Proposition 5.7(4), not a step. The *moment and tail dictionary* crosses with a change of exponent, since $\mathbb{E}|W_T|^{2n} = (2n - 1)!! \mathbb{E}T^n$, so the spatial family has $2n$ finite moments exactly when the causal one has n . The causal article’s results on locality have a counterpart too, in which the substitution $\sigma \mapsto \omega^2/2$ turns a parabolic equation into an elliptic one; it belongs to the next module, on locality, and is not stated here.

5 The corners: Matérn, Student-t, symmetric stable

The two preceding sections described the admissible cone as a whole: its extreme rays, and the part of it that the causal theory reaches. This section is about individual members, the ones a user of scale space would compute with. Three families are treated, each already in use. The *Matérn* kernels are the covariance family of spatial statistics ([63]; [24] for the history; [47]

for the present state). The *Student-t* laws are the classical heavy-tailed relatives of the normal law, and their member with one degree of freedom is the Cauchy kernel of the Poisson scale space ([20]). The *symmetric stable* laws are the kernels the semigroup axiomatics of scale space arrive at ([44]; [16]). What the hemigroup theory adds is one description for all three and a way to compare them. Each family is a curve of profiles: an exponential for the Matérn kernels, the Gaussian mixture of the causal Bessel catalogue for the Student-t laws, a pure power for the stable laws. The profile answers the practical questions about a kernel at once: how many moments it has and how heavy its tail is, whether it is bounded at the origin (Corollary 3.9), and whether it can be computed exactly by recursive filters (§7). Figure 5 shows the three families.

The section has five parts. §5.1 gives the two dials on which the families are compared: the n -th moment is finite exactly when $\int_1^\infty x^{n-1}k(x)dx$ is, and the tail is governed by the support of the profile (Proposition 5.1). §5.2 is the main result: four descriptions single out the Matérn family, and its members of integer index are exactly the families computed by a cascade of identical first-order recursive sections (Theorem 5.4, Corollary 5.5). §5.3 treats the profiles that are mixtures of exponentials, the subcone that contains all three families and on which the bridge of §4 is a bijection (Proposition 5.7). §5.4 and §5.5 place the Student-t and the stable families in it (Propositions 5.10 and 5.12). Table 3 collects what the section and the two around it establish for each family, with the unit-step members of §3 for contrast.

Table 3: The named families compared, at canonical scale t and with no Gaussian part except in the first row. “Lag sections” means an exact realization of every stage by the first-order forward–backward sections of §7; “not a limit” means that no sequence of families so realized converges to the member, by Proposition 7.8 — at its clause (3) for the unit-step members and the Matérn members of non-integer index, and at its clause (5), on the moments alone, for the stable and Student-t members. The tail entries are orders, in the senses made precise in Propositions 5.1, 5.3 and 5.10; for the stable row Proposition 5.12 gives the count of moments, and the order of the density is the classical $p(x) \sim \pi^{-1}\Gamma(\alpha+1)\sin(\pi\alpha/2)|x|^{-\alpha-1}$ of [50], Remark 14.18, pp. 87–88, which no statement here uses. Boundedness at the origin is Corollary 3.9 for the Matérn and unit-step rows, Corollary 3.10 for the stable row, and the explicit density of Proposition 5.10(1) for the Student-t row.

family	profile $k(x)$	finite moments	tail	bounded at 0	lag sections
Gaussian	0, with $a > 0$	all	$e^{-x^2/4at^2}$	yes	a limit only
Matérn, index γ	$2\gamma e^{-x}$	all	$e^{- x /t}$	iff $\gamma > \frac{1}{2}$	iff γ is an integer
Student-t, shape a	a Gaussian mixture	$n < 2a$	$ x ^{-2a-1}$	yes	none, not a limit
stable, index $\alpha < 2$	$C_\alpha^{-1}x^{-\alpha}$	$n < \alpha$	$ x ^{-1-\alpha}$	yes	none, not a limit
unit step, cutoff τ	$\mathbf{1}_{(0,\tau)}$	all	$e^{- x \log x /\tau t}$	no	none, not a limit

5.1 Moments and tails

Two questions one asks of any smoothing kernel are how far it spreads and how fast it falls off. Does it have a variance, and how many moments? Does a distant feature leak into the measurement at an exponential or at a polynomial rate? For an admissible kernel both answers are in the tail of the profile, and the reason is the displacement dictionary of Remark 2.4. The profile describes a catalogue of displacements, $k(x)dx/x$ being the rate of those of size x ; a large displacement of the kernel comes from a large displacement in the catalogue; and the classical theory of infinitely divisible laws makes this exact. A law has an n -th moment exactly when its Lévy measure, restricted to the large jumps, has one ([50], Theorem 25.3, p. 159, in the form with a submultiplicative weight), and when the jumps are bounded by a radius τ the tail of the law is of the order $e^{-|x|\log|x|/\tau}$ and no lighter ([50], Theorem 26.1, p. 168, which states both directions). The proposition reads the two theorems in the folded profile, adds the formula for

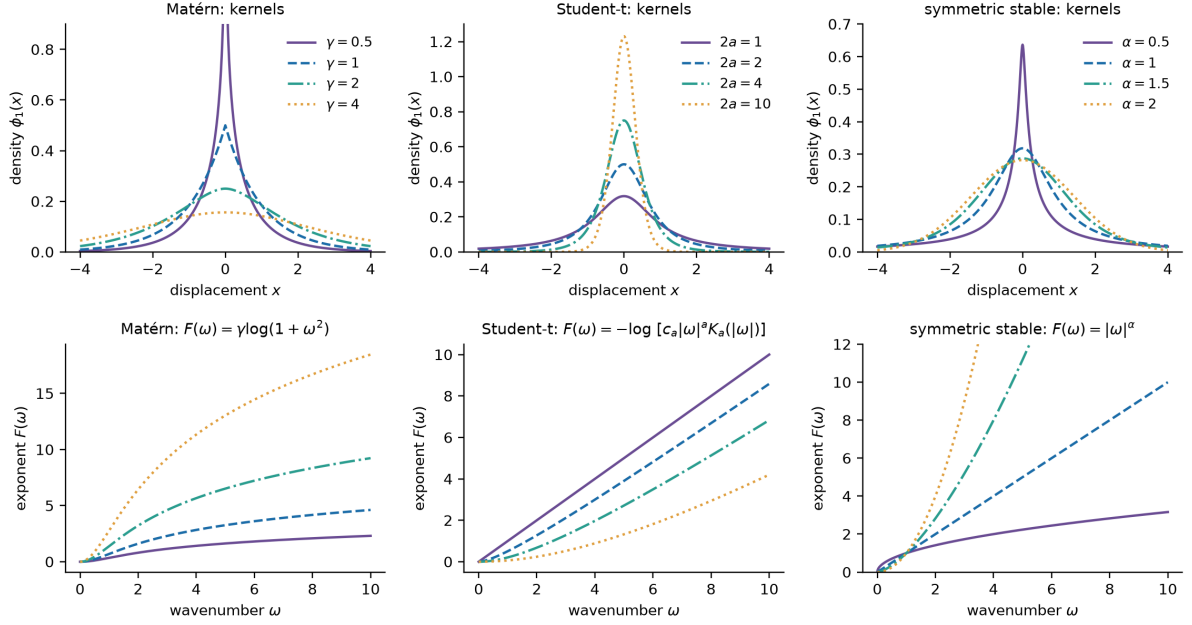


Figure 5: The three corner families at canonical scale 1: the kernels (top) and the exponents $F = -\log \hat{\phi}_1$ (bottom). Left, the Matérn family for four values of the index γ : the Bessel K_0 at $\gamma = \frac{1}{2}$, logarithmically singular at the origin, the Laplace kernel at $\gamma = 1$, and smoother kernels beyond, approaching a Gaussian shape; all have exponential tails, and the exponent grows like a logarithm (Propositions 5.2 and 5.3). Middle, the Student-t family for four values of the degrees of freedom $2a$, in the scale of the proposition, in which a larger a is also a narrower kernel: the Cauchy kernel at $2a = 1$; the tails are polynomial, of order $|x|^{-2a-1}$, and the exponent grows linearly (Proposition 5.10). Right, the symmetric stable family for four values of the index α : the Cauchy kernel again at $\alpha = 1$, the Gaussian at $\alpha = 2$; the exponent is the power $|\omega|^\alpha$ and, for $\alpha < 2$, the tails are of order $|x|^{-1-\alpha}$ (Proposition 5.12). The parameters vary at a fixed canonical scale and the bandwidths are not matched, so the panels show the shapes of the families and are not a comparison of smoothings; Figure 9 compares members at equal bandwidth.

the variance, and draws the conclusion that matters for comparing families: a profile of bounded support gives the lightest tails the cone allows, and they are still heavier than a Gaussian's, so the Gaussians are the only members with Gaussian tails.

Proposition 5.1 (moments and tails). *Let F be admissible with data (a, k) , and let X_t be the displacement at canonical scale t .*

1. (Moments.) *For $n \geq 1$, $\mathbb{E}|X_t|^n < \infty$ if and only if $\int_1^\infty x^{n-1}k(x)dx < \infty$. When finite, the variance is $\mathbb{E}X_t^2 = t^2(2a + \int_0^\infty xk(x)dx)$, the two terms being the jitter and the second moment of the catalogue.*
2. (Tails.) *If $k \not\equiv 0$ and τ is the least radius beyond which k vanishes, then $\mathbb{E} \exp(\alpha|X_t| \log |X_t|) < \infty$ for every $\alpha < 1/(\tau t)$ and for no $\alpha > 1/(\tau t)$: the tail is of order $e^{-|x| \log |x|/(\tau t)}$. If k vanishes beyond no τ , the tail is heavier than every such rate. In particular a completely monotone k , being positive on all of $(0, \infty)$, puts the whole Thorin subclass on the heavy side, and no kernel other than the Gaussians themselves has a Gaussian or lighter tail.*

The criterion of clause (1) and the two rates of clause (2) are the cited theorems read in this paper's convention. The translation into the folded profile, the variance formula and the last sentence of clause (2) are proved below; the variance formula uses no cited fact, the finiteness of $\mathbb{E}X_t^2$ being a conclusion of its argument and not a hypothesis.

Proof of the parts that are this paper's own. The Lévy measure of X_t is the t -dilate of $\nu = k(x)x^{-1}dx$, folded to the two-sided measure ν_2 ; so $\int_{|x|>1} |x|^n \nu_2(dx) = \int_{1/t}^\infty (tx)^n k(x) x^{-1} dx$, which is finite exactly when $\int_1^\infty x^{n-1} k(x) dx$ is, the integrand being locally bounded away from the origin. Applying the moment criterion with the submultiplicative weight $\max(|x|, 1)^n$ gives clause (1)'s criterion: that weight agrees with $|x|^n$ on $\{|x| > 1\}$, the only region the criterion integrates over, and has a finite integral against the law exactly when $|x|^n$ does, since $|x|^n \leq \max(|x|, 1)^n \leq |x|^n + 1$; the weight $|x|^n$ itself is not submultiplicative.

For the variance, what has to be evaluated is a second difference of the transform at the origin and not a second derivative. Write $C = t^2(2a + \int_0^\infty x k(x) dx)$, finite by hypothesis. The half-angle formula $1 - \cos u = 2 \sin^2(u/2)$ gives, for every real c and every $\omega \neq 0$,

$$(1 - \cos c\omega) \frac{2}{\omega^2} = c^2 \left(\frac{\sin(c\omega/2)}{c\omega/2} \right)^2 \leq c^2,$$

an identity rather than an estimate, so the same line supplies the pointwise limit c^2 as $\omega \rightarrow 0$, a dominating function, and the inequality $1 - \cos u \leq u^2/2$. On the exponent side, reading it at $c = tx$ inside (2.3) and dominating by $t^2 x k(x)$ gives $2\omega^{-2} F(t\omega) \rightarrow C$; hence $F(t\omega) \rightarrow 0$, and since $ye^{-y} \leq 1 - e^{-y} \leq y$ for $y \geq 0$, the same limit holds for $2\omega^{-2}(1 - e^{-F(t\omega)})$. On the measure side, μ_t being a probability measure, $1 - \hat{\mu}_t(\omega) = \int (1 - \cos \omega x) \mu_t(dx)$; Fatou's lemma along a sequence $\omega_m \rightarrow 0$ turns the exponent-side limit into $\int x^2 \mu_t(dx) \leq C$, so the second moment is finite — a conclusion of the argument, not a hypothesis of it — and dominated convergence with the dominating function x^2 then gives $2\omega^{-2}(1 - \hat{\mu}_t(\omega)) \rightarrow \int x^2 \mu_t(dx)$. The two limits are limits of the same function, so $\mathbb{E}X_t^2 = C$. The mean vanishes because $|x| \leq 1 + x^2$ makes x integrable against μ_t and μ_t is reflection-symmetric.

Remark, not machine-checked. A second route to the variance evaluates $\mathbb{E}X_t^2 = -\partial_\omega^2 e^{-F(t\omega)}|_{\omega=0} = t^2 F''(0)$ and differentiates (2.2) twice under the integral sign to obtain $F''(0) = 2a + \int_{(0,\infty)} x^2 \nu(dx)$. It reaches the same constant and costs a dominating function for each of the two differentiations, where the second difference costs one and delivers the finiteness of $\mathbb{E}X_t^2$ along the way.

For the last sentence of (2), a completely monotone k is a mixture of exponentials by Bernstein's theorem (§5.3): $k(x) = \int e^{-\theta x} V(d\theta)$ for a positive measure V , which is not the zero measure because $k \not\equiv 0$. Some window $[0, \theta_0]$ then carries positive mass, and on it $e^{-\theta x} \geq e^{-\theta_0 x}$, so $k(x) \geq e^{-\theta_0 x} V([0, \theta_0]) > 0$ at every $x > 0$; hence k vanishes beyond no τ , and the heavy branch of the tail statement applies. Finally, if $k \equiv 0$ then $F = a\omega^2$ and the kernel is Gaussian, while if $k \not\equiv 0$ then F has a nonzero jump catalogue and the tail floor forbids a Gaussian tail: fixing $x_0 > 0$ with $k(x_0) \neq 0$, the catalogue reaches the radius x_0 , so the floor holds at every $\alpha' > 1/(x_0 t)$, and one such α' suffices, since $\alpha'|x| \log|x| \leq \alpha x^2$ once $|x| \geq (2\alpha'/\alpha)^2$ — write $\log|x| = 2 \log \sqrt{|x|} \leq 2\sqrt{|x|}$ — while the law is a probability measure, so the divergence cannot come from the bounded part. No case distinction between a bounded and an unbounded catalogue is needed at this step, one radius the catalogue reaches being enough. $\square$

The two dials are turned by one object, the tail of k : polynomial moment weight by its decay, super-exponential tail order by its support. The Cin rays and their superpositions, with bounded support, are the best-localized members of the cone and have every moment; the price is the echo structure of Lemma 3.5. Completely monotone profiles give kernels with no such echoes, computed by first-order sections when the profile is a finite sum of exponentials with integer weights (§7), at the price of tails no lighter than exponential.

5.2 The Matérn family

The Matérn kernels are the functions on the line with transform $(1 + \theta^2 \omega^2)^{-\gamma}$. The name is that of the covariance functions, which are the members with $\gamma > \frac{1}{2}$, and this paper uses it for the whole range $\gamma > 0$. The family has three histories. It entered statistics as the

covariance functions of the stationary random fields that solve a stochastic Helmholtz equation ([63]). It became the standard covariance family of spatial statistics after Matérn's monograph, where it is the family of correlation functions $2(b\nu/2)^\nu K_\nu(b\nu)/\Gamma(\nu)$ in his notation ([43], (2.4.7), p. 18; the name is Stein's, who defines the class by the spectral density $\phi(\alpha^2 + \omega^2)^{-\nu-1/2}$ and notes that it is rational exactly at half-integer ν , [55], (14), p. 31; [24] for the history; [47] for the present state). And its members are the Green's functions of the operator $(1 - \theta^2 \Delta)^\gamma$, which is the link to Gaussian Markov random fields ([40]). On the line the members of half-integer smoothness $\nu = \gamma - \frac{1}{2}$, which are those of integer index γ , have the spectral density $(\lambda^2 + \omega^2)^{-\gamma}$, and its factorization into $(\lambda + i\omega)^{-\gamma}(\lambda - i\omega)^{-\gamma}$ makes the process the output of a linear stochastic differential equation of order γ . That is how these covariances are computed in Gaussian process regression ([26], §4.1, (8)–(9); [48], §12.3, (12.33)–(12.35)). The factorization of the member of integer index into first-order causal and anticausal factors is therefore classical. What Corollary 5.5 adds is that the same factorization holds between every two scales, a stage being a cascade of such factors with a zero at the earlier scale, and that within the admissible class this property characterizes the family. In probability the same functions are the densities of the symmetric variance-gamma laws, Brownian motion run for a Gamma-distributed time, introduced as a model of asset returns ([41]; [22] for the distribution theory). In image analysis they have appeared as the Bessel potentials of the Bessel scale-space ([9]), composed there along the order γ at a fixed range, which is a semigroup but is not scale invariant. What none of these sources has is the family as a scale space, composed along the *range*: with γ fixed and the range θt as the scale, the kernels form an admissible family, an explicit member of the line paper (Proposition 3.3 of [17]). The stage from scale s to scale t has the quotient of the two endpoint transforms as its transfer function, and it is not itself a member of the family.

The first proposition is the computation that the rest of the subsection reads: the profile of the family is a single exponential, $k(x) = 2\gamma e^{-x/\theta}$, displacements of every size with an exponential cutoff at the range, and the index γ is their overall rate.

Proposition 5.2 (the Matérn family: exponent, transforms, moments). *For $\gamma > 0$:*

1. (Exponent.) $F(\omega) = \gamma \log(1 + \omega^2)$ is admissible, with Gaussian coefficient $a = 0$ and folded profile $k(x) = 2\gamma e^{-x}$; the profile at range θ is $k(x) = 2\gamma e^{-x/\theta}$, with exponent $\gamma \log(1 + \theta^2 \omega^2)$.
2. (Transforms.) $e^{-F(t\omega)} = (1 + t^2 \omega^2)^{-\gamma}$, and the increments have transfer functions $((1 + s^2 \omega^2)/(1 + t^2 \omega^2))^\gamma$, rational in ω for integer γ .
3. (Moments.) Every moment is finite: $X_t \stackrel{d}{=} W_T$ with T Gamma-distributed of shape γ and scale $2t^2$, so $\mathbb{E}X_t^{2n} = (2n - 1)!! (2t^2)^n \Gamma(\gamma + n)/\Gamma(\gamma)$, and in particular $\mathbb{E}X_t^2 = 2\gamma t^2$.

Proof. (1) Fix $\theta > 0$ and write $p = 1/\theta$. The integrand $(1 - \cos \omega x) e^{-px} x^{-1}$ is nonnegative on $(0, \infty)$ and, since $1 - \cos u \leq u^2/2$, is bounded there by $\frac{1}{2} \omega^2 x e^{-px}$, so it is integrable for every ω ; its derivative in ω is $\sin(\omega x) e^{-px}$, whose modulus is at most e^{-px} uniformly in ω . Differentiation under the integral sign is therefore legitimate at every ω with one and the same dominating function, and gives

$$\frac{d}{d\omega} \int_0^\infty (1 - \cos \omega x) e^{-px} \frac{dx}{x} = \int_0^\infty \sin(\omega x) e^{-px} dx = \frac{\omega}{p^2 + \omega^2},$$

which is also the derivative of $\frac{1}{2} \log(1 + \omega^2/p^2)$. The two functions of ω therefore differ by a constant, and both vanish at $\omega = 0$, so

$$\int_0^\infty (1 - \cos \omega x) e^{-x/\theta} \frac{dx}{x} = \frac{1}{2} \log(1 + \theta^2 \omega^2).$$

Hence $F(\omega) = \gamma \log(1 + \theta^2 \omega^2)$ is of the form (2.3) with $a = 0$ and $k(x) = 2\gamma e^{-x/\theta}$, which is non-negative and nonincreasing on $(0, \infty)$. Its two integrability conditions are the two dominating functions above read on the two halves of the range: $\int_0^1 x k(x) dx \leq 2\gamma$ and $\int_1^\infty k(x) x^{-1} dx \leq 2\gamma \int_1^\infty e^{-x/\theta} dx < \infty$. Exhibiting the pair $(0, 2\gamma e^{-1/\theta})$ is what admissibility asks for, so the argument stops there; the range θ is carried by the computation and needs no separate dilation step.

(2) is (1) read through the similarity form of Theorem 2.5: the increment transfer function is the ratio of the endpoint transforms, the denominators being nonvanishing.

(3) Let $\Gamma_{\gamma, 2t^2}$ be the Gamma law of shape γ and scale $2t^2$, that is the law on $(0, \infty)$ with density $u^{\gamma-1} e^{-u/2t^2} / (\Gamma(\gamma)(2t^2)^\gamma)$. Its Laplace transform is $\int_0^\infty e^{-\sigma u} \Gamma_{\gamma, 2t^2}(du) = (1 + 2t^2 \sigma)^{-\gamma}$ for $\sigma \geq 0$, one Gamma integral; so the Gaussian mixture $\int_0^\infty g_u \Gamma_{\gamma, 2t^2}(du)$ — the law of W_T with T so distributed — has cosine transform $(1 + 2t^2 \cdot \omega^2 / 2)^{-\gamma} = (1 + t^2 \omega^2)^{-\gamma}$, which by (2) is the transform of the kernel at scale t . Both are symmetric probability measures, so Proposition 2.7 of [17] identifies them and $X_t \stackrel{d}{=} W_T$. Conditioning on T , $\mathbb{E}X_t^{2n} = \mathbb{E}[T^n] \mathbb{E}[W_1^{2n}] = (2n-1)!! \mathbb{E}T^n$ with $\mathbb{E}[W_v^{2n}] = (2n-1)!! v^n$ and $\mathbb{E}T^n = (2t^2)^n \Gamma(\gamma + n) / \Gamma(\gamma)$; the two are the same Gamma integral read at half-integer and at integer argument. Every moment is then finite, since $|x|^n \leq 1 + x^{2n}$, and $n = 1$ with $\Gamma(\gamma + 1) = \gamma \Gamma(\gamma)$ gives $\mathbb{E}X_t^2 = 2\gamma t^2$.

Remark. Finiteness of every moment also follows from Proposition 5.1(1), the profile decaying exponentially, and the variance from the variance formula there: $\mathbb{E}X_t^2 = t^2 \int_0^\infty x 2\gamma e^{-x} dx = 2\gamma t^2$. That route is machine-checked too and is independent of this one; it does not reach the general even moment. $\square$

The kernel has a closed form in the modified Bessel function K_ν , and it is the form under which the family is known. The next proposition records it, with its two elementary members, the Laplace kernel and the Bessel K_0 , its exponential tail and its Gaussian limit. The closed form and the tail are classical and are cited: the Fourier pair, with the identification of the family as the symmetric variance-gamma laws, is [22], the density (1.1) and the characteristic function (2.12), with self-decomposability in §2.3, p. 6; the asymptotics of the modified Bessel function that give the tail are [15], §10.25(ii), (10.25.3), and [1], §9.7, 9.7.2, p. 378.

Proposition 5.3 (the Matérn family: the kernel and its tail). *For $\gamma > 0$ and $t > 0$ the kernel of the Matérn family at canonical scale t has the density*

$$\phi_t(x) = \frac{1}{\sqrt{\pi} \Gamma(\gamma) t} \left(\frac{|x|}{2t} \right)^{\gamma - \frac{1}{2}} K_{\gamma - \frac{1}{2}} \left(\frac{|x|}{t} \right),$$

for $\gamma > \frac{1}{2}$ the Matérn kernel of smoothness $\gamma - \frac{1}{2}$ and range t normalised to unit mass; for every $\gamma > 0$ these are the symmetric variance-gamma laws. Here $\gamma = 1$ gives the Laplace kernel $e^{-|x|/t} / (2t)$ at every $x \neq 0$, and $\gamma = \frac{1}{2}$ gives $K_0(|x|/t) / (\pi t)$ at every x . The origin is excluded from the first of those two values because the display reads there as the product of $(|x|/2t)^{1/2} = 0$ with $K_{1/2}(0)$ and does not return $(2t)^{-1}$. The difference is on a null set and leaves the kernel, a density, unchanged. As $|x| \rightarrow \infty$, $\phi_t(x) \sim c_\gamma t^{-\gamma} |x|^{\gamma-1} e^{-|x|/t}$: exponential tails, the heavy side of Proposition 5.1(2), as for every completely monotone profile. As $\gamma \rightarrow \infty$ with $2\gamma t^2 = v$ held fixed, ϕ_t converges weakly to the Gaussian of variance v .

The displayed function is a probability density for every $\gamma > 0$, but a Matérn covariance function is bounded at the origin, which needs the smoothness $\gamma - \frac{1}{2}$ to be positive, and Corollary 3.9 says the same from the side of the profile. Of the proposition's content, the two special values, that the kernel is a density at all (Proposition 2.8) and the Gaussian limit, from $(1 + v\omega^2/2\gamma)^{-\gamma} \rightarrow e^{-v\omega^2/2}$ and Proposition 2.8 of [17], are proved in this paper; the closed form and the tail are the two cited facts.

The theorem says that four descriptions single out the same family. The first is the profile, a single exponential. The second is the Thorin measure of §5.3, a single atom: among the mixtures

of exponentials, the Matérn members are the pure ones. The third is the kernel, through its transform. The fourth is the engineering one: for integer γ , every stage of the family, from scale s to scale t , is a rational filter with a single pole pair, which is what lets the family be computed by recursive filters with no approximation (§7). The equivalence of the fourth with the first is the point. A stage that is a cascade of identical first-order sections is what one would ask for on practical grounds. Within the cone that property singles out one family, the one spatial statistics had already chosen on grounds of its own.

Theorem 5.4 (the Matérn theorem). *For an admissible family in its canonical gauge and $\gamma, \theta > 0$, clauses (1), (2) and (3) are equivalent, and when γ is a positive integer they are equivalent to clause (4).*

1. *The profile is a single exponential, $k(x) = 2\gamma e^{-x/\theta}$, and $a = 0$.*
2. *The family lies in the symmetric Thorin subclass of Proposition 5.7 with Gaussian coefficient 0 and the one-point Thorin measure $2\gamma \delta_{1/\theta}$.*
3. *The kernels are Matérn: $\hat{\phi}_t(\omega) = (1 + \theta^2 t^2 \omega^2)^{-\gamma}$.*
4. *The stage transfer functions are $\hat{\mu}_{s,t}(\omega) = ((1 + \theta^2 s^2 \omega^2)/(1 + \theta^2 t^2 \omega^2))^\gamma$ for all $s \leq t$, rational in ω , with a single pole pair when $s < t$.*

The parameter θ is a change of gauge and may be normalised to 1.

The equivalences hold at a given pair (γ, θ) , and the one cited fact the proof uses is the uniqueness of the Lévy pair (Proposition 2.3 of [17](3)), which turns an equality of exponents into an equality of profiles. Clause (4) is stated for integer γ only, the single-pole-pair case: at a non-integer exponent the stage transfer function is not a ratio of polynomials. The canonical gauge is a hypothesis because clauses (3) and (4) speak of the kernels and the stages at the parameter t : a Matérn family read in another gauge χ has the stage transfer function with $\chi(s)$ and $\chi(t)$ in place of s and t , and satisfies clause (1) all the same. Proposition 7.5 gives a class of admissible stage transfer functions; which rational functions in general are admissible stages is open (§8), and Remark 7.6 shows that the first clause of that proposition has no converse.

Proof. (1) if and only if (2). With $k(x) = 2\gamma e^{-x/\theta}$ and $a = 0$, take $U = 2\gamma \delta_{1/\theta}$. Then (5.1) reads $\frac{1}{2} \cdot 2\gamma \log(1 + \omega^2 \theta^2) = \gamma \log(1 + \theta^2 \omega^2)$, which is the exponent computed in Proposition 5.2(1); so (1) implies (2) with that one-point measure, the factor 2 of the folded profile cancelling the $\frac{1}{2}$ of (5.1). Conversely, a one-point $U = 2\gamma \delta_{1/\theta}$ makes the right-hand side of (5.1) that same exponent, so $F(\omega) = \gamma \log(1 + \theta^2 \omega^2)$, and the next paragraph's uniqueness argument returns the single exponential.

(1) implies (3) is Proposition 5.2(1)–(2) after the gauge change $t \mapsto \theta t$, which Proposition 5.2(1) supplies directly at range θ .

(3) implies (1). Reading the transform at the canonical scale $t = 1$ and taking logarithms turns clause (3) into $F(\omega) = \gamma \log(1 + \theta^2 \omega^2)$; the other scales say nothing further. The computation in Proposition 5.2(1) shows that this exponent has the pair $(0, 2\gamma e^{-x/\theta} x^{-1} dx)$, and the pair of an exponent is unique (Proposition 2.3 of [17](3)), so $a = 0$ and the Lévy measure is that one. Uniqueness of the pair is an identity of measures, so it gives $k(x) = 2\gamma e^{-x/\theta}$ for almost every $x > 0$; the pointwise identity on $(0, \infty)$ follows because k is nonincreasing there while $2\gamma e^{-x/\theta}$ is continuous. Indeed the set on which the two agree is co-null, hence meets every subinterval of $(0, \infty)$; approaching a given $x > 0$ from the left through that set gives $k(x) \leq 2\gamma e^{-y/\theta}$ for points $y < x$ arbitrarily close to x , whence $k(x) \leq 2\gamma e^{-x/\theta}$, and approaching from the right gives the reverse inequality. (Monotonicity of the comparison function is not used; continuity of one side and monotonicity of the other suffice.)

(3) if and only if (4), for integer γ , is Proposition 5.2(2) read in both directions. Forwards, the cascade law makes the increment transfer function the ratio of the two endpoint transforms,

the denominators being nonvanishing; at the collapsed pair $s = t = 0$ the same law reads $\hat{\mu}_{0,1} = \hat{\mu}_{0,1}\hat{\mu}_{0,0}$ with $\hat{\mu}_{0,1} > 0$, so $\hat{\mu}_{0,0} = 1$, which is clause (4) there. Backwards, clause (4) at $s = 0$ is the endpoint transform of clause (3); no limit in s is needed. $\square$

The engineering reading of the theorem is the next corollary. One of its terms belongs to §7: a first-order forward–backward section is a first-order recursive filter run forward along the signal, together with its mirror image run backward, and a cascade of γ identical such pairs has the transfer function $((1 + p^2\omega^2)/(1 + q^2\omega^2))^\gamma$, with q the range of the pole and p that of the zero. The realization half of the corollary is proved in that section, where the sections are constructed. What the corollary does *not* prescribe is the pair: the pole and the zero are not tied in advance to the two scales of the stage, and the third clause asks for one kernel from the origin and nothing else. That the zero then vanishes is the content of the argument.

Corollary 5.5 (the Matérn families of integer index are the ones realized by identical first-order sections). *Let $(\Phi_{s,t})$ be an admissible family in its canonical gauge and let γ be a positive integer. Call a symmetric kernel a cascade of γ identical forward–backward pairs of first-order sections when its transform is $((1 + p^2\omega^2)/(1 + q^2\omega^2))^\gamma$ for some $p, q \geq 0$, the pair being the first-order forward–backward section pair of Proposition 7.2 with pole range q and zero range p . Then the following are equivalent.*

1. *The family is Matérn of index γ : its exponent is $F(\omega) = \gamma \log(1 + \theta^2\omega^2)$ for some $\theta > 0$.*
2. *Every stage is such a cascade: for all $0 \leq s < t$ the increment $\mu_{s,t}$ is a cascade of γ identical pairs, with p and q depending on the pair of scales.*
3. *One kernel from the origin is such a cascade: for some $t > 0$ the kernel ϕ_t is.*

In (1) the pair of the stage from s to t is $(p, q) = (\theta s, \theta t)$; in (3) the zero necessarily vanishes, $p = 0$ and $q = \theta t$. Such a family has every moment finite and exponential tails.

The word *identical* is what carries the corollary, and it cannot be dropped. An admissible family whose kernel from the origin is a cascade of first-order forward–backward pairs of different ranges need not be Matérn: the second example of Remark 7.6, of exponent $2 \log(1 + \omega^2) - \log(1 + \omega^2/4)$, has the kernel transform $[(1 + t^2\omega^2/4)/(1 + t^2\omega^2)] \cdot (1 + t^2\omega^2)^{-1}$ from the origin at scale t , a cascade of two pairs, one carrying a zero of range $t/2$ and one a pure pole of range t . The same happens with three pairs at $k(x) = 6e^{-x} - 2e^{-2x}$, which is positive ($6e^{-x} \geq 2e^{-x} > 2e^{-2x}$) and nonincreasing ($k'(x) = -2e^{-x}(3 - 2e^{-x}) < 0$) and has $F(\omega) = 3 \log(1 + \omega^2) - \log(1 + \omega^2/4)$ by Proposition 5.2(1) at the two rates.

Proof. (1) *implies* (2). A Matérn family of positive integer index γ and range θ has, by Proposition 7.2 read in the gauge $t \mapsto \theta t$, the stage transfer functions $((1 + \theta^2 s^2 \omega^2)/(1 + \theta^2 t^2 \omega^2))^\gamma$ at every pair of scales, realized by γ identical forward–backward pairs; so every stage is a cascade of the kind named, with $(p, q) = (\theta s, \theta t)$ and $p < q$.

(2) *implies* (3) is the stage from $s = 0$, which in the canonical gauge is the kernel $\phi_t = \mu_{0,t}$ from the origin, at any $t > 0$.

(3) *implies* (1). Suppose $\hat{\phi}_t(\omega) = ((1 + p^2\omega^2)/(1 + q^2\omega^2))^\gamma$ for some $t > 0$ and some $p, q \geq 0$. First $q > p$: the transform of a probability measure is bounded by 1 in modulus, so the ratio $(1 + p^2\omega^2)/(1 + q^2\omega^2)$ is bounded on the line, which forces $q \geq p$, and $q = p$ would make $\hat{\phi}_t \equiv 1$, hence $\phi_t = \delta_0$ and $\Phi_{0,t} = \text{Id}$, against (ND). Next $p = 0$: by Proposition 2.8 the kernel from the origin is absolutely continuous, so its transform tends to 0 at infinity — the Fourier transform of an integrable function does — while

$$\left(\frac{1 + p^2\omega^2}{1 + q^2\omega^2}\right)^\gamma \longrightarrow \left(\frac{p}{q}\right)^{2\gamma} \quad (|\omega| \rightarrow \infty),$$

a positive number unless $p = 0$. So $p = 0$ and $q > 0$, and the canonical gauge reads $e^{-F(t\omega)} = (1 + q^2\omega^2)^{-\gamma}$, that is $F(t\omega) = \gamma \log(1 + q^2\omega^2)$; substituting $\omega \mapsto \omega/t$ gives clause (1) with $\theta = q/t$.

The index is not free. If the kernels from the origin at two scales are cascades of γ and γ' identical pairs, the paragraph just proved gives $\gamma \log(1 + \theta^2\omega^2) = \gamma' \log(1 + \theta'^2\omega^2)$; dividing by $\log|\omega|$ and letting $|\omega| \rightarrow \infty$ gives $\gamma = \gamma'$. So the number of pairs is the index, at every scale.

The consequences. Every moment is finite by Proposition 5.2(3), and the tail is exponential by Proposition 5.3. $\square$

Remark 5.6 (the literature of the Matérn corner). The kernels of Theorem 5.4 are the Matérn covariance functions of spatial statistics, defined there by the spectral density $(1 + \theta^2\omega^2)^{-\gamma}$ or as the Green's functions of a fractional Helmholtz operator ([63]; [40]; [47]). In image analysis the same kernels appear as Bessel potentials, and the Bessel scale-space built from them ([9]) composes the family along the *order* γ at fixed range — an additive semigroup in γ that is explicitly not scale invariant — with no composition law stated for the range family at fixed order. In natural-image statistics they appear as Bessel K forms, a statistical model of filter responses rather than a kernel family (see [45]). In none of these is the family derived from axioms on measurement, and the hemigroup is what makes the derivation possible: composed along the range the family is a two-parameter cascade whose increments are not stationary in the scale parameter, so no semigroup axiomatics can reach it, which is why the literature composed along the order instead. The probabilistic identity, the symmetric variance-gamma laws ([41]), is what the classification actually sees.

5.3 The symmetric Thorin subclass

A function on $(0, \infty)$ is *completely monotone* when it is smooth and its derivatives alternate in sign, $(-1)^n k^{(n)} \geq 0$ for every n . By Bernstein's theorem, cited below, these are the mixtures of exponentials, $k(x) = \int e^{-\theta x} U(d\theta)$ with a positive measure U . Each exponential is a Matérn profile, so these are the superpositions of Matérn members over their ranges, and the mixing measure U is called the *Thorin measure*. The class is classical. On the half-line it is Thorin's class of generalized gamma convolutions, introduced to prove the infinite divisibility of the Pareto and the lognormal laws ([59]; [58]) and developed into a theory by Bondesson ([8]); the class here is its symmetric extension to the line.

Three things make it the organising object of this section. All three named families lie in it, the Matérn family as its atoms and the Student-t family under the hypothesis of Remark 5.11. It is the part of the cone on which the bridge of §4 is a bijection: a causal family with a completely monotone delay profile and its image determine each other, with an explicit map between their Thorin measures. And it contains the families realized by first-order sections, as the members whose Thorin measure is a finite sum of atoms of even integer mass (Proposition 7.7). The proposition gives the representation of the exponent that goes with the mixture, (5.1), in which every atom of U contributes one Matérn exponent; clause (3) is the bijection, and clause (4) shows that the subclass is strictly smaller than the image of the bridge, so that the chain from the Thorin subclass through the subordinated slice to the cone is strict at both steps. With the Matérn members at its start, the chain and the members that make each step strict are

$$\begin{array}{ccccccc}
 \text{Matérn members} & \subsetneq & \text{Thorin subclass} & \subsetneq & \mathcal{S} & \subsetneq & \text{admissible cone} \\
 & & \uparrow & & \uparrow & & \uparrow \\
 & & \text{stable} & & k(x) = 2 \operatorname{erfc}(x/\sqrt{2\tau}) & & \text{Cin rays}
 \end{array}$$

where each member named in the lower row lies in the class above it and not in the class to the left of that one. The Student-t members sit beside the stable ones under the hypothesis of Remark 5.11, and unconditionally nowhere in this chain. The first step is strict because

the Matérn members are the atoms of the subclass, the second by clause (4), and the third by Proposition 4.6(3). Figure 4 divides the last step further, by dimension. The one cited fact is Bernstein's theorem, that a completely monotone function is the Laplace transform of a positive measure ([52], Theorem 1.4, p. 3; [19], Vol. 2, §XIII.4, Theorems 1 and 1a, pp. 439–440).

Proposition 5.7 (the symmetric Thorin subclass). *For an admissible F with data (a, k) the following are equivalent.*

1. k is completely monotone.
2. F has a symmetric Thorin representation

$$F(\omega) = a\omega^2 + \frac{1}{2} \int_{(0,\infty)} \log\left(1 + \frac{\omega^2}{\theta^2}\right) U(d\theta), \quad U \geq 0, \quad \int_{(0,\infty)} \log\left(1 + \frac{1}{\theta^2}\right) U(d\theta) < \infty. \quad (5.1)$$

A third member of the equivalence — that the kernels are symmetric extended generalized gamma convolutions in Thorin's sense — is stated apart, as Lemma 5.9, because it rests on the reading of Thorin's definition rather than on Bernstein's theorem.

Moreover:

- (3) (The bridge on Thorin subclasses.) *If F_1 is causally admissible with completely monotone profile $k_1 = \int e^{-\theta u} U_1(d\theta)$, then $F_1(\cdot^2/2)$ is a symmetric Thorin member with $a = \frac{1}{2}b_0$ and U equal to twice the image of U_1 under $\theta \mapsto \sqrt{2}\theta$; the map $U_1 \mapsto U$ is linear and injective, and carries the causal Thorin measures onto the Thorin measures of the symmetric Thorin members, so the subordination bridge restricts to a linear bijection of the causal Thorin subclass onto the symmetric one.*
- (4) (Strictness the other way.) *The subordinated slice $\mathcal{S}$ of Proposition 4.6 is strictly larger than the symmetric Thorin subclass: the image of the causal ray with profile $k_1 = \mathbf{1}_{(0,\tau)}$ has the explicit profile $k(x) = 2 \operatorname{erfc}(x/\sqrt{2\tau})$, which is not completely monotone. So the symmetric Thorin subclass is strictly inside $\mathcal{S}$, which is strictly inside the admissible cone.*

Clause (3) is classical. Both directions, with the correspondence of the measures, are Bondesson's Theorem 7.3.1, p. 115: if φ is the moment generating function of a generalized gamma convolution then $\psi(s) = \varphi(s^2/2)$ is that of a symmetric extended generalized gamma convolution, and conversely; and the proof on pp. 115–116 computes the spectral measure of ψ as the image of that of φ under $t \mapsto \pm\sqrt{2}t$, the left extremity of φ becoming half the coefficient of s^2 in ψ . Read at imaginary argument — which is Lemma 5.9 — that is the clause. What belongs to this paper is the folded normalisation, the linearity and injectivity of the measure map, and the verification that the constructed causal datum is causally admissible.

What is taken from Bernstein's theorem is its existence half. The uniqueness of the Thorin measure is not taken from it: it is Proposition 2.9. The computation that turns the mixture into (5.1), the correspondence between the two integrability conditions, and clauses (3) and (4), which are computations on the bridge profile (4.2), are proved below. An atom of U at $\theta = 0$ is excluded because it would make k bounded below and $\int_1^\infty k(x)x^{-1}dx$ infinite.

Proof of the parts that are this paper's own. (1) implies (2). By Bernstein's theorem, $k(x) = \int_{[0,\infty)} e^{-\theta x} U(d\theta)$ for a positive measure U on $[0, \infty)$; an atom at $\theta = 0$ would make k bounded below by a positive constant, so that $\int_1^\infty k(x)x^{-1}dx = \infty$, against admissibility. So U lives on $(0, \infty)$, and it is σ -finite, its transform being finite at $x = 1$. Tonelli with the computation of Proposition 5.2(1) gives

$$\int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x} = \int_{(0,\infty)} \left(\int_0^\infty (1 - \cos \omega x) e^{-\theta x} \frac{dx}{x} \right) U(d\theta) = \frac{1}{2} \int_{(0,\infty)} \log\left(1 + \frac{\omega^2}{\theta^2}\right) U(d\theta),$$

which is (5.1). The integrability condition of (5.1) is not a further hypothesis but a consequence of that identity: read at $\omega = 1$ the identity says that the integral is $2(F(1) - a)$, and $F(1)$ is finite for every admissible F by Lemma 2.10 of [17]. Three regimes read off from it what the classical form of the condition states separately: $\log(1 + \theta^{-2}) \geq 2 \log(1/\theta)$ near 0, $\geq \frac{1}{2}\theta^{-2}$ at infinity, and $\geq \log 2$ on $(0, 1]$, so that $\int_{(0,1]} \log(1/\theta) U(d\theta)$ and $\int_{(1,\infty)} \theta^{-2} U(d\theta)$ are finite and U is finite on $(0, 1]$, hence on every compact subset of $(0, \infty)$.

(2) *implies (1)*. Given (5.1) with U carried by $(0, \infty)$, set $k'(x) := \int_{(0,\infty)} e^{-\theta x} U(d\theta)$, which is finite for every $x > 0$: on $(0, 1]$ the integrand is at most 1 and U is finite there, and on $(1, \infty)$ one has $e^{-\theta x} \leq C_x \theta^{-2}$ against the second integral, both facts being the three regimes of the previous paragraph. It is the finiteness of U on $(0, 1]$ that is doing the work here, and it is why the displayed condition is the one printed: the two integrals alone leave U free to put infinite mass just below $\theta = 1$. The displayed Tonelli, read backwards, says that k and k' have the same compensated integral $\int_0^\infty (1 - \cos \omega x) \cdot dx/x$ at every frequency. That k' is the profile of a symmetric Lévy pair at all — that is, that $\int_0^\infty (1 \wedge x^2) k'(x) dx/x < \infty$ — follows by averaging that identity over ω against a standard Gaussian law: Tonelli replaces the weight $1 - \cos \omega x$ by $1 - e^{-x^2/2}$, which lies between $\frac{1}{3}(1 \wedge x^2)$ and $1 \wedge x^2$, so the Lévy condition transfers from k to k' . Uniqueness of the pair (Proposition 2.3 of [17](3)) then identifies the two Lévy measures, hence k and k' almost everywhere on $(0, \infty)$, and hence everywhere there, k being nonincreasing and k' continuous. So k is a Laplace transform, and Bernstein's theorem read the other way returns complete monotonicity.

Uniqueness of U . If U' on $(0, \infty)$ also satisfies (5.1), the previous paragraph applied to U' gives $k(x) = \int e^{-\theta x} U'(d\theta)$ as well, so U and U' have the same Laplace transform on $(0, \infty)$ and it is finite there; by Proposition 2.9, $U' = U$. This does not spend the uniqueness half of Bernstein's theorem: the Laplace uniqueness the argument needs is proved in the line paper.

(3). With $k_I(u) = \int e^{-\theta u} U_I(d\theta)$, (4.2) and Tonelli give

$$k(x) = \int U_I(d\theta) 2x \int_0^\infty g_u(x) e^{-\theta u} \frac{du}{u} = 2 \int e^{-x\sqrt{2\theta}} U_I(d\theta),$$

by the first-passage identity used in Proposition 4.5(2); substituting $\theta' = \sqrt{2\theta}$ this is $\int e^{-\theta' x} U(d\theta')$ with U twice the image of U_I , so k is completely monotone, and the Gaussian coefficient is $\frac{1}{2}b_0$ by Lemma 4.2. That U_I is σ -finite, which the Tonelli needs, is not a further hypothesis: $k_I(1)$ is a real number, so the transform of U_I is finite at $u = 1$ against a weight vanishing nowhere.

The identification of k deserves a sentence, because the exponent determines it only up to a null set. What Proposition 2.3 of [17](3) returns, applied to the subordinated exponent and to any admissible F with that exponent, is equality of the two Lévy measures, hence of the two profiles almost everywhere on $(0, \infty)$; the conclusions of this clause — complete monotonicity, which is a statement about every derivative at every point — are pointwise. The passage is the one Theorem 5.4 makes: the two profiles are nonincreasing and the Laplace transform $\int e^{-\theta' x} U(d\theta')$ is continuous on $(0, \infty)$ by dominated convergence, and a nonincreasing function almost everywhere equal to a continuous one equals it everywhere, the good set meeting every subinterval on each side of a given point.

The factor 2 is the folding convention and is checked on the Matérn member: the causal Gamma data are $U_I = \gamma \delta_1$, whence $U = 2\gamma \delta_{\sqrt{2}}$ and (5.1) returns $\frac{1}{2} \cdot 2\gamma \log(1 + \omega^2/2) = \gamma \log(1 + \omega^2/2)$, which is Proposition 4.5(2); the image without the factor would return half of it.

The bridge is onto the symmetric Thorin subclass. Let F be admissible with data (a, k) and Thorin measure U on $(0, \infty)$ as in (5.1). Put $b_0 := 2a$, let U_I be half the image of U under $\theta' \mapsto \theta'^2/2$, and let $k_I(u) := \int e^{-\theta u} U_I(d\theta)$ for $u > 0$. The transform is finite: the exponents satisfy $u\theta'^2/2 - (u\theta'/2 - u/2) = \frac{u}{2}((\theta' - \frac{1}{2})^2 + \frac{3}{4}) \geq 0$, so $k_I(u) \leq \frac{1}{2}e^{u/2} \int e^{-u\theta'/2} U(d\theta')$, which the three regimes of (1) implies (2) make finite. Hence k_I is nonnegative and nonincreasing on

$(0, \infty)$, and U_I is σ -finite. The exponent of (b_0, k_I) is computed rather than identified. For $u > 0$ and $\sigma \geq 0$ one has $(1 - e^{-\sigma u})/u = \int_0^\sigma e^{-su} ds$, and integrating out u first,

$$\int_0^\infty (1 - e^{-\sigma u}) k_I(u) \frac{du}{u} = \int U_I(d\theta) \int_0^\sigma \frac{ds}{s + \theta} = \int \log\left(1 + \frac{\sigma}{\theta}\right) U_I(d\theta)$$

by Tonelli, twice. At $\sigma = \omega^2/2$ the substitution $\theta = \theta'^2/2$ turns $\log(1 + \sigma/\theta)$ into $\log(1 + \omega^2/\theta'^2)$ and U_I into $\frac{1}{2}U$, and $b_0\sigma = a\omega^2$, so $F_I(\omega^2/2)$ is the right-hand side of (5.1), which is $F(\omega)$. At $\sigma = 1$ the display gives admissibility. The two integrability conditions of the causal cone together say $\int_0^\infty \min(1, u) k_I(u) du/u < \infty$, and $\min(1, u) \leq e(1 - e^{-u})$ for $u \geq 0$: for $u \leq 1$ because $e(1 - e^{-u}) \geq e^u(1 - e^{-u}) = e^u - 1 \geq u$, and for $u \geq 1$ because $e(1 - e^{-u}) \geq e - 1 \geq 1$. So the conditions are bounded by $e \int \log(1 + 1/\theta) U_I(d\theta) = \frac{e}{2} \int \log(1 + 2/\theta'^2) U(d\theta')$, and $\log(1 + 2x) \leq 2\log(1 + x)$, which is $1 + 2x \leq (1 + x)^2$, bounds that by e times the Thorin integral of (5.1) at $\omega = 1$, finite because $F(1)$ is. Finally $\sqrt{2 \cdot \theta'^2/2} = \theta'$ on $(0, \infty)$, so U is twice the image of U_I under $\theta \mapsto \sqrt{2\theta}$. So every symmetric Thorin member is the bridge image of a causal Thorin member, with Gaussian coefficient and Thorin measures related as in (3); no uniqueness theorem and no part of Bernstein's theorem is used.

The measure map is a linear bijection. It is linear because taking an image measure and multiplying by 2 both are. It is injective: if twice the images of U_I and U_I' under $\theta \mapsto \sqrt{2\theta}$ agree, so do the images, and the image of either under $\theta' \mapsto \theta'^2/2$, which inverts $\theta \mapsto \sqrt{2\theta}$ on $(0, \infty)$, returns the measure itself. Surjectivity onto the symmetric Thorin measures is the previous paragraph.

A second route to the correspondence. The two conditions also correspond directly, without passing through an exponent. Write $C_I := \int \log(1 + \theta^{-1}) U_I(d\theta)$ for the causal condition and $C := \int \log(1 + \theta'^{-2}) U(d\theta')$ for (5.1)'s. Since U is twice the image of U_I under $\theta \mapsto \sqrt{2\theta}$, the substitution $\theta'^2 = 2\theta$ gives $C = 2 \int \log(1 + v) U_I(d\theta)$ with $v := 1/2\theta$, while $C_I = \int \log(1 + 2v) U_I(d\theta)$; and $\log(1 + v) \leq \log(1 + 2v) \leq 2\log(1 + v)$, the second inequality being $1 + 2v \leq (1 + v)^2$. So $C_I \leq C \leq 2C_I$, and each condition holds exactly when the other does. The route above does not compare the conditions at all: it reads both off the finiteness of one exponent value.

(4). The causal ray has $k_I = \mathbf{1}_{(0, \tau)}$, and (4.2) gives $k(x) = 2x \int_0^\tau g_u(x) du/u$. Substituting $v = x^2/2u$, so that $du/u = -dv/v$ and $g_u(x) = \sqrt{v/\pi} x^{-1} e^{-v}$,

$$\int_0^\tau g_u(x) \frac{du}{u} = \frac{1}{\sqrt{\pi}x} \int_{x^2/2\tau}^\infty v^{-1/2} e^{-v} dv = \frac{1}{x} \operatorname{erfc}\left(\frac{x}{\sqrt{2\tau}}\right),$$

so $k(x) = 2 \operatorname{erfc}(x/\sqrt{2\tau})$, identified pointwise by the passage of clause (3) — erfc is continuous, being an integral over a moving half-line of an integrable function.

It is not completely monotone, and one bound settles it. Suppose it were; then by Bernstein's theorem it is $\int e^{-\theta x} V(d\theta)$ for a positive V on $[0, \infty)$, and $V \neq 0$ because $\operatorname{erfc} > 0$. Some window carries mass: $V(-\infty, n] > 0$ for an integer n , and that mass is finite because the transform is. On that window $e^{-\theta x} \geq e^{-nx}$, so

$$k(x) \geq c e^{-nx}, \quad c := V(-\infty, n] > 0,$$

for every $x > 0$. Against that, for $y \geq 1$ the comparison $e^{-v^2} \leq e^{-yv}$ on (y, ∞) gives $\int_y^\infty e^{-v^2} dv \leq e^{-y^2}/y \leq e^{-y^2}$, so $\operatorname{erfc} y \leq 2e^{-y^2}$ and $k(x) \leq 4e^{-x^2/2\tau}$ once $x \geq \sqrt{2\tau}$. Take $x \geq \max(\sqrt{2\tau}, 2\tau(n+1), 4/c+1)$. The middle bound gives $x^2/2\tau \geq (n+1)x$ and the last gives $4e^{-x} \leq 4/x < c$, whence

$$k(x) \leq 4e^{-x^2/2\tau} \leq 4e^{-x} e^{-nx} < c e^{-nx} \leq k(x),$$

which is absurd. Its image is admissible by Lemma 4.2, so it lies in $\mathcal{S}$ and not in the Thorin subclass; the other strict inclusion is Proposition 4.6(4).

Remark, not machine-checked. The direct route reads the same conclusion off the asymptotic $\operatorname{erfc}(x/\sqrt{2\tau}) \sim \sqrt{2\tau/\pi} x^{-1} e^{-x^2/2\tau}$, which says more than the contradiction needs: an upper bound of Gaussian type at one sufficiently large x is enough, and the bound above costs a monotone comparison where the asymptotic costs Watson's lemma. $\square$

Where the named families sit in the chain of (4) is said by their own statements. The Matérn family is the one-atom case of (2), so it lies in the symmetric Thorin subclass. The Thorin density of the stable family is Proposition 5.12. The membership of the Student-t family is Proposition 5.10(3), under the hypothesis stated there. The Cin rays lie outside $\mathcal{S}$ by Proposition 4.6(4).

The bijection in clause (3) is why the causal corners of [18] land on named spatial families with the same organisation. The causal Gamma family is the single atom of the causal Thorin subclass, and the Matérn family is the single atom of the symmetric one. The stable families correspond under $\theta \mapsto \sqrt{2\theta}$. The causal Bessel family maps to a Student-t family that is Thorin too, under the hypothesis of Remark 5.11. Clause (4) adds that the bridge is a bijection of the Thorin subclasses and only an injection of the whole cones.

It remains to connect (5.1) with Thorin's own definition, so that the name is earned. Bondeson defines the extended generalized gamma convolutions by a two-sided exponent: a moment generating function with a measure V on the whole line and a compensating term for each θ ([8], Chapter 7, the definition on p. 105 with (7.1.1) and the integrability condition (7.1.2); the generalized gamma convolutions themselves are Theorem 3.1.1, §3.1, p. 30). For a symmetric law the contributions of θ and $-\theta$ pair up, the compensating terms cancel, and what is left at imaginary argument is the right-hand side of (5.1) with U the fold of V . The first lemma is that computation, and it is elementary.

Lemma 5.8 (Thorin's two-sided exponent at $\sigma = i\omega$). *Let $V \geq 0$ be a measure on $\mathbb{R}$ with $V\{0\} = 0$, invariant under $\theta \mapsto -\theta$, and let $U := 2V|_{(0,\infty)}$ be its fold onto the positive half-line. For $\theta \neq 0$ and $\omega \in \mathbb{R}$,*

$$\frac{\theta}{\theta - i\omega} \cdot \frac{-\theta}{-\theta - i\omega} = \left(1 + \frac{\omega^2}{\theta^2}\right)^{-1}, \quad -\log\left|\frac{\theta}{\theta - i\omega}\right| = \frac{1}{2} \log\left(1 + \frac{\omega^2}{\theta^2}\right), \quad \frac{i\omega\theta}{1 + \theta^2} + \frac{i\omega(-\theta)}{1 + (-\theta)^2} = 0,$$

and consequently

$$-\int_{\mathbb{R}} \log\left|\frac{\theta}{\theta - i\omega}\right| V(d\theta) = \frac{1}{2} \int_{(0,\infty)} \log\left(1 + \frac{\omega^2}{\theta^2}\right) U(d\theta).$$

So Thorin's two-sided exponent, read at $\sigma = i\omega$ with $b = 0$ and V symmetric, is the right-hand side of (5.1) at (a, U) with $a = \frac{1}{2}c$: the first identity pairs $\pm\theta$ without choosing a branch of the logarithm, the third cancels the compensating terms, and the second says that the real part alone already supplies the folded integrand.

Proof. The three identities are pointwise computations at a fixed $\theta \neq 0$ and $\omega \in \mathbb{R}$. For the first, the two denominators multiply to $(\theta - i\omega)(-\theta - i\omega) = -(\theta - i\omega)(\theta + i\omega) = -(\theta^2 + \omega^2)$, and the two numerators to $-\theta^2$, so the product is $\theta^2/(\theta^2 + \omega^2) = (1 + \omega^2/\theta^2)^{-1}$, a positive real number: no branch of a logarithm is chosen because none is taken. For the second, $|\theta - i\omega| = \sqrt{\theta^2 + \omega^2}$, so $-\log|\theta/(\theta - i\omega)| = \frac{1}{2} \log((\theta^2 + \omega^2)/\theta^2) = \frac{1}{2} \log(1 + \omega^2/\theta^2)$. The third is $\theta + (-\theta) = 0$ over the common denominator $1 + \theta^2$, which is even in θ .

For the fold, the integrand of the left-hand side is, by the second identity, $\frac{1}{2} \log(1 + \omega^2/\theta^2)$, an even function of θ that is finite off the origin; V gives the origin no mass and is invariant under $\theta \mapsto -\theta$, so the integral over $\mathbb{R}$ is twice the integral over $(0, \infty)$, and twice $V|_{(0,\infty)}$ is U . That is the displayed identity. Read at Thorin's exponent with $b = 0$, the compensating terms cancel in the pairs $\pm\theta$ by the third identity, the Gaussian term $\frac{1}{2}c\sigma^2$ at $\sigma = i\omega$ is $-\frac{1}{2}c\omega^2$, and the pairs of logarithms contribute $-\log(1 + \omega^2/\theta^2)$ each by the first identity, which is twice the

modulus identity; so the Lévy exponent, minus Thorin's, is $\frac{1}{2}c\omega^2 + \frac{1}{2}\int_{(0,\infty)} \log(1+\omega^2/\theta^2) U(d\theta)$, the right-hand side of (5.1) at $(\frac{1}{2}c, U)$. $\square$

The second lemma is the reading of the definition through that computation, and nothing in the paper uses it. The definition as cited is a moment generating function on real σ , which the laws in question do not have at any $\sigma \neq 0$, so what is missing is a formal statement of the definition at imaginary argument, which is a reading of the source and not a proof.

Lemma 5.9 (the Thorin subclass is the symmetric generalized gamma convolutions). *A symmetric law is an extended generalized gamma convolution in Thorin's sense exactly when its exponent has the two-sided form of Lemma 5.8 — that is, when it is $-\frac{1}{2}c\sigma^2 - \int_{\mathbb{R}\setminus\{0\}} (\log \frac{\theta}{\theta-\sigma} - \frac{\sigma\theta}{1+\theta^2}) V(d\theta)$ with $c \geq 0$ and $V \geq 0$ symmetric, read at $\sigma = i\omega$. The two minus signs are the convention used here, in which the exponent is minus the logarithm of the transform; the source writes the logarithm of the moment generating function, without them. Consequently, for an admissible F with data (a, k) , the two equivalent conditions of Proposition 5.7 hold if and only if the kernels of the family are symmetric extended generalized gamma convolutions.*

Proof. By the cited definition a law belongs to the class exactly when its exponent is $-b\sigma - \frac{1}{2}c\sigma^2 - \int (\log \frac{\theta}{\theta-\sigma} - \frac{\sigma\theta}{1+\theta^2}) V(d\theta)$ for a positive V on $\mathbb{R} \setminus \{0\}$, and among these the symmetric ones are those with $b = 0$ and V symmetric. Read at $\sigma = i\omega$, Lemma 5.8 turns that into $\frac{1}{2}c\omega^2 + \frac{1}{2}\int_{(0,\infty)} \log(1+\omega^2/\theta^2) U(d\theta)$ with U the fold of V : the compensating terms cancel in pairs, and the pairing of $\pm\theta$ contributes $-\log(1+\omega^2/\theta^2)$, which is twice the modulus identity there. That is (5.1) with $a = \frac{1}{2}c$.

The source's integrability condition on V and (5.1)'s on U are equivalent, and the comparison is not term by term. The source asks for $\int_{\mathbb{R}\setminus\{0\}} (1+\theta^2)^{-1} V(d\theta) < \infty$ and $\int_{|\theta|\leq 1} |\log \theta^2| V(d\theta) < \infty$; folded, these read $\int_{(0,\infty)} (1+\theta^2)^{-1} U(d\theta) < \infty$ and $\int_{(0,1]} 2\log(1/\theta) U(d\theta) < \infty$. Each direction is an elementary comparison of integrands. Forwards: on $(0, 1]$ one has $\log(1+\theta^{-2}) \leq \log 2 + 2\log(1/\theta)$ and $(1+\theta^2)^{-1} \geq \frac{1}{2}$, so the first condition bounds $U((0, 1])$ and the second the logarithm; on $(1, \infty)$, $\log(1+\theta^{-2}) \leq \theta^{-2} \leq 2(1+\theta^2)^{-1}$. Backwards: $\log(1+\theta^{-2}) \geq \log 2$ on $(0, 1]$ and $\geq \frac{1}{2}\theta^{-2} \geq \frac{1}{2}(1+\theta^2)^{-1}$ on $(1, \infty)$, while $2\log(1/\theta) \leq \log(1+\theta^{-2})$ on $(0, 1]$. So membership in the class is Proposition 5.7(2), which is (1) by that proposition. $\square$

5.4 The Student-t family

Student's t laws are the classical heavy-tailed relatives of the normal law: a normal variable whose variance is itself random, drawn from an inverse-gamma law. Read through §4 that representation is a subordination, Brownian motion run for an inverse-gamma time, and the inverse-gamma laws are the kernels of the causal Bessel family, so the Student- t family is the image of that causal corner under the bridge (Proposition 4.5(3)). That the t laws are infinitely divisible for every number of degrees of freedom was proved by Grosswald ([23]; see [29] for the role of quotients of Bessel functions). Halgreen proved that they are self-decomposable, through the corresponding property of the mixing law ([25]; [3] for the mixtures in general), which is the fact that makes them members of the admissible cone. In the scale-space literature the family appears at one point. With one degree of freedom the kernel is the Cauchy kernel, which is the Poisson scale space of Felsberg and Sommer ([21]; [20]) and the member of the α -scale spaces that Duits et al. single out ([16]); it is also the symmetric stable member of index 1 of §5.5, the point where two corners meet.

The proposition gives the kernel and its transform in closed form, places the family in the Thorin subclass, and counts its moments. The tail is a power, of order $|x|^{-2a-1}$, and the shape parameter a is a dial on the number of finite moments, those of order below $2a$, twice the count of the causal ancestor. So this is the corner of the cone with polynomial tails of any prescribed

order, where the Matérn corner has exponential tails and every moment, and the stable corner fewer than two moments.

Proposition 5.10 (the Student-t family). *For $a > 0$ let T_1 be the scaled inverse-gamma law of the causal Bessel family, with density $p_{T_1}(u) = u^{-a-1}e^{-1/(2u)}/(2^a\Gamma(a))$, and let $X_1 := W_{T_1}$. Then:*

1. *the kernel at canonical scale 1 is $\phi_1(x) = \frac{\Gamma(a+\frac{1}{2})}{\sqrt{\pi}\Gamma(a)}(1+x^2)^{-a-\frac{1}{2}}$, the Student-t law with $2a$ degrees of freedom in the scale in which the density is proportional to $(1+x^2)^{-a-1/2}$, and $\phi_t = t^{-1}\phi_1(\cdot/t)$;*
2. *the transform is $e^{-F(\omega)} = \frac{2^{1-a}}{\Gamma(a)}|\omega|^a K_a(|\omega|)$;*
3. *if T_1 is a generalized gamma convolution — Remark 5.11 — then the law is self-decomposable and lies in the symmetric Thorin subclass;*
4. *$\mathbb{E}|X_t|^n < \infty$ if and only if $n < 2a$, and the tail is $\phi_t(x) \sim ct^{2a}|x|^{-2a-1}$, polynomial.*

Throughout, a is the shape parameter of the causal Bessel family — the letter that family carries in the literature — and not the Gaussian coefficient of (2.3), which is a different number and vanishes for these members, as the proof of clause (3) computes under that clause's hypothesis.

The scaling convention, so that the degrees of freedom come out as $2a$ and not a , is fixed by the density in clause (1) and the rate in p_{T_1} , and the two should be read together whenever the family is quoted. The shape parameter is therefore a dial on the number of finite moments, and the count is twice the causal one, $n < 2a$ against $n < a$ for the delay. Clause (3) is conditional: that the inverse-gamma law is a generalized gamma convolution is Halgreen's theorem, recorded as Remark 5.11 and taken here as a hypothesis, and under it the clause follows from the bridge on Thorin subclasses. Clause (2) is obtained from clause (1) and the Laplace transform of the delay law, which after an exponential change of variables is the integral by which K_a is defined, so that no asymptotic of the Bessel function enters.

Proof of the parts that are this paper's own. (1) Conditioning on T_1 , $\phi_1(x) = \int_0^\infty g_u(x) p_{T_1}(u) du$; the substitution $w = 1/u$ turns this into

$$\frac{1}{\sqrt{2\pi} 2^a \Gamma(a)} \int_0^\infty w^{a-\frac{1}{2}} e^{-w(1+x^2)/2} dw = \frac{\Gamma(a+\frac{1}{2})}{\sqrt{2\pi} 2^a \Gamma(a)} \left(\frac{2}{1+x^2} \right)^{a+\frac{1}{2}} = \frac{\Gamma(a+\frac{1}{2})}{\sqrt{\pi} \Gamma(a)} (1+x^2)^{-a-\frac{1}{2}}.$$

The scaling clause is Proposition 4.5. (2) is the causal transform $e^{-F_1(\sigma)} = \frac{2^{1-a}}{\Gamma(a)}(\sqrt{2\sigma})^a K_a(\sqrt{2\sigma})$ evaluated at $\sigma = \omega^2/2$; the checked proof computes that Laplace integral rather than quoting it, and the computation is one substitution: with $A = 1/2$, $B = \sigma$ and $u = ce^v$, $c = \sqrt{A/B}$, the integral $\int_0^\infty u^{-a-1} e^{-1/(2u)-\sigma u} du$ becomes $2c^{-a} \int_0^\infty \cosh(av) e^{-2\sqrt{AB} \cosh v} dv$, which is the integral by which K_a is defined here. No asymptotic of K_a enters, at this or any order.

(3). Let U_I be a Thorin measure of T_1 , as the hypothesis provides: $\mathbb{E}e^{-\sigma T_1} = \exp\{-\int_{(0,\infty)} \log(1+\sigma/\theta) U_I(d\theta)\}$ for $\sigma \geq 0$, with $\int_{(0,\infty)} \log(1+\theta^{-1}) U_I(d\theta) < \infty$. That exponent is causally admissible with drift 0 and delay profile $k_I(u) = \int e^{-\theta u} U_I(d\theta)$: a Laplace transform is nonnegative and nonincreasing, and the causal Frullani identity $\int_0^\infty (1-e^{-\sigma u})e^{-\theta u} du/u = \log(1+\sigma/\theta)$ with Tonelli turns the exponent of Definition 4.1 into the Thorin integral, so that the two integrability windows follow from its finiteness at $\sigma = 1$ — for $1 \wedge u \leq e(1-e^{-u})$ bounds the windows by the exponent there. By Lemma 4.2 the image $F_I(\omega^2/2)$ is admissible, and by the mixture identity with clause (1) it is an exponent of the Student-t family: the law is self-decomposable. Complete monotonicity of the spatial profile — membership in the symmetric Thorin subclass by Proposition 5.7(1) iff (2) — is Proposition 5.7(3) read at that datum, which also returns the Thorin measure, twice the image of U_I under $\theta \mapsto \sqrt{2\theta}$, and the vanishing Gaussian coefficient.

(4). The tail of the density in (1) is $|x|^{-2a-1}$, so $\mathbb{E}|X|^n < \infty$ exactly when $n < 2a$, which also follows from $\mathbb{E}|W_T|^{2n} = (2n-1)!! \mathbb{E}T_1^n$ and $\mathbb{E}T_1^n < \infty$ exactly when $a > n$. $\square$

Remark 5.11 (the inverse-gamma delay law is a generalized gamma convolution). The hypothesis that Lemma 4.8 and clause (3) above quantify over is a theorem of the literature, and it is the one thing about this corner that the paper does not prove. For $a > 0$ the inverse-gamma law T_1 of shape a is a generalized gamma convolution: there is a positive measure U_1 on $(0, \infty)$ with $\int_{(0, \infty)} \log(1 + \theta^{-1}) U_1(d\theta) < \infty$ and

$$\mathbb{E}e^{-\sigma T_1} = \exp\left\{-\int_{(0, \infty)} \log\left(1 + \frac{\sigma}{\theta}\right) U_1(d\theta)\right\}, \quad \sigma \geq 0. \quad (5.2)$$

That is [25], §2, pp. 14–15, read at $\lambda = -a < 0$, $\chi = 1$, $\psi = 0$, the parameters at which the generalized inverse Gaussian density of his (1) is the density p_{T_1} of Proposition 5.10. His (5) exhibits the measure: it has density $u(y) = g_a(2y)$ on $(0, \infty)$, where $g_\nu(x) = 2\{\pi^2 x(J_\nu^2(\sqrt{x}) + Y_\nu^2(\sqrt{x}))\}^{-1}$ is Grosswald's ([23]), and his integrability conditions (3) — $\int_0^1 |\log y| U_1(dy) < \infty$ and $\int_1^\infty y^{-1} U_1(dy) < \infty$ — are the displayed finiteness for a measure finite on the compact subsets of $(0, \infty)$; the displayed form is what finiteness of the exponent at $\sigma = 1$ says, and it is the form both consumers read. Being a generalized gamma convolution, T_1 is in particular self-decomposable, which is the weaker hypothesis Lemma 4.8 quantifies over.

Nothing in the paper depends on this remark: both consumers quantify over (5.2) or over its consequence. The measure Halgreen exhibits is built from the Bessel functions J_ν and Y_ν , and no route to (5.2) through the elementary theory is known to the author.

Which principle cuts the Student-t family out of the cone is a question about locality, and it is the subject of the next module. Nothing is stated about it here.

5.5 The symmetric stable family

The symmetric stable laws are the laws with exponent $c|\omega|^\alpha$, $0 < \alpha \leq 2$ ([50], Theorem 14.14, p. 86, for $\alpha < 2$, the Gaussian being the case $\alpha = 2$): the laws that reproduce themselves under addition and rescaling, and the attractors of the generalized central limit theorem. They are the earliest non-Gaussian scale spaces in the axiomatic literature. Pauwels et al. obtained the kernels with transform $e^{-t|\omega|^\alpha}$, for every $\alpha > 0$, from recursivity and scale invariance, those with $\alpha \leq 2$ being the positive ones ([44]). Duits et al. axiomatized them as the α -scale spaces, generated by the fractional powers $(-\Delta)^{\alpha/2}$ of the Laplacian in the present index, with the Poisson scale space $\alpha = 1$ as the case they single out ([16]; [20]), and Pedersen et al. read them as stable Lévy motions and compared them on natural images ([45]). In the hemigroup theory the family is the semigroup corner: Corollary 2.6 says that it is exactly what the axioms give when the stages are required to depend only on the difference of their scales.

The proposition adds the coordinates of the family in this paper's terms. The profile is a pure power, and the pure powers are the profiles that a dilation changes only by a constant, which is the stationarity of the increments seen in the profile. The family lies in the Thorin subclass, with a power-law Thorin density, and it is the image under the bridge of the causal stable family of half the index. And the moment count is what the semigroup costs: for a pure power, $\int_1^\infty x^{n-1-\alpha} dx$ is finite only for $n < \alpha$, so no member but the Gaussian has a variance, whereas every non-power profile with $\int_1^\infty x k(x) dx$ finite has one. So the heavy tails of the stable scale spaces come from the semigroup requirement: the same axioms without it admit members with every moment finite. The extended semigroup class beyond $\alpha = 2$ is not in the cone: positivity fails there.

Proposition 5.12 (the symmetric stable family). *For $0 < \alpha < 2$ the exponent $F(\omega) = |\omega|^\alpha$ is admissible with $a = 0$ and profile $k(x) = C_\alpha^{-1} x^{-\alpha}$, where $C_\alpha = \int_0^\infty (1 - \cos u) u^{-1-\alpha} du$; its symmetric Thorin density is proportional to $\theta^{\alpha-1} d\theta$; the kernels are the symmetric α -stable*

densities; $\mathbb{E}|X_t|^n < \infty$ if and only if $n < \alpha$; and the family is the only admissible one-parameter semigroup. It is the image under the bridge of the causal stable family of index $\alpha/2$. The Gaussian ray closes the family at $\alpha = 2$, with $k \equiv 0$ and $a = 1$.

The phrase “the kernels are the symmetric α -stable densities” names the classical family whose exponent is $|\omega|^\alpha$. No closed form of the density is asserted, and outside a few values of α none exists; absolute continuity is Proposition 2.8. The moment clause is the cited moment criterion of Proposition 5.1(1) applied to the pure power.

Proof. The exponent, the profile and the constant are Corollary 2.6, as is the statement that this is the only admissible one-parameter semigroup. The moment criterion applied to $k(x) = C_\alpha^{-1}x^{-\alpha}$ gives $\int_1^\infty x^{n-1-\alpha}dx < \infty$ exactly when $n < \alpha$ (Proposition 5.1(1)). The causal stable family of index $\alpha/2$ has $F_1(\sigma) = \sigma^{\alpha/2}$, whose image is $2^{-\alpha/2}|\omega|^\alpha$ by Proposition 4.5(4), a positive multiple of F ; the causal Thorin density is proportional to $\theta^{\alpha/2-1}d\theta$, and its image under $\theta' = \sqrt{2\theta}$, with the Jacobian, is proportional to $\theta'^{\alpha-1}d\theta'$ by Proposition 5.7(3).

The Thorin density is also reached directly, without the bridge and without Bernstein’s theorem: for $c > 0$ the Gamma integral $\int_0^\infty e^{-\theta x} c \theta^{\alpha-1} d\theta = c\Gamma(\alpha)x^{-\alpha}$ exhibits the stable profile as a Laplace transform at $c = (C_\alpha\Gamma(\alpha))^{-1}$, and the Tonelli of Proposition 5.7 turns that into (5.1). Only the positivity of C_α is used, not its value: the integrand is nonnegative, integrable — at most $u^{1-\alpha}/2$ near the origin and at most $2u^{-1-\alpha}$ beyond 1 — and nonzero on $(0, 1)$. $\square$

6 The scale evolution in generator form

Linear scale space is usually met as an equation: the scale space of a signal is the solution of the heat equation with the signal as initial value, and the kernel is what solving the equation produces. The classification of the line paper runs the other way. It is stated on the kernels, and it says nothing about what equation, if any, a member of the class solves. This section supplies the equation. For every admissible family, in its canonical gauge, the scale space $u(t, x)$ of a Schwartz signal satisfies

$$\partial_t u = \mathcal{A}_t u,$$

with a generator $\mathcal{A}_t$ that is read off the pair (a, k) . The Gaussian coefficient contributes a second derivative. The profile contributes, for every displacement size, the amount by which the average of the signal’s two neighbours at that distance exceeds the signal (Figure 6). After a reparametrization of scale this is the heat equation for the Gaussian and the fractional heat equation of the α -scale spaces for the stable family, so the classical equations are recovered. The other members have equations of their own. The generator of the Matérn family is a *bounded* operator, which is how its exact implementation (§7) looks at the level of the generator. The generator of a Cin ray is a single second difference.

The equation matters for a second reason. Locality of the evolution and the non-creation of local extrema are the requirements from which the Gaussian axiomatics select one member ([33]; [36]). They are conditions on the evolution in scale, and this is the form in which they are asked. They are the subject of the next module. The present section states the form and computes the generators of the corners, and stops there. The rest of the infinitesimal theory is outside this paper: the existence and uniqueness of solutions of the evolution equation, the decomposition of a generator over the generators of the extreme rays of §3, and the infinitesimal form of the bridge of §4.

The signals are taken in the Schwartz class. It is the simplest choice that makes every step of the evolution identity unconditional. Enlarging it is a question this paper does not answer: to the Sobolev class $W^{2,1}$, two derivatives in L^1 , for the Gaussian part, or to the full domain of the generator in general.

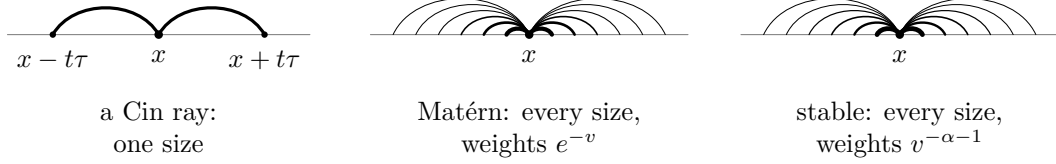


Figure 6: The jump part of the generator (6.2), schematically. At a position x and a scale t the generator compares $g(x)$ with the average of its two neighbours at distance tv , for every displacement size v . It sums the differences with the weight $\varpi(dv)$, drawn here as line width. A Cin ray has one size, so its generator is a single second difference. The Matérn member has every size with exponentially decaying weight, and the sum is a bounded operator, $2\gamma/t$ times the difference between convolution by the Laplace kernel of range t and the identity. A stable member has every size with power-law weight, not summable at small sizes, and the sum is the fractional Laplacian. The Gaussian part of a generator is not drawn: it is the limit of the same comparison at vanishing size, a multiple of the second derivative.

Definition 6.1 (the signal class). $\mathcal{D} := \mathcal{S}(\mathbb{R})$ is the Schwartz class on the line: smooth functions all of whose derivatives decay faster than every polynomial. It is contained in L^1 , is stable under differentiation, and its Fourier image is itself.

The equation is simplest on the Fourier side. In the canonical gauge the kernel at scale t has transform $e^{-F(t\omega)}$, and differentiating in t brings down the symbol $B = \omega F'$ of (2.4), evaluated at $t\omega$ and divided by t . Since $B \geq 0$, every wavenumber is damped, at the rate $B(t\omega)/t$. The symbol is a classical object of the probability literature. A self-decomposable law is the limit law of a process of Ornstein–Uhlenbeck type, its exponent being $\int_0^\infty \psi(e^{-s}\omega) ds$ for the exponent ψ of a Lévy process, the background driving process, written with the sign of this paper, so that the process at time 1 has the transform $e^{-\psi}$ ([50], Theorem 17.5, pp. 108–109), and differentiating that representation gives $\psi = \omega F' = B$. What this section adds to it is the reading as an evolution equation for the signal, with its domain and its constants. The signal side is the same identity read through the displacement dictionary of Remark 2.4. Here ω^2 is the symbol of $-\partial_x^2$, and $1 - \cos(\omega h)$ is the symbol of $g \mapsto g - \frac{1}{2}(g(\cdot - h) + g(\cdot + h))$, the negative of the symmetric second difference at distance h . So the two terms of B become a Laplacian and a superposition of second differences over the displacement sizes. The differences are weighted by the tail measure ϖ and dilated to the current scale. The factor t^{-1} says that the natural clock is log-scale: in $\theta = \log t$ the generators at different scales are dilates of one operator.

Proposition 6.2 (the scale evolution in generator form). *Let F be admissible with data (a, k) and $\varpi = -dk$, and let $u(t, \cdot) := \mu_{0,t} * f$ be the scale space of $f \in \mathcal{D}$ in the canonical gauge. Then on the Fourier side*

$$\partial_t \hat{u}(t, \omega) = -\frac{1}{t} B(t\omega) \hat{u}(t, \omega), \quad B(\omega) = \omega F'(\omega) = 2a\omega^2 + \int_{(0, \infty)} (1 - \cos \omega v) \varpi(dv), \quad (6.1)$$

for every ω and every $t > 0$, and on the signal side

$$\partial_t u(t, x) = \mathcal{A}_t u(t, x), \quad \mathcal{A}_t g(x) := 2at g''(x) - \frac{1}{t} \int_{(0, \infty)} \left[g(x) - \frac{1}{2}(g(x - tv) + g(x + tv)) \right] \varpi(dv), \quad (6.2)$$

the integral converging absolutely for $g \in \mathcal{D}$. The generators at different scales are conjugate, $\mathcal{A}_t = t^{-1} D_t \mathcal{A}_1 D_t^{-1}$, with $\mathcal{A}_1 = -B(D)$ and $D = -i\partial_x$; in log-scale $\theta = \log t$ the evolution reads $\partial_\theta u = D_t \mathcal{A}_1 D_t^{-1} u$.

The operator $\mathcal{A}_1 = -B(D)$ is the generator of the symmetric Lévy process with Gaussian part $2a$ and folded jump measure ϖ , which is the probabilistic reading of the symbol and is used nowhere below.

The signal-side identity is a pointwise derivative in the scale at a fixed position. Its right-hand side contains a second space derivative of the scale space. That derivative exists: the scale space of a Schwartz signal is twice differentiable. The one step of the proof that is not a computation on symbols comes from the tails. The scale space $u(t, \cdot)$ is not a Schwartz function, since it inherits the tails of the kernel, and those are polynomial for the Student-t and stable families. So the generator is first moved off u and onto the signal, by translation invariance, and the multiplier identity is applied there.

Proof. $\hat{u}(t, \omega) = e^{-F(t\omega)} \hat{f}(\omega)$, and F is C^1 off the origin with $\omega F'(\omega) = B(\omega)$ by Lemma 7.1 of [17] and (2.4); differentiating in t gives (6.1), since $\partial_t F(t\omega) = \omega F'(t\omega) = t^{-1} B(t\omega)$.

For (6.2): $B(t\omega) = 2at^2\omega^2 + \int (1 - \cos t\omega v) \varpi(dv)$, where ω^2 is the symbol of $-\partial_x^2$ and $1 - \cos(\omega tv)$ is the symbol of $g \mapsto g - \frac{1}{2}(T_{tv} + T_{-tv})g$. So the Fourier multiplier $B(tD)$ acts on $g \in \mathcal{D}$ as $-2at^2 g'' + \int [g - \frac{1}{2}(g(\cdot - tv) + g(\cdot + tv))] \varpi(dv)$, the integrand being bounded by $\min(2\|g\|_\infty, \frac{1}{2}t^2 v^2 \|g''\|_\infty)$ and hence ϖ -integrable, since $\int (1 \wedge v^2) \varpi < \infty$. Thus $\mathcal{A}_t = -t^{-1} B(tD)$ on $\mathcal{D}$, which is (6.2) read as a Fourier multiplier.

That $\partial_t u$ equals $\mathcal{A}_t u$ pointwise is the inverse transform of (6.1), and the inversion is used twice. Write $\langle \kappa_\varphi, h \rangle = \int_{\mathbb{R}} h(z) \cos(\omega z - \varphi) dz$ for the pairing of h with the character of frequency ω and phase φ . For a symmetric probability measure ν and an h that is continuous and integrable with integrable transform,

$$(\nu * h)(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\nu}(\omega) \langle \kappa_{\omega x}, h \rangle d\omega,$$

by Fourier inversion for h together with one exchange of integrals against the finite ν ; the symmetry of ν is what turns the ν -average of $\cos(\omega(x-y))$ into $\hat{\nu}(\omega) \cos(\omega x)$, and no other step of the proof uses it.

Read at $h = f$ and $\nu = \mu_{0,s}$, this represents the scale space as $u(s, x) = (2\pi)^{-1} \int e^{-F(s\omega)} \langle \kappa_{\omega x}, f \rangle d\omega$. Differentiating under that integral at $s = t$ is licensed on $s \in (t/2, 3t/2)$ by the dominating function $s^{-1} B(s\omega) |\langle \kappa_{\omega x}, f \rangle|$, which is integrable because $B(s\omega) \leq C(1 + s^2 \omega^2)$ by Lemma 2.10 of [17] while $|\langle \kappa_\varphi, f \rangle| \leq (\|f\|_1 + \|f''\|_1)/(1 + \omega^2)$ uniformly in the phase — the two crude bounds $|\langle \kappa_\varphi, f \rangle| \leq \|f\|_1$ and $\omega^2 |\langle \kappa_\varphi, f \rangle| = |\langle \kappa_\varphi, f'' \rangle| \leq \|f''\|_1$, combined — and iterating that estimate on f'' improves the decay to $(1 + \omega^2)^{-2}$, which is one power more than the symbol's growth consumes. So $\partial_t u(t, x) = -(2\pi)^{-1} \int t^{-1} B(t\omega) e^{-F(t\omega)} \langle \kappa_{\omega x}, f \rangle d\omega$.

Read at $h = \mathcal{A}_t f$ and $\nu = \mu_{0,t}$, it represents the right-hand side — but only after the generator has been moved off $u(t, \cdot)$, which is *not* in $\mathcal{D}$: the scale space inherits the kernel's tails, and the Student-t and stable kernels have polynomial ones, so the multiplier identity above does not apply to it. Translation invariance supplies the step: $\mathcal{A}_t$ commutes with convolution by a finite measure, $\mathcal{A}_t(\nu * f) = \nu * (\mathcal{A}_t f)$, the exchange of the ϖ -integral with the ν -integral being licensed by the same bound $\min(2\|f\|_\infty, \frac{1}{2}t^2 v^2 \|f''\|_\infty)$ that makes the generator's own integral converge. Hence $\mathcal{A}_t u(t, \cdot) = \mu_{0,t} * (\mathcal{A}_t f)$, whose argument is continuous and integrable, and $\langle \kappa_{\omega x}, \mathcal{A}_t f \rangle = -t^{-1} B(t\omega) \langle \kappa_{\omega x}, f \rangle$ by the multiplier identity at the scale t . The two frequency integrands agree, which is (6.2).

Conjugation: D_t has symbol map $\omega \mapsto t\omega$, so $D_t B(D) D_t^{-1} = B(tD)$; and $\partial_\theta = t \partial_t$. $\square$

Remark 6.3 (relation to the causal theory). The causal article [18] states its scale evolution in the same form. In place of the multiplier identity it uses Phillips' formula, which gives the generator of a subordinated semigroup from that of the original one ([52], Chapter 13). The causal generator carries the memory kernel of the half-line, an integral against the past of the signal. This one carries symmetric second differences, the average of the two neighbours at every displacement size. The bridge of §4 maps the one to the other for the subordinated members. Its infinitesimal form, Phillips' formula read on the Fourier side, is one of the parts of the infinitesimal theory that this paper leaves out.

The next proposition evaluates the generator on the members that have names. Five kinds of operator appear. The Gaussian has a differential operator, and its equation is the heat equation in the variance. The stable family has a fractional differential operator: in the variable $v = t^\alpha$ its equation is the α -scale-space equation of [16] (p. 288, (66)), whose generator $-(-\Delta)^{\alpha'}$, $0 < \alpha' \leq 1$, is the present one at $\alpha' = \alpha/2$, their exponent being half of the stable index used here. The Matérn family has a bounded integral operator, the Laplace kernel minus the identity, which is the generator of a compound Poisson process: jumps of Laplace-distributed size arriving at a finite rate. The members of the Thorin subclass have mixtures of such operators over the Thorin measure. And the extreme rays of §3 have finite differences. Which of these kinds locality admits is the subject of the next module.

Proposition 6.4 (the generators of the corners). *In the canonical gauge, and with Lap_θ denoting convolution by the Laplace kernel $\frac{1}{2\theta}e^{-|x|/\theta}$ of range θ :*

1. Gaussian: $\mathcal{A}_t = 2at \partial_x^2$, the heat equation $\partial_v u = \partial_x^2 u$ in the variance gauge $v = at^2$.
2. Symmetric stable, $F = |\omega|^\alpha$ with $0 < \alpha < 2$: $\mathcal{A}_t = -\alpha t^{\alpha-1}(-\partial_x^2)^{\alpha/2}$, the fractional Laplacian; in $v = t^\alpha$ the equation is $\partial_v u = -(-\partial_x^2)^{\alpha/2} u$, and the jump measure is $\varpi(dv) = C_\alpha^{-1} \alpha v^{-\alpha-1} dv$.
3. Matérn, $F = \gamma \log(1 + \omega^2)$: $\mathcal{A}_t = \frac{2\gamma}{t} (\text{Lap}_t - I)$, a bounded operator of norm at most $4\gamma/t$. It is the generator of a compound Poisson process, whose jumps are Laplace-distributed of range t and arrive at the rate $2\gamma/t$.
4. Thorin members, with Thorin measure U as in (5.1) and $a = 0$:

$$\mathcal{A}_t = \frac{1}{t} \int_{(0,\infty)} (\text{Lap}_{t/\theta} - I) U(d\theta),$$

the Thorin mixture of Matérn generators, as an identity at every g that is twice continuously differentiable with g and g'' bounded. Under the hypothesis of Remark 5.11 the Student- t family is the case of the Bessel Thorin measure: U is twice the image of the causal U_1 of (5.2) under $\theta \mapsto \sqrt{2\theta}$, by Proposition 5.7(3).

5. Cin ray of step τ : $\mathcal{A}_t = \frac{1}{t} \delta_{t\tau}^2$, a single symmetric second difference, where $\delta_h^2 g := \frac{1}{2}(g(\cdot - h) + g(\cdot + h)) - g$.

In clause (2) the fractional Laplacian is understood as the Fourier multiplier $|\omega|^\alpha$, so the clause is a statement about the symbol $B(\omega) = \alpha|\omega|^\alpha$ and the jump measure, read through the multiplier identity of Proposition 6.2. The class named in clause (4) is the class on which the integral of that proposition converges, by the bound in its proof; it is larger than $\mathcal{D}$, the class the proposition itself is stated on. It is needed because for a Thorin member with $U((0,\infty)) = k(0+) = \infty$ — the symmetric stable members among them — the tail measure is infinite near the origin, and boundedness of g alone does not make the integral converge. For a member whose Thorin measure is finite, the Matérn one and every finite quadrature among them, boundedness of g suffices, as the proof records.

Proof. (1) is Proposition 6.2 with $\varpi = 0$.

(2) $B(\omega) = \omega F'(\omega) = \alpha|\omega|^\alpha$, so $-t^{-1}B(t\omega) = -\alpha t^{\alpha-1}|\omega|^\alpha$, and $|\omega|^\alpha$ is the symbol of $(-\partial_x^2)^{\alpha/2}$; the change of variable $v = t^\alpha$ gives $\partial_v = (\alpha t^{\alpha-1})^{-1} \partial_t$. The jump measure is $-dk$ for $k(x) = C_\alpha^{-1} x^{-\alpha}$ (Proposition 5.12).

(3) and (4) are one computation. For $\theta > 0$, $\int_0^\infty [g - \frac{1}{2}(g(\cdot - v) + g(\cdot + v))] \theta e^{-\theta v} dv = g - \text{Lap}_{1/\theta} g$, because $\int_0^\infty \theta e^{-\theta v} dv = 1$ and $\int_0^\infty \frac{1}{2}(g(x - v) + g(x + v)) \theta e^{-\theta v} dv = \int_{\mathbb{R}} g(x - y) \frac{1}{2} \theta e^{-\theta|y|} dy = (\text{Lap}_{1/\theta} g)(x)$.

A Thorin member with $a = 0$ has $k(x) = \int e^{-\theta x} U(d\theta)$ (Proposition 5.7), and its Choquet measure is the U -mixture of exponential laws, $\varpi = \int \theta e^{-\theta v} U(d\theta) dv$ — equivalently, the image of the product of U with the standard exponential law under $(\theta, u) \mapsto u/\theta$. This is an identification of measures and not a differentiation of the profile: ϖ is only given by its tails, and almost everywhere at that, so what is checked is that the mixture has the right ones, $\int_y^\infty \int \theta e^{-\theta v} U(d\theta) dv = \int e^{-\theta y} U(d\theta) = k(y)$ by Tonelli, after which two folded measures with the same tails on $(0, \infty)$ agree. Writing the mixture as an image measure rather than as a density is what makes the exchange below the ordinary Fubini on a product.

Substituting into (6.2) and exchanging the U -integral with the ϖ -integral — licensed by the absolute convergence of Proposition 6.2 for g bounded with bounded second derivative — and dilating $v \mapsto tv$, which sends $\text{Lap}_{1/\theta}$ to $\text{Lap}_{t/\theta}$, gives $\mathcal{A}_t g = \frac{1}{t} \int (\text{Lap}_{t/\theta} - I)g U(d\theta)$, which is (4). For (3) the Matérn member has $U = 2\gamma\delta_1$ by Proposition 5.2(1) and Proposition 5.7, so $\mathcal{A}_t = \frac{2\gamma}{t}(\text{Lap}_t - I)$; the norm bound is $\|\text{Lap}_t\| \leq 1$ on L^1 and on L^∞ . There $\varpi = 2\gamma e^{-v} dv$ is finite and boundedness of g alone suffices, as it does whenever U is finite, since $\varpi((0, \infty)) = U((0, \infty)) = k(0+)$; a Thorin member with $k(0+) = \infty$ has ϖ infinite near the origin, which is why clause (4) names the class it does.

(5) is Proposition 6.2 with $a = 0$ and $\varpi = \delta_\tau$, the profile being $\mathbf{1}_{(0, \tau)}$ (Lemma 3.1(1)). $\square$

7 Implementation: exact cascades and recursive sections

A scale-space representation is computed at finitely many scales, and the usual way to compute it is to treat each scale separately: convolve the signal with the kernel of that scale, truncated to a finite filter, or replace the kernel by a recursive filter that approximates it ([13, 65]). Both are approximations of the kernel. The discretization of the signal has a theory of its own ([34]), and this section does not enter it: the statements below are for signals on the line, and sampling the signal is a separate approximation. This section is about the scale direction: for an admissible family nothing in that direction has to be approximated. The family is a cascade. The record at one scale, the smoothed signal there, is obtained from the record at the previous scale by one stage, and the hemigroup axiom says that the result is the scale space itself. So the axiom that defines the class is also its algorithm, and what has to be implemented is a stage.

The section follows that thought, and Figure 7 is the picture. The cascade reproduces the scale space exactly at every knot of any finite ladder of scales (§7.1, Proposition 7.1). For the Matérn family with integer index a stage is a rational filter, realized by first-order recursive sections run forward and backward along the signal, with positive coefficients and a cost per sample that does not depend on the scale (§7.2, Proposition 7.2), and a criterion on the pole-zero diagram says which rational filters — among them the recursive filters used for the Gaussian — are themselves scale spaces (Proposition 7.3, Remark 7.4). Finite mixtures of Matérn members with integer weights have such stages too, a rational filter generates a ladder of scales and one candidate family between its levels, and a family with rational stages need not be of either kind (§7.3, Proposition 7.5, Remark 7.6). Every member with a completely monotone profile is a limit of members with finitely many ranges, but the families realized by the first-order sections reach only the members whose Thorin measure consists of atoms of even integer mass, possibly with a Gaussian part, and so neither the stable nor the Student-t members, which lack moments that every such limit has (§7.4, Propositions 7.7 and 7.8). §7.5 collects the costs, and §7.6 runs the cascade on a sampled signal, checks the statements on it, and applies the criterion of Proposition 7.3 to two published recursive Gaussians (Example 7.12). The causal theory has the same structure on the time axis, with one-sided first-order sections ([18]). What is new on the line is that a section becomes a forward-backward pair, the form of recursive filter that image processing has long used for the Gaussian, and that the theory says which of those filters are scale spaces.

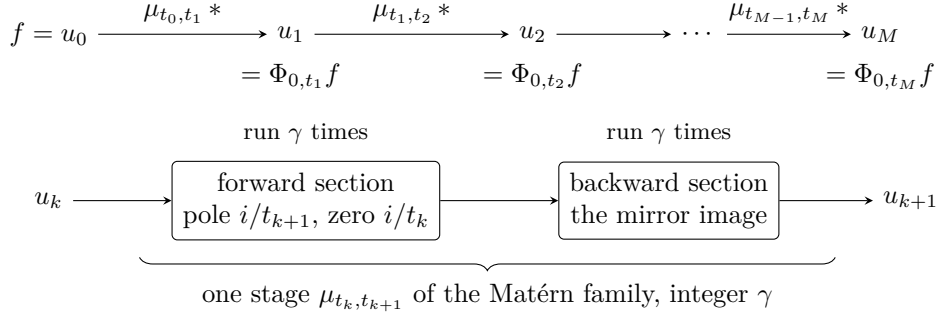


Figure 7: The cascade as an algorithm. Top: over any ladder of scale knots the cascade of stages reproduces the scale space exactly at every knot (Proposition 7.1). Bottom: for the Matérn family with integer γ , a stage is γ first-order recursive sections run forward along the signal followed by γ run backward (the label above a box says how often its section is run), each with one state and with positive coefficients (Proposition 7.2).

7.1 Exactness at the knots

Choose the scales at which the representation is wanted, $0 = t_0 < t_1 < \dots < t_M$, start from the signal, and compute each record from the previous one. The proposition says that this computes the scale space, and that adding knots later changes nothing at the old ones.

Proposition 7.1 (exactness at the knots). *Let $0 = t_0 < t_1 < \dots < t_M$ be any finite set of scale knots and define the discrete cascade $u_0 := f$, $u_{k+1} := \mu_{t_k, t_{k+1}} * u_k$. Then $u_k = \Phi_{0, t_k} f$ exactly for every k , and refining the knot set changes nothing at the old knots. On the transform side the increment transforms telescope, $\prod_{k < M} e^{-F(t_{k+1}\omega) + F(t_k\omega)} = e^{-F(t_M\omega)}$.*

The cascade is the family sampled at the knots, so there is no discretization error in scale to control. Since the family exists at every scale, a ladder of knots can also be shifted at will, and an arbitrary dilation is handled by instantiating the family at the shifted knots.

Proof. Induction on (A6): $\mu_{0, t_{k+1}} = \mu_{t_k, t_{k+1}} * \mu_{0, t_k}$, by Lemma 5.1 of [17], and $u_0 = \Phi_{0, t_0} f = f$. $\square$

7.2 The Matérn cascade

Exactness at the knots is useful only if a stage can be computed, and in general a stage is a convolution with a kernel that has no closed form. The Matérn family with integer index is the case where a stage costs one state and a few operations per sample. The transform of a stage is a ratio of two polynomials in ω^2 of the same degree, and each of its factors splits into a part with its pole in the upper half-plane and a part with its pole in the lower one. The first is a causal first-order recursive filter along the signal, a *lag section* as the proposition names it, and the second is its mirror image. Running the one forward and the other backward realizes their product, which is the symmetric stage. Each section is a convex combination of the identity and a one-sided exponential average, so positivity and unit mass hold already at the level of the realization.

Proposition 7.2 (the Matérn cascade: forward–backward first-order sections). *For the Matérn family with integer γ , so that $e^{-F(t\omega)} = (1 + t^2\omega^2)^{-\gamma}$, the increment between adjacent knots has transform*

$$\hat{\mu}_{t_k, t_{k+1}}(\omega) = \left(\frac{1 + t_k^2 \omega^2}{1 + t_{k+1}^2 \omega^2} \right)^\gamma = \left(\frac{1 + it_k \omega}{1 + it_{k+1} \omega} \cdot \frac{1 - it_k \omega}{1 - it_{k+1} \omega} \right)^\gamma, \quad (7.1)$$

a cascade of γ identical causal first-order pole-zero sections in x , called lag sections below, and γ identical anticausal ones, each with the one-state realization

$$w' = \frac{y - w}{t_{k+1}} \quad (\text{forward}), \quad v = \frac{t_k}{t_{k+1}} y + \left(1 - \frac{t_k}{t_{k+1}}\right) w,$$

and its mirror image run backward. Which solution of the state equation is meant is part of the realization: its solutions differ by a multiple of $e^{-x/t_{k+1}}$, and the one intended is the solution in L^1 , equivalently the one with $w(-\infty) = 0$ for the forward pass and $w(+\infty) = 0$ for the backward one. The zero of the transfer function costs no differentiation of the input: the output is a convex combination of the input and the state. The section's kernel is the mixture $\frac{t_k}{t_{k+1}} \delta_0 + (1 - \frac{t_k}{t_{k+1}}) \text{Exp}_+(t_{k+1})$ of no displacement and a one-sided exponential displacement, $\text{Exp}_\pm(t)$ denoting the one-sided exponential kernels $t^{-1}e^{-|x|/t}$ carried by $\pm(0, \infty)$; so the two-sided increment for $\gamma = 1$ is the four-term mixture

$$r^2 \delta_0 + r(1 - r)(\text{Exp}_+ + \text{Exp}_-) + (1 - r)^2 \text{Lap}_{t_{k+1}}, \quad r := \frac{t_k}{t_{k+1}} \in (0, 1), \quad (7.2)$$

where $\text{Exp}_\pm$ are taken at the range t_{k+1} and Lap_t is the Laplace kernel of range t ; positivity and unit mass are manifest at the realization level. Sampled on a grid, a full scale space over M knots costs $2\gamma M$ first-order states and $O(\gamma M)$ multiply-adds per sample, in a forward and a backward pass at each knot. Sampling the signal is a separate approximation and is not treated here.

Proof. The increment transform is the ratio of the endpoint transforms, the denominators being nonvanishing (Lemma 4.2 of [17]), which is Proposition 5.2(2); the factorization is $1 + t^2\omega^2 = (1 + it\omega)(1 - it\omega)$. The transfer identity of the section is

$$\frac{t_k}{t_{k+1}} + \left(1 - \frac{t_k}{t_{k+1}}\right) \frac{1}{1 + it_{k+1}\omega} = \frac{1 + it_k\omega}{1 + it_{k+1}\omega},$$

and the convex-combination clause is $0 < t_k/t_{k+1} < 1$. In particular the output is a convex combination of the input and the state, so the section differentiates nothing. For (7.2), convolve the causal and anticausal mixtures and use $\text{Exp}_+(t) * \text{Exp}_-(t) = \text{Lap}_t$, whose transform identity is $(1 + it\omega)^{-1}(1 - it\omega)^{-1} = (1 + t^2\omega^2)^{-1}$. $\square$

The proposition bears on the approximation of the Gaussian by recursive filters, and it turns into a test. A recursive filter is given by its transfer function, which is a rational function of ω ; the question is whether that function is the transform of the kernel of an admissible family, in which case the filter is not an approximation of a scale space but a scale space. The next proposition answers it from the pole-zero diagram alone, in three conditions. The first is a glance at the real axis. The other two are read off one explicit exponential sum, built by giving every pole a positive exponential term and every zero a negative one: that sum is the candidate profile, and the filter is a scale space exactly when it is a profile at all, nonnegative and nonincreasing. Example 7.12 applies the test to two published designs.

Proposition 7.3 (which rational transfer functions are admissible kernels). *Let R be an even rational function with $R(0) = 1$, written in lowest terms. Write $\pm i\theta_1, \dots, \pm i\theta_M$ for its poles and $\pm i\zeta_1, \dots, \pm i\zeta_L$ for its zeros, listed with multiplicity and normalised by $\text{Re } \theta_m > 0$ and $\text{Re } \zeta_l > 0$; both lists are closed under conjugation, so the signed profile*

$$k(x) := 2 \sum_{m \leq M} e^{-\theta_m x} - 2 \sum_{l \leq L} e^{-\zeta_l x}, \quad x > 0, \quad (7.3)$$

is real-valued. Then R is the transform of the kernel at canonical scale 1 of an admissible family — equivalently, $-\log R$ is an admissible exponent — if and only if

1. R has neither a pole nor a zero on the real axis, so that $R > 0$ there;
2. $k \geq 0$ on $(0, \infty)$; and
3. k is nonincreasing on $(0, \infty)$.

The family is then the one with Gaussian coefficient 0 and folded profile k . Condition (i) alone excludes every R with a real zero, since the transform of an infinitely divisible law has no zero at all (Lemma 4.2 of [17]).

A Gaussian factor is not covered and needs none: a rational R forces the Gaussian coefficient to vanish, which is the growth argument in the proof, and a design with a Gaussian factor is that factor times a rational one. The criterion is for the transfer function on the line; a sampled filter is a different object, and the discrete classification is a different theory ([34]).

Proof. The factorisation. R is even and rational, so $R(\omega) = N(\omega^2)/D(\omega^2)$ for polynomials N and D with no common factor, real because R is real on the real axis. Its poles and zeros are therefore closed under $\omega \mapsto -\omega$ and under conjugation, and under (i) none of them is real; choosing from each pair $\pm\omega_0$ the one in the upper half-plane and writing it as $i\theta$ with $\operatorname{Re} \theta > 0$ gives the two lists of the statement, each closed under conjugation. Then $D(\omega^2) = c_D \prod_m (\omega^2 + \theta_m^2)$ and $N(\omega^2) = c_N \prod_l (\omega^2 + \zeta_l^2)$, and $R(0) = 1$ fixes the constant:

$$R(\omega) = \frac{\prod_{l \leq L} (1 + \omega^2/\zeta_l^2)}{\prod_{m \leq M} (1 + \omega^2/\theta_m^2)}. \quad (7.4)$$

The exponential-profile identity at a complex rate. For $\operatorname{Re} \theta > 0$ and ω real,

$$\int_0^\infty (1 - \cos \omega x) e^{-\theta x} \frac{dx}{x} = \frac{1}{2} [\operatorname{Log}(\theta - i\omega) + \operatorname{Log}(\theta + i\omega)] - \operatorname{Log} \theta, \quad (7.5)$$

with Log the principal logarithm, which is what Proposition 5.2(1) says at a real rate. Indeed $1 - \cos \omega x = \frac{1}{2}(e^{i\omega x} + e^{-i\omega x})$, so the integral is half the sum of $\int_0^\infty (e^{-\theta x} - e^{-(\theta \mp i\omega)x}) dx/x$, and for two points α, β of the open right half-plane

$$\int_0^\infty (e^{-\alpha x} - e^{-\beta x}) \frac{dx}{x} = \int_0^\infty \int_{[\alpha, \beta]} e^{-sx} ds dx = \int_{[\alpha, \beta]} \frac{ds}{s} = \operatorname{Log} \beta - \operatorname{Log} \alpha,$$

the segment $[\alpha, \beta]$ lying in that half-plane by convexity: the exchange is Fubini, the modulus of the inner integrand being at most e^{-cx} with $c > 0$ the least real part on the segment, which is integrable against dx on $[1, \infty)$ while on $(0, 1)$ the inner integral is bounded by $|\beta - \alpha|$; and $1/s$ is analytic on a neighbourhood of the segment, where Log is a primitive.

The exponent of the signed profile. k of (7.3) is continuous on $(0, \infty)$, bounded there by $2(M + L)$, and $|k(x)| \leq 2(M + L)e^{-cx}$ with $c := \min_m \operatorname{Re} \theta_m \wedge \min_l \operatorname{Re} \zeta_l > 0$; so $\int_0^\infty (1 - \cos \omega x)|k(x)| dx/x$ is finite, the integrand being $O(x)$ at the origin and exponentially small at infinity. Summing (7.5) over the two lists,

$$\Psi(\omega) := \int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x} = \sum_m \operatorname{Log}\left(1 + \frac{\omega^2}{\theta_m^2}\right) - \sum_l \operatorname{Log}\left(1 + \frac{\omega^2}{\zeta_l^2}\right),$$

where each summand is $\operatorname{Log}(\theta - i\omega) + \operatorname{Log}(\theta + i\omega) - 2\operatorname{Log} \theta$ and no branch is chosen for a product: the value is real, since conjugate rates contribute conjugate summands and a real rate a real one. Exponentiating kills the branches, $e^{\operatorname{Log} z} = z$ for every $z \neq 0$, and (7.4) gives $e^{-\Psi(\omega)} = R(\omega)$. So $\Psi = -\log R$ with the real logarithm of a positive number, at every real ω .

Sufficiency. Assume (i), (ii) and (iii). Then k is nonnegative, nonincreasing, bounded on $(0, 1]$ and exponentially small at infinity, so $\int_0^1 x k(x) dx$ and $\int_1^\infty k(x) x^{-1} dx$ are finite: by

Lemma 2.3 the pair $(0, k)$ is of the form (2.3), and by Theorem 2.5 it is the pair of an admissible family, whose exponent is $\Psi = -\log R$. The transform of its kernel at canonical scale 1 is $e^{-\Psi} = R$.

Necessity. Assume $R = e^{-F}$ with F admissible, of data (a, k_F) . Then $R > 0$ and $R \leq 1$ on the real axis, so R has there neither a zero nor a pole, which is (i). Next $a = 0$: by (7.4) $|R(\omega)|$ is bounded above and below by multiples of $|\omega|^{2(L-M)}$ for large $|\omega|$, so $F = -\log R$ grows at most like a multiple of $\log |\omega|$, while $F(\omega) \geq a\omega^2$ for every ω . So $F = \Psi$ by the previous paragraph, that is

$$\int_0^\infty (1 - \cos \omega x) k_F(x) \frac{dx}{x} = \int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x}, \quad \omega \in \mathbb{R},$$

an identity between an exponent with a positive Lévy measure and one with a signed density. The uniqueness of the Lévy pair (Proposition 2.3 of [17](3)) is stated for positive measures, so it is applied after a recombination, as in the proof of Lemma 3.5: writing $k = k^+ - k^-$ for the positive and negative parts, both $(k_F + k^-)(x) dx/x$ and $k^+(x) dx/x$ are positive measures on $(0, \infty)$ satisfying the Lévy condition — each of $k^\pm$ is dominated by $|k|$, which does — and the displayed identity says that the two pairs $(0, (k_F + k^-) dx/x)$ and $(0, k^+ dx/x)$ have the same exponent. Hence they are equal, and $k_F = k$ almost everywhere on $(0, \infty)$. Since k is continuous and k_F is nonnegative and nonincreasing, the identity holds everywhere and k inherits both properties: the agreement set is conull, hence meets every interval, and approaching a given x from each side through it gives $k(x) \geq k(y)$ for $x < y$ and $k \geq 0$. That is (ii) and (iii).

The last sentence. If R vanishes at a real ω_0 then R is the transform of no infinitely divisible law, the transform of such a law having no zero (Lemma 4.2 of [17]); in particular it is the transform of no admissible kernel. $\square$

Remark 7.4 (which recursive Gaussians are scale spaces). Read at the designs of the image-processing literature, the criterion says the following. If the transfer has only real poles and no zeros, then $R = \prod_i (1 + t_i^2 \omega^2)^{-1}$, whose signed profile $2 \sum_i e^{-x/t_i}$ is positive and decreasing, so the three conditions hold and the filter is the kernel of the symmetric Thorin member with atoms at $1/t_i$ (Proposition 5.7), the multi-range Matérn; its dilates form a scale space whose increments are again forward-backward lag sections. The classical all-pole approximations of the Gaussian with real poles — in particular the cascade of γ identical first-order sections at $t = \sigma/\sqrt{2\gamma}$, whose transfer is $(1 + \sigma^2 \omega^2 / 2\gamma)^{-\gamma} \rightarrow e^{-\sigma^2 \omega^2 / 2}$ — are therefore not approximations of a scale space but scale spaces, Matérn members converging to the Gaussian along γ at matched variance, the variance being $2\gamma t^2$ by Proposition 5.2(3). Neither complex poles nor zeros off the real axis are excluded in advance, and neither is harmless. A complex pair contributes a term $e^{-cx} \cos(dx)$ to the signed profile, which changes sign by itself, so condition (ii) has to be checked against the other terms; a real pole of sufficient weight can carry such a pair, and Remark 7.6 exhibits an admissible family that does, of exponent $3 \log(1 + \omega^2) + \log(1 + \omega^4/4)$. A zero of the transfer on the real axis fails (i) outright, while a zero off it is a term subtracted in (7.3) and may leave (ii) and (iii) standing or not: the second example of that remark, of exponent $2 \log(1 + \omega^2) - \log(1 + \omega^2/4)$, has all its poles and zeros on the imaginary axis, is admissible, and is neither a Thorin member nor realized by lag sections. The two standard recursive Gaussians of the image-processing literature are optimized for accuracy and are decided by different conditions. The design of [13] adds a causal and an anticausal half (§5.1, (38), two complex-conjugate pole pairs), so its transfer is not a modulus squared at all, and at the published coefficients that sum has zeros on the real axis: it fails (i). The design of [65] is a forward-backward cascade with one real pole and one conjugate pair (p. 141, Table 1 with (6a)–(6b)); it passes (i), and at the published coefficients its signed profile goes negative, so it fails (ii). Each verdict is a computation on the printed coefficients of one design and not a theorem about a class of designs; Example 7.12 carries both out. Whatever their accuracy as Gaussian approximations, a filter whose R fails the test is the kernel of no exact cascade. The

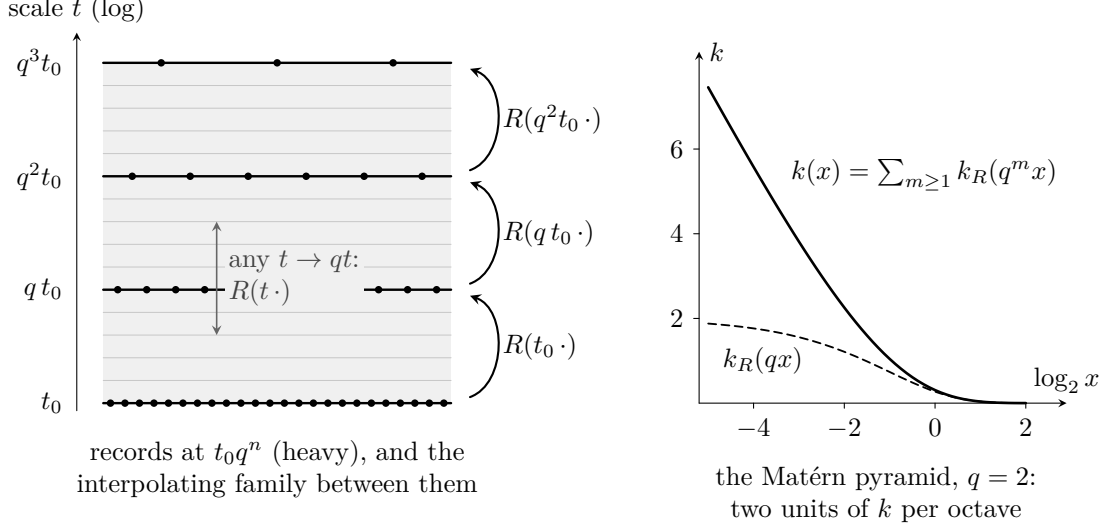


Figure 8: The pyramid of Proposition 7.5(2), schematically. Left: a ladder of scales at ratio q . The stage from one level to the next is one rational filter R , dilated with the level, and the proposition gives the one family that has these stages and exists at every scale between the levels. For that family the stage from any scale t to qt is $R(t \cdot)$, so the ladder can start at any t_0 . The marks on the levels, sparser by q from level to level, show the subsampling that a pyramid in multiresolution practice adds to such a ladder. They are drawn for orientation: the statement is about convolution ladders on the line and involves no subsampling. Right: the folded profile of the interpolating family for $R = (1 + \omega^2)^{-1}$ and $q = 2$, with its first term dashed. Each level of the ladder adds one copy of k_R , dilated by q , so the profile grows by $k_R(0+) = 2$ per octave as $x \rightarrow 0$ and is unbounded at the origin, where the profile $2\gamma e^{-x}$ of a Matérn member is bounded.

theory's recommendation is therefore the opposite of the literature's optimization: approximate the Gaussian *from inside the class*, by real-pole cascades with no zeros, and accept the Matérn's exponential tails and its kink at the origin for $\gamma = 1$ as the price of exactness in scale.

7.3 Rational increments and the pyramid they interpolate

Two questions follow. Which other families have rational stages, and what can be said of a rational filter that is given in advance, as a recursive filter usually is? The first clause of the proposition answers the first: a finite mixture of Matérn members with integer weights w_i in the exponent has rational stages of order $2 \sum_i w_i$, realized by lag sections as before. The second clause starts from a single rational filter R that is the transform of an infinitely divisible law, and asks for a scale space in which R is the stage between scales at a fixed ratio q . Such a ladder of scales at ratio q , with covariance under dilation by q alone, is a *pyramid* in the sense of multiresolution practice ([11]). The clause says that R and q determine one pyramid. They also determine one candidate for a fully covariant family that interpolates the pyramid, and the candidate is admissible exactly when its profile is nonincreasing. The Matérn pyramid is the case $R = (1 + \omega^2)^{-1}$. The family that interpolates it has the exponent $\sum_{m \geq 1} \log(1 + q^{-2m} \omega^2)$, a mixture of Matérn members over the ranges q^{-m} , and it is a different member from the Matérn family: its stages are rational only between scales at ratio q^n , whereas those of the Matérn family are rational between every pair of scales. Figure 8 shows the ladder, the family between its levels, and the profile of that family.

Proposition 7.5 (rational increments and the pyramid they interpolate). 1. If

$$F(\omega) = \sum_{i=1}^N w_i \log\left(1 + \frac{\omega^2}{\theta_i^2}\right)$$

with positive integer weights w_i and no Gaussian part, then every increment is rational,

$$\hat{\mu}_{s,t} = \prod_i \left(\frac{1 + s^2 \omega^2 / \theta_i^2}{1 + t^2 \omega^2 / \theta_i^2} \right)^{w_i},$$

of order $2 \sum_i w_i$, realized by $\sum_i w_i$ forward and $\sum_i w_i$ backward lag sections, each pole-zero pair at the common ratio s/t .

2. Let R be a symmetric rational function that is the transform of an infinitely divisible law, with folded Lévy profile k_R . Then for each $q > 1$ there is exactly one function F that is continuous with $F(0) = 0$ and satisfies $F(q\omega) - F(\omega) = -\log R(\omega)$, namely

$$e^{-F(\omega)} = \prod_{m \geq 1} R(q^{-m}\omega), \quad \text{with folded profile } k(x) = \sum_{m \geq 1} k_R(q^m x);$$

it is admissible exactly when this k is nonincreasing, in particular whenever k_R is. The Matérn pyramid is the case $R(\omega) = (1 + \omega^2)^{-1}$.

The index in the profile sum runs from $m = 1$. Write $F = \sum_{m \geq 1} \psi(q^{-m} \cdot)$ with $\psi = -\log R$. The dilate $\psi(c \cdot)$ has profile $k_R(x/c)$, so the term with $c = q^{-m}$ contributes $k_R(q^m x)$. An $m = 0$ term would make the q -step increment $\psi(q \cdot)$ in place of ψ .

Proof. (1) The increment transform telescopes by Proposition 7.1, and each factor $((1 + s^2 \omega^2 / \theta^2) / (1 + t^2 \omega^2 / \theta^2))$ is realized as in Proposition 7.2 after the gauge change $t \mapsto t/\theta$; admissibility is Proposition 5.7 with $U = \sum_i 2w_i \delta_{\theta_i}$.

(2) *Uniqueness.* If F and $\tilde{F}$ both solve the equation and are continuous with value 0 at the origin, then $D := F - \tilde{F}$ satisfies $D(q\omega) = D(\omega)$, so $D(\omega) = D(q^{-m}\omega) \rightarrow D(0) = 0$ as $m \rightarrow \infty$; hence $D \equiv 0$. *Existence.* Write $\psi := -\log R$, which is a symmetric Lévy exponent. Near the origin $\psi(\omega) = O(\omega^2)$: R is rational and even with $R(0) = 1$, so it is an analytic function of ω^2 near the origin with value 1 there, and so is its logarithm, with value 0. Then $\sum_{m \geq 1} \psi(q^{-m}\omega)$ converges, the terms being $O(q^{-2m}\omega^2)$, and its value F satisfies $F(q\omega) - F(\omega) = \psi(\omega)$ by telescoping. *The pair of ψ .* It has no Gaussian coefficient: R is rational, so ψ grows like a multiple of $\log |\omega|$, while a positive Gaussian coefficient forces quadratic growth. Its Lévy measure is $k_R(x) dx/x$ with k_R the folded profile named in the hypothesis, which is what the substitution below reads; the hypothesis asks for that profile, and for a rational R with real poles it is a finite combination of exponentials. *The profile.* Substituting $u = cx$ in the representation shows that $\psi(c \cdot)$ has folded profile $k_R(u/c)$; taking $c = q^{-m}$ and summing over $m \geq 1$ gives $k(x) = \sum_{m \geq 1} k_R(q^m x)$, the exchange of the sum with the integral being Tonelli, the summands being nonnegative. That F is of the form (2.3) at the pair $(0, k)$ — that is, that k satisfies the two integrability conditions of Lemma 2.3 — follows from the finiteness of $F(1)$, which the convergence of the series gives. *Admissibility* is Theorem 2.5 through clause (3) of Lemma 7.1 of [17], the representation (2.3) — and not that lemma's differentiability clause, which is a different statement: F is of that form with k nonincreasing exactly when the family is admissible, and k inherits monotonicity from k_R when k_R is nonincreasing, each summand being so. In the other direction, that admissibility forces k to be nonincreasing is the uniqueness of the Lévy pair (Proposition 2.3 of [17](3)), and it fixes k only up to a null set, so the monotonicity is monotonicity of a representative. $\square$

The time-causal counterpart. The construction of clause (2) has a counterpart in the time-causal literature. Lindeberg's time-causal limit kernel is the infinite cascade of first-order recursive temporal filters, truncated exponential kernels, whose time constants form a geometric progression, with the transform

$$\hat{\Psi}(\omega; \tau, c) = \prod_{k \geq 1} \frac{1}{1 + i c^{-k} \sqrt{c^2 - 1} \sqrt{\tau} \omega}, \quad c > 1$$

([37], §5, (38), p. 62; [38]). It is built so that the temporal scale space is covariant under scaling by the powers of the ratio c , and the cascade of first-order temporal smoothing steps that it is the limit of goes back to [39]. Clause (2) read at the one-pole transfer function $R(\omega) = (1 + \omega^2)^{-1}$ and the ratio $q = c$ gives $e^{-F(\omega)} = \prod_{m \geq 1} (1 + q^{-2m} \omega^2)^{-1}$, which is the squared modulus of that transform after the dilation of ω by $\sqrt{(c^2 - 1)\tau}$: the two-sided kernel is the limit kernel run forward and backward. The clause differs from that construction in three respects. It is symmetric in space and not causal in time. It allows any rational R that is the transform of an infinitely divisible law, and says when the result is admissible. And it proves that the pyramid determines the family, which the axioms of this paper then interpolate between the levels.

Remark 7.6 (the converse fails: a rational family with a complex pole pair). The converse of Proposition 7.5(1) is false: an admissible family all of whose increments are rational, of bounded order, need not have an exponent of the form stated there, and need not lie in the Thorin subclass. Take $a = 0$ and

$$k(x) = 2e^{-x} (3 + 2 \cos x), \quad x > 0.$$

It is positive, and $k'(x) = -2e^{-x} (3 + 2 \cos x + 2 \sin x) < 0$ because $2 \cos x + 2 \sin x \geq -2\sqrt{2} > -3$, so k is nonincreasing; both integrability conditions are immediate. The identity of Proposition 5.2(1) holds for a complex rate of positive real part, by the same differentiation under the integral sign, and the two rates $1 \pm i$ of the oscillating term combine to a real exponent:

$$F(\omega) = 3 \log(1 + \omega^2) + \log\left(1 + \frac{\omega^4}{4}\right), \quad e^{-F(\omega)} = \frac{1}{(1 + \omega^2)^3 (1 + \omega^4/4)}.$$

So every increment is rational, of order at most ten. But $k^{(4)}(x) = 2e^{-x} (3 - 8 \cos x)$ is negative near the origin, so k is not completely monotone and the family is not a Thorin member (Proposition 5.7). Monotonicity of the profile therefore does not exclude oscillating terms, a real pole of sufficient weight being enough to carry a complex pair. Such a family has rational stages and no realization by the lag sections of Proposition 7.2, each of which is positive by itself: the factor of the complex pair, $1/(1 + \omega^4/4)$, is the transform of a kernel proportional to $e^{-|x|}(\cos x + \sin |x|)$, which changes sign, so the positivity of the member comes from the product only.

Complex poles are not what fails, and a second example shows it. Take $a = 0$ and

$$k(x) = 4e^{-x} - 2e^{-2x}, \quad x > 0.$$

It is positive, since $e^{-x} \leq 1 < 2$; it is nonincreasing, since $k'(x) = -4e^{-x}(1 - e^{-x}) \leq 0$; and both integrability conditions are immediate, k being bounded and exponentially decaying. The identity of Proposition 5.2(1) at the two rates gives

$$F(\omega) = 2 \log(1 + \omega^2) - \log\left(1 + \frac{\omega^2}{4}\right), \quad e^{-F(\omega)} = \frac{1 + \omega^2/4}{(1 + \omega^2)^2},$$

so every increment is rational, of order at most six, and every pole and every zero of the kernel's transform is on the imaginary ω -axis: the transfer has no oscillating factor at all. Yet

$k''(x) = 4e^{-x} - 8e^{-2x}$ is negative at the origin, so k is not completely monotone and the family is not a Thorin member, hence not in the class of Proposition 7.8. Nor is it realized by lag sections: the stage from s to t has pole ranges $t, t, s/2$ and zero ranges $t/2, s, s$, and the pole at range $s/2$ has no zero of smaller range to pair with, so any factorization into first-order sections has a section with a negative coefficient. So “real poles” implies neither membership in the Thorin subclass nor a realization by lag sections, and the real-pole case of Remark 7.4 is the case of real poles *and no zeros*.

7.4 Finite Thorin approximations and their realization

The Student-t and stable families have no rational stages, and neither has a Matérn member of non-integer index, as their transforms show (Propositions 5.10(2), 5.12 and 5.2(2)). They are all members of the Thorin subclass of §5.3 — the Student-t family under the hypothesis of Remark 5.11 — however, and a member of that subclass is a mixture of Matérn members over its Thorin measure U . Replacing U by a finite sum of atoms gives a family with finitely many ranges, and rounding the weights to even integers gives one with rational stages. The proposition says that both steps stay inside the admissible class, and that the kernels of the quadrature converge weakly to those of the given member at every scale. So the approximation is made once, on the measure, and what is then computed is the exact scale space of another member of the class, with positive kernels and an exact cascade. The convergence is that of the quadratures with real weights. The rounding is a second step, which the proposition shows to stay inside the class. It does not converge to the given member in general, by Proposition 7.8 below, and the run of §7.6 measures what it costs. The proposition also reads the structure on the generator side, where the mixture becomes a parallel sum of bounded blocks.

Proposition 7.7 (finite Thorin approximations and their realization). *Let F be admissible with completely monotone profile and Thorin measure U as in (5.1). Every measure $U_N = \sum_{i \leq N} w_i \delta_{\theta_i}$ with positive weights gives, with the same Gaussian coefficient, an admissible F_N . A sequence of such measures is a quadrature of U when $F_N(\omega) \rightarrow F(\omega)$ for every ω , and the kernels of a quadrature converge weakly to those of F at every scale. Every such U has a quadrature, and it has one in the variant that carries the mass above the last node as a Gaussian coefficient: there are cutoffs $\Theta_N \uparrow \infty$ and finite measures $U_N = \sum_i w_i \delta_{\theta_i}$ with positive weights, carried by $(0, \Theta_N]$, such that both the exponents F_N at Gaussian coefficient a and the exponents $\tilde{F}_N$ at Gaussian coefficient*

$$a_N := a + \frac{1}{2} \int_{(\Theta_N, \infty)} \theta^{-2} U(d\theta) < \infty$$

converge to $F(\omega)$ at every ω ; $\tilde{F}_N$ is admissible as well. Rounding the weights to even integers $w_i = 2n_i$ keeps F_N admissible and makes the increments of its jump part rational, realized by $\sum_i n_i$ forward-backward lag sections per knot, in series; a Gaussian part is a separate factor $e^{-a(t^2-s^2)\omega^2}$ of every increment, and is not rational. The Thorin sum lives in the exponent, so the stages compose as a product of factors and the generators add: the generator of Proposition 6.2 is the sum

$$\mathcal{A}_t = 2at \partial_x^2 + \frac{1}{t} \sum_i w_i (\text{Lap}_{t/\theta_i} - I)$$

of a Laplacian and finitely many bounded operators, one per atom.

The weights are those of the quadrature U_N , so with the factor $\frac{1}{2}$ of (5.1) the exponent weights are $w_i/2$, which is why the rounding is to even integers. The constants in the parallel generator are those of Proposition 6.4(4), the Gaussian term carrying the factor t and the sum the factor $1/t$. The existence clause is what makes the rest of the proposition about something: a quadrature is defined by its convergence, and without it the definition could be empty. Its proof

places the nodes by hand and gives no rate, and no optimality of the placement is claimed. The variant with the Gaussian coefficient is the one the run of §7.6 computes, the Thorin measures there having unbounded support.

Proof. Admissibility of F_N is closure of the cone under positive sums (Lemma 2.7, Proposition 3.3), each summand $\frac{1}{2}w_i \log(1 + \omega^2/\theta_i^2)$ being admissible with profile $w_i e^{-\theta_i x}$; the same closure, with the Gaussian ray, gives admissibility of $\tilde{F}_N$. Weak convergence of a quadrature is the convergence of the exponents that defines it, together with Proposition 2.8 of [17]. Even weights $w_i = 2n_i$ make the exponent $\sum_i n_i \log(1 + \omega^2/\theta_i^2)$, so every increment is rational and has the realization by Proposition 7.5(1). The generator is Proposition 6.4(4) with $U = \sum_i w_i \delta_{\theta_i}$, plus the Gaussian term of Proposition 6.4(1).

A quadrature exists. Three facts about (5.1)'s condition are used, and all three read the comparison $\log(1 + \lambda x) \leq \lambda \log(1 + x)$, valid for $\lambda \geq 1$ and $x \geq 0$ because the two sides agree at $x = 0$ and the derivative of the left is $\lambda/(1 + \lambda x) \leq \lambda/(1 + x)$: that U is finite on $(0, \Theta]$ for every Θ , since $\log(1 + \theta^{-2}) \geq \log(1 + \Theta^{-2}) > 0$ there; that $\int_{(\Theta, \infty)} \theta^{-2} U(d\theta) < \infty$, since $\log(1 + \theta^{-2}) \geq \frac{1}{2}\theta^{-2}$ for $\theta \geq 1$; and that $\theta \mapsto \log(1 + \omega^2/\theta^2)$ is U -integrable at each fixed ω , being at most $\max(1, \omega^2) \log(1 + \theta^{-2})$.

Take $\Theta_N := N$. Partition $[1/N, N]$ into finitely many cells of width δ_N , and let U_N put the mass $U(\text{cell})$ at a point of each nonempty cell; the weights are positive and finitely many. On $[1/N, N]$,

$$\left| \frac{\partial}{\partial \theta} \log\left(1 + \frac{\omega^2}{\theta^2}\right) \right| = \frac{2\omega^2}{\theta(\theta^2 + \omega^2)} \leq \frac{2}{\theta} \leq 2N,$$

a bound free of ω , so moving each cell's mass to a point of that cell changes the Thorin integral by at most $2N\delta_N U([1/N, N])$, and δ_N may be chosen to make that at most $1/N$. What is left is the mass of U off $[1/N, N]$, whose contribution $\frac{1}{2} \int_{(0, 1/N) \cup (N, \infty)} \log(1 + \omega^2/\theta^2) U(d\theta)$ tends to 0 at each fixed ω by dominated convergence, the integrand being dominated by the U -integrable $\max(1, \omega^2) \log(1 + \theta^{-2})$ and the domain shrinking to the empty set. So $F_N(\omega) \rightarrow F(\omega)$ for every ω . For the variant, the difference $\tilde{F}_N - F_N$ is $(\frac{1}{2} \int_{(N, \infty)} \theta^{-2} U(d\theta))\omega^2$, and the integral is the tail of a finite one, so it tends to 0. $\square$

The Thorin measures of the three named families are those of their causal ancestors under the bijection $\theta \mapsto \sqrt{2}\theta$ of Proposition 5.7(3): one atom for the Matérn family, a power law for the symmetric stable family (Proposition 5.12), and the density of Remark 5.11 for the Student-t family. For a power law, logarithmically spaced nodes are a natural choice. Any quadrature with positive weights gives a member of the class, by the proposition.

The proposition leaves one question open. The quadratures that converge have real weights, and only even integer weights give a realization by lag sections. Which members can the families realized by lag sections reach? The next proposition answers it: exactly the members whose Thorin measure is made of atoms of even integer mass, together with a Gaussian part. So rounding the weights changes the member, and for a member with a spread-out Thorin measure no refinement of the nodes undoes the change. The reason is on the half-line. Under the bridge a family realized by lag sections is a finite convolution of gamma laws with integer shapes. A limit of such convolutions is a generalized gamma convolution whose Thorin measure is the limit of theirs, and a limit of measures with integer atoms has integer atoms. Mass that escapes to infinity becomes a drift on the half-line, which is a Gaussian part on the line.

The cited facts. Bondesson's closure theorem says that a weak limit of generalized gamma convolutions is one, and that the Thorin measures then converge vaguely on $(0, \infty)$ ([8], Theorem 3.1.5, pp. 34–35, with the definition of the class at (3.1.1)–(3.1.2), p. 29). The continuity theorem for Laplace transforms says that probability laws on the half-line whose transforms converge at every point, to a limit that tends to 1 at the origin, converge weakly to a probability law with that transform ([19], Vol. 2, §XIII.1, Theorem 2, p. 431). Clause (5) uses one

further cited fact, the moment criterion of §5.1 read at the weight $e^{\beta|x|}$, which is submultiplicative ([50], Theorem 25.3, p. 159). The proposition is proved in prose from those and is not machine-checked.

Proposition 7.8 (the closure of the families realized by lag sections). *Let $\mathcal{R}$ be the set of exponents $F(\omega) = \sum_{i \leq N} n_i \log(1 + \omega^2/\theta_i^2)$ with $N \geq 0$, positive integers n_i and $\theta_i > 0$. These are exactly the families realized by lag sections in the sense of Proposition 7.2, that is by cascades of first-order sections whose pole and zero sit at the two scales of the stage: each member of $\mathcal{R}$ is admissible with no Gaussian part and has every increment a cascade of such sections, by Proposition 7.5(1); and conversely, an admissible family one of whose kernels from the origin is such a cascade has its exponent in $\mathcal{R}$.*

1. (Limits.) *Let $F_n \in \mathcal{R}$, let F be admissible, and suppose that the kernels of F_n at canonical scale 1 converge weakly to that of F . Then F has a symmetric Thorin representation (5.1) whose Thorin measure is a sum of atoms of even integer mass, $U = \sum_i 2n_i \delta_{\theta_i}$ with positive integers n_i and with finitely many θ_i in every compact subset of $(0, \infty)$. Its Gaussian coefficient may be positive.*
2. (Every such member is a limit.) *Conversely, every admissible F with a symmetric Thorin representation of that form, with any Gaussian coefficient $a \geq 0$, is the limit in that sense of a sequence in $\mathcal{R}$.*
3. (What is not a limit.) *Consequently an admissible F is not such a limit if its profile is not completely monotone, or if its Thorin measure has a part without atoms or an atom whose mass is not an even integer. In particular no symmetric stable member with $0 < \alpha < 2$ is such a limit, and no Matérn member of non-integer index.*
4. (The class of limits is closed.) *If every F_n is such a limit, F is admissible, and the kernels of F_n at canonical scale 1 converge weakly to that of F , then F is such a limit as well.*
5. (Every limit has light tails.) *If F is such a limit then its displacement X_t has finite moments of every order, and an exponential moment: for every $t > 0$ there is $\beta > 0$ with $\mathbb{E}e^{\beta|X_t|} < \infty$. Consequently an admissible family with a divergent absolute moment of some order is not such a limit. No symmetric stable member with $0 < \alpha < 2$ is one, and no Student- t member is one either — the second exclusion with no hypothesis on that family's Thorin measure.*

The proposition gives no rate and no distance, and the numerical run of §7.6 measures the distance for two members. Clause (5) is what settles the Student- t family: the exclusion is read off its moments, which Proposition 5.10(4) gives from the explicit density, and not off its Thorin measure, which is known only through Remark 5.11. So neither of the two families with heavy tails is reachable, and neither exclusion rests on a hypothesis. The Gaussian, at the other end, is a limit, as clause (2) says with $U = 0$, and this is the convergence of the Matérn members along γ at matched variance. Clause (5) has no converse: light tails do not make a member a limit, the Matérn member of non-integer index having every moment and an exponential moment and being excluded by clause (3).

Proof. The identification of $\mathcal{R}$. One direction is Proposition 7.5(1) at $U = \sum_i 2n_i \delta_{\theta_i}$. For the other, let the kernel of an admissible family at some scale $t > 0$ be a cascade of lag sections in that sense. A section of the stage from s to t has transfer $(1 + s^2\omega^2/\theta^2)/(1 + t^2\omega^2/\theta^2)$, so at $s = 0$ its zero is gone and the cascade's transfer is $\prod_j (1 + t^2\omega^2/\theta_j^2)^{-n_j}$ with positive integer multiplicities. Taking logarithms, the exponent at that scale is $\sum_j n_j \log(1 + t^2\omega^2/\theta_j^2)$, a member of $\mathcal{R}$ read at the ranges θ_j/t ; and $\mathcal{R}$ is invariant under $\omega \mapsto \lambda\omega$, so the family's exponent lies in it whatever the gauge.

(1) *The transforms.* Weak convergence of the kernels gives convergence of their cosine transforms, $e^{-F_n(\omega)} \rightarrow e^{-F(\omega)}$ for every ω , and the limit is positive, so $F_n(\omega) \rightarrow F(\omega)$.

The causal side. Write $F_n = \sum_i n_i^{(n)} \log(1 + \omega^2/\theta_{i,n}^2)$ and put $G_n(\sigma) := F_n(\sqrt{2\sigma})$ for $\sigma \geq 0$. Then $G_n(\sigma) = \int_{(0,\infty)} \log(1 + \sigma/t) V_n(dt)$ with $V_n := \sum_i n_i^{(n)} \delta_{\theta_{i,n}^2/2}$, and e^{-G_n} is the Laplace transform of the law T_n of a finite sum of independent gamma variables, of shapes $n_i^{(n)}$ and rates $\theta_{i,n}^2/2$, since a gamma variable of shape n and rate t has the Laplace transform $(1 + \sigma/t)^{-n}$. In the parametrization of the cited definition T_n is the generalized gamma convolution with left extremity 0 and measure V_n , which is finite with compact support in $(0, \infty)$ and so satisfies the integrability conditions asked there. Put $G(\sigma) := F(\sqrt{2\sigma})$. Then $G_n \rightarrow G$ pointwise on $[0, \infty)$, and G is continuous with $G(0) = 0$, so $e^{-G(\sigma)} \rightarrow 1$ as $\sigma \downarrow 0$. By the continuity theorem for Laplace transforms the laws T_n converge weakly to a probability law T on $[0, \infty)$ with Laplace transform e^{-G} . By the closure theorem T is a generalized gamma convolution, say with left extremity $b \geq 0$ and measure V , so that $G(\sigma) = b\sigma + \int_{(0,\infty)} \log(1 + \sigma/t) V(dt)$, and $V_n \rightarrow V$ vaguely on $(0, \infty)$.

A vague limit of integer-valued point measures is one. Each V_n is a finite sum of atoms of positive integer mass. Let I be an open interval with compact closure in $(0, \infty)$ and $V(\partial I) = 0$. Squeezing the indicator of I between continuous functions of compact support whose integrals against V differ by as little as desired shows $V_n(I) \rightarrow V(I)$, so $V(I)$ is a limit of nonnegative integers and is one. The atoms of V are countably many, so for every $x \in (0, \infty)$ the intervals $(x - \varepsilon, x + \varepsilon)$ with $V(\{x \pm \varepsilon\}) = 0$ have ε arbitrarily small, and along them the integers $V((x - \varepsilon, x + \varepsilon))$ decrease to $V(\{x\})$ and are therefore eventually equal to it. So every x has a neighbourhood in which V is an atom of integer mass at x , or zero, and nothing else: $V = \sum_i n_i \delta_{t_i}$ with positive integers n_i and with finitely many t_i in every compact subset of $(0, \infty)$.

Back to the line. Reading G at $\sigma = \omega^2/2$,

$$F(\omega) = \frac{1}{2}b\omega^2 + \int_{(0,\infty)} \log\left(1 + \frac{\omega^2}{2t}\right) V(dt) = a\omega^2 + \frac{1}{2} \int_{(0,\infty)} \log\left(1 + \frac{\omega^2}{\theta^2}\right) U(d\theta),$$

with $a = \frac{1}{2}b$ and U twice the image of V under $t \mapsto \sqrt{2t}$, which is the correspondence of Proposition 5.7(3). The integrability condition of (5.1) is the finiteness of the Thorin integral at $\omega = 1$, that is $2(G(\frac{1}{2}) - \frac{1}{2}b)$, and G is finite on $[0, \infty)$. So F has the representation (2) of Proposition 5.7, and $U = \sum_i 2n_i \delta_{\sqrt{2t_i}}$.

(2) Let $F(\omega) = a\omega^2 + \sum_i n_i \log(1 + \omega^2/\theta_i^2)$ be of that form, the sum countable and convergent for every ω by the integrability conditions of (5.1). Put $F_N(\omega) := N \log(1 + a\omega^2/N) + \sum_{i \leq N} n_i \log(1 + \omega^2/\theta_i^2)$, omitting the first term when $a = 0$. Then $F_N \in \mathcal{R}$, its first term being the Matérn exponent of index N and range $\sqrt{a/N}$. Since $N \log(1 + x/N) \rightarrow x$ for $x \geq 0$, $F_N(\omega) \rightarrow F(\omega)$ for every ω , so the transforms of the kernels converge pointwise and the kernels converge weakly by Proposition 2.8 of [17].

(3) If F is such a limit, clause (1) gives it a representation (5.1) with a Thorin measure made of atoms of even integer mass. By Proposition 5.7 its profile is then completely monotone, and the Thorin measure of an admissible F is unique, by the uniqueness of the data (a, k) with Proposition 2.9. So an F whose profile is not completely monotone, or whose Thorin measure is of another kind, is not such a limit. The symmetric stable member with $0 < \alpha < 2$ has a Thorin measure with a density (Proposition 5.12), and the Matérn member of index γ has the Thorin measure $2\gamma\delta_1$, the exponent $\gamma \log(1 + \omega^2)$ being (5.1) at that measure (Proposition 5.2(1)), whose atom is an even integer only for integer γ .

(4) By clauses (1) and (2) the limits are exactly the admissible F with a symmetric Thorin representation $U = \sum_i 2n_i \delta_{\theta_i}$ of the stated kind, at any Gaussian coefficient, so what has to be shown is that this class is closed under the convergence in question. The argument of clause (1) shows it, read at a sequence from the class rather than from $\mathcal{R}$: what it uses of the approximating

sequence is that $G_n(\sigma) := F_n(\sqrt{2\sigma})$ is $b_n\sigma + \int_{(0,\infty)} \log(1 + \sigma/t) V_n(dt)$ with $b_n \geq 0$ and V_n a measure whose atoms have positive integer mass, satisfying the integrability conditions of the cited definition, so that e^{-G_n} is the Laplace transform of a generalized gamma convolution. A member of the class supplies that too: $b_n = 2a_n$ and V_n is half the image of U_n under $\theta \mapsto \theta^2/2$, whose atoms have mass n_i , and the two integrability conditions follow from the finiteness of $G_n(1)$ by the three regimes in the proof of Proposition 5.7, (1) implies (2). Everything after that reads nothing else.

(5) Let F be such a limit, with Gaussian coefficient $a \geq 0$ and Thorin measure $U = \sum_i 2n_i \delta_{\theta_i}$ as in clause (1), and let k be its profile. If $U = 0$ the kernels are Gaussian and every moment and every exponential moment is finite. Otherwise the atoms have a smallest range: (5.1)'s condition gives $\log 2 \cdot U((0, 1]) \leq \int_{(0,1]} \log(1 + \theta^{-2}) U(d\theta) < \infty$, so $U((0, 1])$ is finite while every atom has mass at least 2, whence there are finitely many atoms in $(0, 1]$, and finitely many in every compact subset of $(0, \infty)$ by clause (1); so $\theta_{\min} := \min_i \theta_i$ exists and is positive. By Proposition 5.7, (1) if and only if (2), the profile is $k(x) = \sum_i 2n_i e^{-\theta_i x}$, finite at $x = 1$; hence for $x \geq 1$

$$k(x) = \sum_i 2n_i e^{-\theta_i} e^{-\theta_i(x-1)} \leq e^{-\theta_{\min}(x-1)} \sum_i 2n_i e^{-\theta_i} = k(1) e^{\theta_{\min}} e^{-\theta_{\min}x}.$$

Every moment is then finite by Proposition 5.1(1), since $\int_1^\infty x^{n-1} k(x) dx$ is dominated by a multiple of $\int_1^\infty x^{n-1} e^{-\theta_{\min}x} dx$. For the exponential moment, fix $t > 0$ and $\beta < \theta_{\min}/t$. The Lévy measure of X_t is the t -dilate of the folded $k(x)x^{-1}dx$, so, as in the proof of Proposition 5.1, $\int_{|x|>1} e^{\beta|x|} \nu_2(dx) = \int_{1/t}^\infty e^{\beta tx} k(x) x^{-1} dx$, which the bound above makes finite: near infinity the integrand is dominated by a multiple of $e^{(\beta t - \theta_{\min})x}$, and on the compact stretch $[1/t, 1]$ it is bounded, k being finite and nonincreasing there. The weight $x \mapsto e^{\beta|x|}$ is submultiplicative, so the moment criterion of [50], Theorem 25.3, p. 159, applies to it and gives $\mathbb{E}e^{\beta|X_t|} < \infty$.

The consequence is the contrapositive. A symmetric stable member with $0 < \alpha < 2$ has $\mathbb{E}|X_t|^n = \infty$ for $n \geq \alpha$ (Proposition 5.12), and the Student-t member of shape a has $\mathbb{E}|X_t|^n = \infty$ for $n \geq 2a$ (Proposition 5.10(4)); so neither is a limit of families realized by lag sections. $\square$

Remark 7.9 (the limit class is the symmetric Pólya frequency functions). The class of limits identified above is classical under another name. A probability density f on the line is a *Pólya frequency function of order ∞* when for every n and all $x_1 < \dots < x_n, y_1 < \dots < y_n$ the matrix $(f(x_i - y_j))_{i,j}$ has nonnegative determinant. Schoenberg's representation theorem, as printed in [31], Theorem 3.2(a), p. 345, with the class $\mathcal{E}_2^*$ of p. 336, says that f is one exactly when the reciprocal of its two-sided Laplace transform φ is the entire function

$$\frac{1}{\varphi(s)} = e^{-\gamma s^2 + \delta s} \prod_i (1 + a_i s) e^{-a_i s}, \quad \gamma \geq 0, \quad \delta, a_i \text{ real}, \quad \sum_i a_i^2 < \infty, \quad (7.6)$$

with the nondegeneracy $\gamma + \sum_i a_i^2 > 0$; Karlin attributes the theorem to Schoenberg (1951) in the notes to his chapter, p. 391. The primary source is [53], Theorem I, pp. 333–334, read at the page: it states the representation for the class of his (3), p. 331, without the normalization to a density that Karlin's statement makes explicit, and the same paper proves the variation-diminishing property of these functions.

Read at a member of the class, the two agree. Let $F(\omega) = a\omega^2 + \sum_i n_i \log(1 + \omega^2/\theta_i^2)$ be as in Proposition 7.8(1). By clause (5) its kernel from the origin at canonical scale 1 has an exponential moment, so its two-sided Laplace transform converges on a strip about the imaginary axis and is there the analytic continuation of the cosine transform: $\varphi(s) = e^{-F(-is)} = e^{as^2} \prod_i (1 - s^2/\theta_i^2)^{-n_i}$, whence

$$\frac{1}{\varphi(s)} = e^{-as^2} \prod_i \left[\left(1 + \frac{s}{\theta_i}\right) \left(1 - \frac{s}{\theta_i}\right) \right]^{n_i}.$$

That is (7.6) with $\gamma = a$, $\delta = 0$ and the numbers $\pm 1/\theta_i$ for the a_i , each with multiplicity n_i : the factors $e^{-a_i s}$ cancel in pairs, and $\sum_i a_i^2 = \sum_i 2n_i \theta_i^{-2}$, which (5.1)'s condition makes finite and which is what makes the product entire. Conversely, if f is a symmetric probability density whose reciprocal transform is (7.6), then evenness makes the zero set of $1/\varphi$ symmetric, so $\delta = 0$ and the a_i pair as $\pm 1/\theta_j$ with multiplicities m_j ; the paired factors collapse to $\prod_j (1 - s^2/\theta_j^2)^{m_j}$, and reading at $s = i\omega$ gives the cosine transform $e^{-\gamma\omega^2} \prod_j (1 + \omega^2/\theta_j^2)^{-m_j}$, an exponent of the form (5.1) at $U = \sum_j 2m_j \delta_{\theta_j}$ — admissible by Proposition 5.7, with finitely many θ_j below any bound because $\sum_j m_j \theta_j^{-2} < \infty$. So the kernels from the origin of the members of the class are exactly the symmetric Pólya frequency functions of order ∞ , the nondegeneracy condition corresponding to $F \not\equiv 0$, which (ND) supplies. Bondesson states the half-line form of the same identification, for Thorin's class on $[0, \infty)$, two pages after the closure theorem used above ([8], p. 36): “Obviously the $[PF_\infty]$ -class equals the subclass of [Thorin's class] for which the U -measure is discrete with atoms with integral mass. The closure theorem for GGC's guarantees in particular that also the PF_∞ -class is closed with respect to weak limits; note that atoms with mass 1 cannot be split in the limit.”

What this does not say. The identification is of the kernels *from the origin*. A stage between two positive scales is not of Schoenberg's form at all: the reciprocal of its transform is $(1 + t^2\omega^2)/(1 + s^2\omega^2)$ read at imaginary argument, which has poles, while (7.6) is an entire function. Nothing is therefore claimed here about the families — neither that their stages are Pólya frequency functions nor that they diminish variation. And the question the identification suggests, whether the scale space of such a member creates no new local extremum, is not treated in this paper; it is the subject of a further module.

7.5 What cannot be stepped in scale, and what it costs

The first remark is a limit of the cascade picture: one corner has a local equation that cannot be used as an algorithm, because it cannot be solved by stepping in scale from the signal. The second collects the costs.

Remark 7.10 (the local equation of the Student-t corner cannot be marched). To *march* an equation in scale is to step its solution forward from the signal by a local update, as an explicit scheme does with the heat equation. On the line the scale-locally generated corner is the heat equation, marched routinely. The jointly local corner, the Student-t family, is of a different kind: its scale space solves an elliptic boundary-value problem in the half-plane of scale and position rather than a parabolic evolution in scale. That equation is the subject of the next module, on locality and the selection of the Gaussian, and no statement of this paper asserts it.

What this section does establish about that corner is the following. The family can be stepped in scale, as every admissible family can (Proposition 7.1); its stages are nonlocal operators, those of the Poisson semigroup for the member $a = \frac{1}{2}$. What the family lacks is a realization by lag sections. The Thorin quadrature of Proposition 7.7 with the Bessel spectral density has real weights and therefore no such realization, and by Proposition 7.8(5) no family realized by lag sections converges to it either, its moments being finite only below the order $2a$ while every such limit has all of them. So the stages of this corner are computed from its kernels and not from a recursion in x .

What is announced and not proved here is that the Cauchy problem of the local equation in t is ill-posed, so that the equation cannot be marched in scale from the signal. At the member $a = \frac{1}{2}$ the equation is the Laplace equation, whose solutions at frequency ω combine $e^{-t|\omega|}$ with $e^{+t|\omega|}$ — a perturbation of the data at high frequency amplified without bound, Hadamard's classical instability — and the Poisson stages select the decaying solution; for the other members it is a description and not a claim of this paper.

Remark 7.11 (costs). Per sample and per knot: the Matérn family costs 2γ first-order states and 4γ multiply-adds — two per section, one for the state and one for the output, and 2γ

sections, γ forward and γ backward — in a forward and a backward pass at that knot; a Thorin approximant $U_N = \sum_i 2n_i \delta_{\theta_i}$ costs $2 \sum_i n_i$ states and $4 \sum_i n_i$ multiply-adds; the Gaussian by real-pole recursion costs 2γ states at an accuracy in the exponent that is $O(1/\gamma)$ at each fixed frequency (the expansion of $\gamma \log(1 + \sigma^2 \omega^2 / 2\gamma)$ against $\sigma^2 \omega^2 / 2$, the two transfers of Remark 7.4), and by a direct finite-impulse-response filter of length L costs L ; the stable and Student-t families have no such realization and are not limits of families that have one (Proposition 7.8(5), on their moments alone and with no hypothesis on either family's Thorin measure), so their stages are computed from their kernels or their transforms, at the cost of the method chosen. The counts above are totals over the ladder; the stages can be run one after another, so the live state of an implementation is that of one stage. This paper makes no empirical claim about natural images.

7.6 A numerical run

The statements of this section are about signals on the line, and an implementation works on samples. This subsection runs the cascade on one sampled signal and checks numbered results of the paper on it, in the manner of the numerical example of the causal article ([18]). It then reads two published recursive approximations of the Gaussian in the coordinates of this paper. The signal is synthetic and every number quoted is a check of a statement, so the run adds no claim about natural images. Both figures and all the numbers are produced, from a fixed seed, by the scripts `make-fig-experiment-b.py` and `make-fig-recursive-gaussians.py` that accompany the paper. The signal is a sum of four Gaussian bumps, of heights 0.95, 0.70, 0.90 and 0.12 and widths 1.5, 30, 8 and 8 samples, centred at the samples 300, 900, 1500 and 1700, a plateau of height 0.5 between the samples 2500 and 3800 with edges $\tanh(\cdot/3)$, and uniform noise of amplitude ± 0.175 on the samples 3000 to 3299 from NumPy's default generator with the seed 20260918, clipped to $[0, 1]$. The numbers of checks 1 and 2 depend on it, the fine bump most of all, and the others do not. The scripts fix the rest of what a reproduction needs: the ladder with its first stage from scale 0, the periodic solution of each recursion, the transforms used for the Gaussian and the Poisson members, the mesh of the L^1 distances, and the optimizer with its start and its stopping rule. The L^1 distances are computed on the window $|x| \leq 400$ at canonical scale 1, and are distances on that window. The Poisson kernel has the mass $1.6 \cdot 10^{-3}$ outside it, so a printed distance from the Poisson kernel to a member whose tail is negligible there, the members with weights 2 among them, is lower than the distance on the line by that amount.

The sampled section. The signal has 4096 samples at unit spacing and is continued periodically. A lag section of range τ , measured in samples, is realized by the recursion $w_n = p w_{n-1} + (1 - p) y_n$ with the pole

$$p(\tau) = \frac{2\tau^2}{2\tau^2 + 1 + \sqrt{4\tau^2 + 1}}, \quad \text{chosen so that} \quad \frac{p}{(1-p)^2} = \tau^2.$$

Write z^{-1} for the unit delay. The stage between the knots $s < t$ is the section $g(1 - p_s z^{-1}) / (1 - p_t z^{-1})$ with $p_s = p(s)$, $p_t = p(t)$ and $g = (1 - p_t) / (1 - p_s)$, run once forward and once backward. Its zero is the pole of the previous knot. Its impulse response is g at $n = 0$ and $g(p_t - p_s) p_t^{n-1}$ for $n \geq 1$, positive with unit mass, so the sampled section is again a convex combination of the identity and a one-sided average. The forward-backward pair has the transfer function $(1 + s^2 \Omega^2) / (1 + t^2 \Omega^2)$ with $\Omega^2 = 4 \sin^2(\omega/2)$, the symbol of minus the second difference. This is the transfer function of Proposition 7.2 with Ω in place of ω . Three properties of the continuum family therefore hold for the sampled cascade on the infinite lattice without error: it telescopes as in Proposition 7.1, its kernels have the variance $2\gamma t^2$ of Proposition 5.2(3), and it solves the generator equation of Proposition 6.4(3) with the sampled Laplace kernel. On the lattice

the kernels from the origin are convolution products of two-sided geometric kernels, powers of one such kernel for the Matérn members, and these are among the discrete scale-space kernels of [34], whose classification allows the factors $1/(1 - \beta z)$ and $1/(1 - \gamma z^{-1})$ in the generating function. A stage between two positive scales is not among them: it has the factor $1 - p_s z^{-1}$, a zero on the positive real axis, which that classification excludes. So the stages of the sampled cascade preserve positivity and mass, which is what this paper's axioms ask, and not the stronger property of creating no new extrema, which belongs to the kernels from the origin and is the subject of the next module. On the periodic signal of the run the kernels are wrapped around the 4096 samples, which changes their variance by less than 10^{-8} of its value at the scales used. The sampling is a choice made for the run and the paper states nothing about it. It is a different spatial operator from the continuum one, by the difference between Ω and ω , and the first check reports how far the two are apart.

Example 7.12 (the machine run end to end). *The run.* The signal of Figure 9 has one specimen for each thing the members do differently: a fine and a broad bump, a strong bump with a weak one 200 samples away, a plateau with two edges, and a burst of noise on the plateau. Three members are computed by the cascade of lag sections over a geometric ladder of 48 knots: the Matérn members $\gamma = 1$ and $\gamma = 4$, and a member with 24 ranges, the one with nodes of weight 2 at $\theta_i = \pi(i - \frac{1}{2})$, which the last check explains. The Poisson scale space, which is the stable member $\alpha = 1$ and also the Student-t member $a = \frac{1}{2}$ ([20]), and the Gaussian are computed by their transforms, since neither has rational stages. The members are compared at equal bandwidth: at the display scale $s \in [1, 64]$, in samples, a member runs at the canonical scale $t = cs$ with $F(c) = 1$, so that the transfer function of every continuum member equals $1/e$ at $\omega = 1/s$; the sampled members are run at the same t , and at $s = 1$ the sampled Matérn member $\gamma = 1$ has the value 0.388 there against $1/e = 0.368$. The factors are $c = 1.311$ and 0.533 for the two Matérn members, 1.670 for the member with 24 ranges, 1 for the Poisson member and $\sqrt{2}$ for the Gaussian with $F = \omega^2/2$.

In gray levels the scale spaces of the five members cannot be told apart, so the figure shows one of them and the differences of three from the Gaussian. The differences are largest along the axes of the narrow features and at the edges, where the Matérn member $\gamma = 1$ keeps the cusp of its kernel at every scale, and for the Poisson member they extend far from every feature, which is its tail. The largest differences are 0.040 and 0.014 for the Matérn members $\gamma = 1$ and $\gamma = 4$ and 0.111 for the Poisson member, on a signal with values in $[0, 0.95]$.

The checks.

1. *Exactness at the knots* (Propositions 7.1, 7.2 and 7.5(1)). Refining the ladder from 48 to 95 knots moves the records at the old knots by at most 4.8×10^{-14} for the three members, and the last record agrees with the stage from 0 to the last knot computed in one step to 2.9×10^{-14} . Both are rounding error. A forward Euler march of the generator equation of Proposition 6.4(3) and (4) over the same ladder errs by 5.6×10^{-3} , 3.1×10^{-3} and 3.7×10^{-3} for the three members, and by half of that on the refined ladder. That is the first-order error in the scale step which the cascade does not have. The sampled cascade differs from the continuum member by 9.7×10^{-3} at $s = 1$ and by 1.1×10^{-5} at $s = 64$, for $\gamma = 1$. That is the sampling of the signal, which the section does not treat.
2. *Positivity and unit mass in the realization* (Proposition 7.2, (7.2)). The output of every section of the three cascades, 96, 384 and 2304 of them, lies between the least and the largest value of the signal, and the mean of the record changes by less than 10^{-15} over the ladder.
3. *Moments and tails* (Propositions 5.2(3), 5.1 and 5.12). The variance of the impulse response of the cascade equals $2\gamma t^2$ to 1.2×10^{-9} at all 48 knots for the Matérn members, and equals $t^2 \int_0^\infty x k(x) dx = 0.9916 t^2$ to 5×10^{-12} for the member with 24 ranges. The

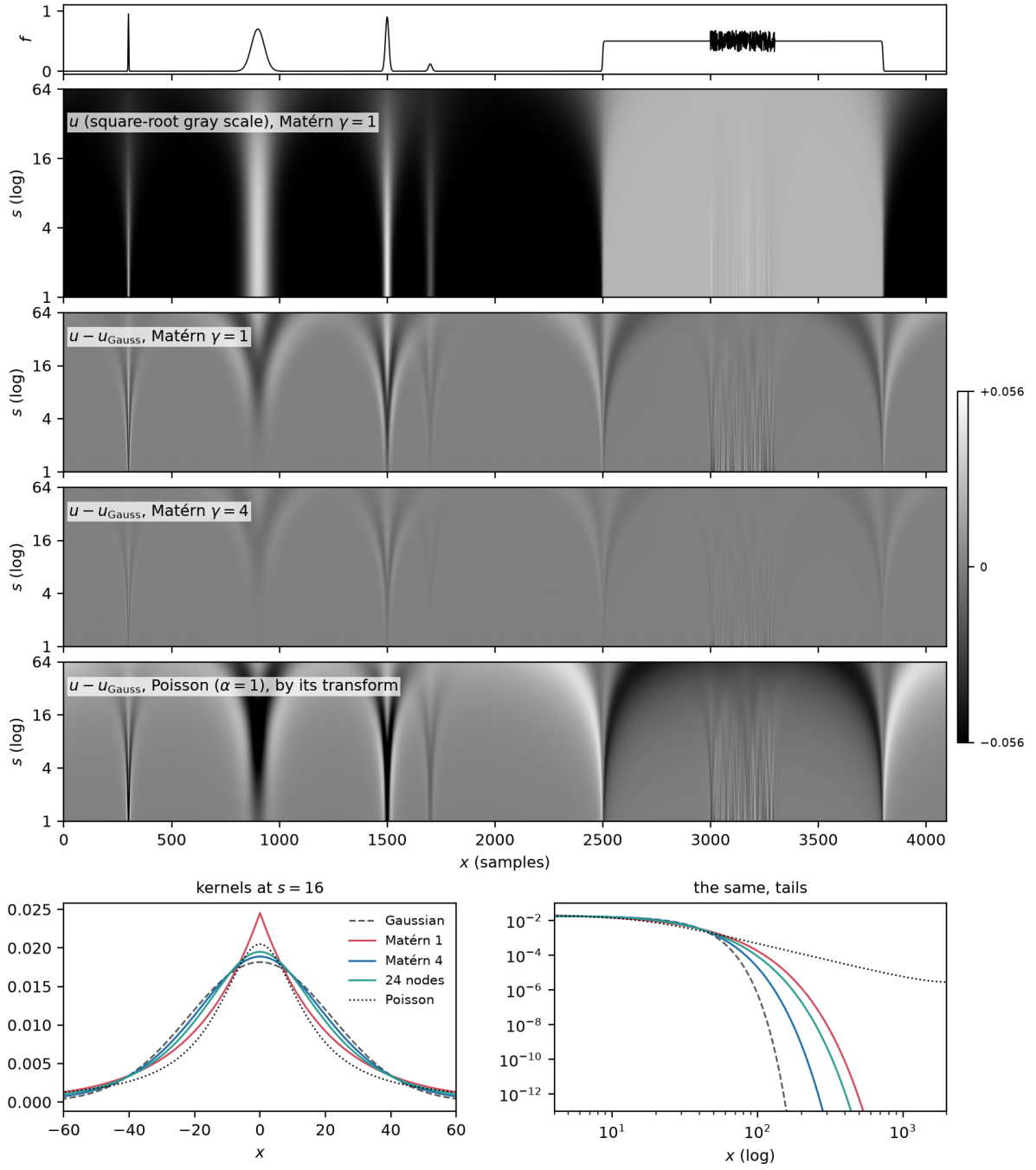


Figure 9: The run of Example 7.12. Top: the signal. Second panel: the scale space of the Matérn member $\gamma = 1$, computed by the cascade of lag sections over 48 knots, position horizontal and the display scale s vertical on a logarithmic axis, the signal entering at the bottom edge; its gray scale is the square root of the value, so that the low levels are visible in print. Next three panels: the difference from the Gaussian scale space at the same bandwidth, for the Matérn members $\gamma = 1$ and $\gamma = 4$ and for the Poisson scale space, on the one gray scale of the key at the right. The key saturates at ± 0.056 , half the largest difference, which occurs in the Poisson panel, so that panel is clipped around the fine event. The members are matched in bandwidth: at the display scale s the transfer function of each continuum member equals $1/e$ at $\omega = 1/s$. Bottom: the kernels at $s = 16$ on linear and on logarithmic axes.

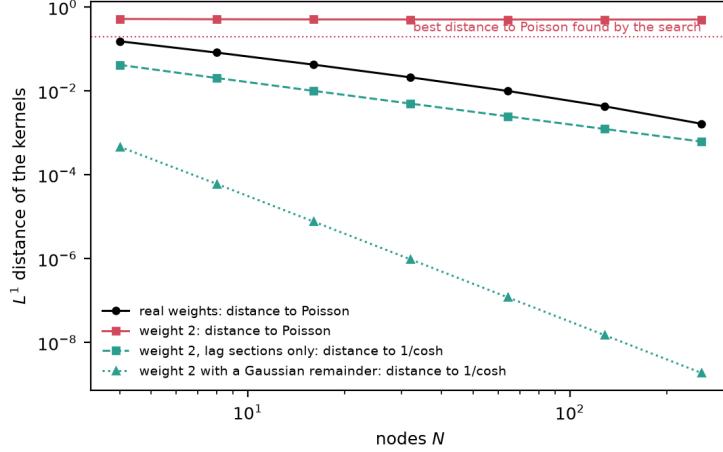


Figure 10: The Poisson member and its quadratures with N nodes (Example 7.12, check 5): L^1 distances between kernels at canonical scale 1. Black: real weights, which converge to the Poisson kernel. Red: weight 2 at the nodes $\pi(i - \frac{1}{2})$, which stay at distance 0.507 from it on the window of the computation (0.509 on the line). Teal: the distance of the same members from the member with transform $1/\cosh \omega$, for the members realized by lag sections alone (dashed) and with the part of the measure above the last node carried as a Gaussian coefficient (dotted). The red dotted line is the smallest distance from the Poisson kernel found by the search described in the text, which is an upper bound for the best distance and not a separation.

tail of a Matérn kernel of integer index is $x^{\gamma-1}e^{-x/t}$ up to a constant, so the rate is fitted to $\log h - (\gamma - 1) \log x$, a straight-line fit over $12 < x/t < 20$. Multiplied by t it is 1.000 for $\gamma = 1$ and 1.02 for $\gamma = 4$, against 1. The excess at $\gamma = 4$ is the bias of the window, the lower-order terms of the polynomial factor, and a fit of $\log h$ alone would lie below 1 on every window. For the member with 24 ranges, whose tail has no polynomial factor, the rate over $8 < x/t < 14$ is 1.5706 against its smallest node $\pi/2$. The Poisson kernel obtained by inverting its transform has the tail power -1.99 against -2 . The right panel of the bottom row shows the three regimes on logarithmic axes.

4. *The boundedness threshold* (Corollary 3.9). The kernel at the origin is the limit of $\pi^{-1} \int_0^\Lambda e^{-F(\omega)} d\omega$ as the truncation frequency Λ grows. Table 4 gives that truncated inversion integral at three values of Λ for six members at canonical scale 1, grouped by $c = k(0+)$; the members with two atoms have the Thorin measure $w_1\delta_1 + w_2\delta_3$, so their ranges are 1 and $\frac{1}{3}$ and their profile is $w_1e^{-x} + w_2e^{-3x}$ with $c = w_1 + w_2$. For $c < 1$ the

Table 4: The truncated inversion integral $\pi^{-1} \int_0^\Lambda e^{-F(\omega)} d\omega$ of check 4 of Example 7.12, for six members at canonical scale 1 grouped by the value $c = k(0+)$ of the profile at the origin. It diverges like a power of Λ for $c < 1$, like its logarithm at $c = 1$, and converges for $c > 1$ (Corollary 3.9).

member	c	$\Lambda = 10^2$	$\Lambda = 10^4$	$\Lambda = 10^6$
Matérn, $\gamma = 0.4$	0.8	2.64	8.68	23.87
two atoms, weights 0.3 and 0.5	0.8	4.17	14.64	40.94
Matérn, $\gamma = 0.5$	1	1.69	3.15	4.62
Cin ray	1	1.20	2.03	2.85
Matérn, $\gamma = 0.6$	1.2	1.17	1.55	1.70
two atoms, weights 0.6 and 0.6	1.2	1.84	2.58	2.87

integrals grow like a power of Λ . For $c = 1$ they grow by the same amount over each

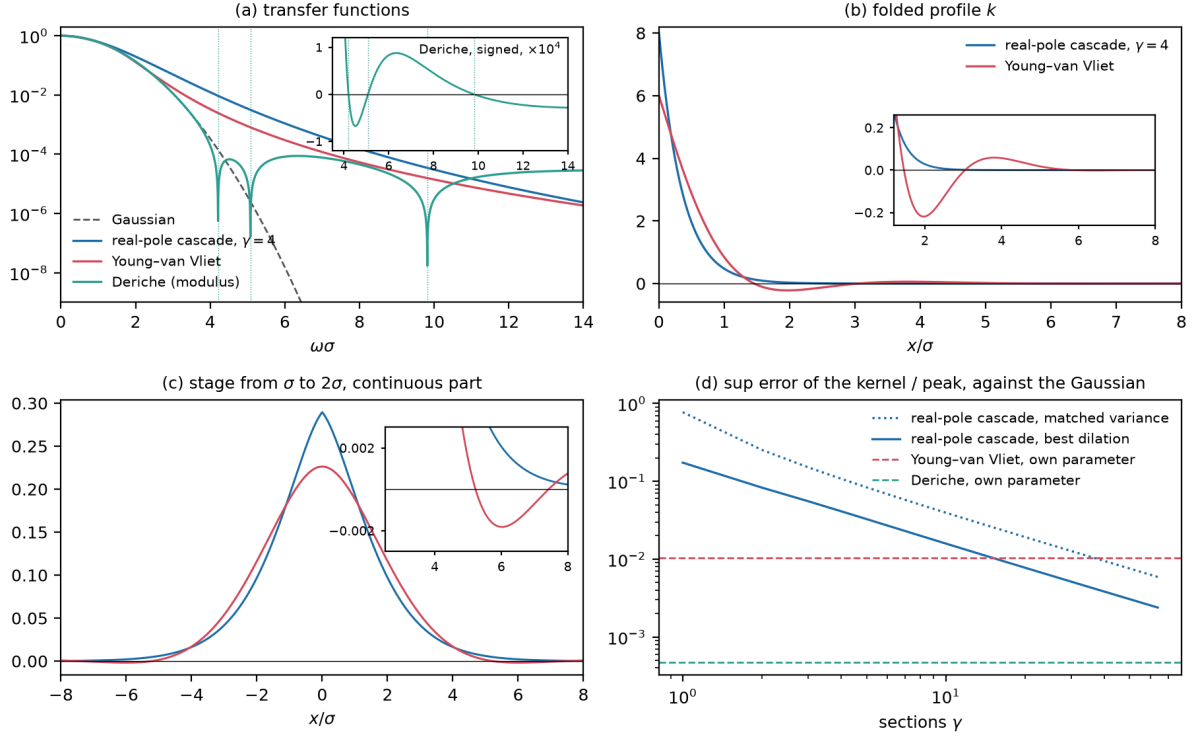


Figure 11: Two recursive approximations of the Gaussian and the real-pole cascade, at $\sigma = 1$ (Example 7.12); all three are the continuous prototypes, with the published coefficients. (a) The transfer functions, that of Deriche's design in modulus; the inset shows it with its sign, magnified, and the dotted lines mark its three zeros. (b) The folded profile of the cascade of four identical sections and that of the design of Young and van Vliet, which is negative from $x = 1.46\sigma$. (c) The continuous part of the stage from σ to 2σ for the same two, with the negative lobe of the second in the inset. (d) The largest error of the kernel against the Gaussian, relative to the peak: for the cascade of γ identical sections at matched variance (dotted) and at the dilation that minimizes it (solid), and for the two designs at their own scale parameter.

factor of 100, which is a logarithm. For $c > 1$ the growth falls from one step to the next, by factors of 2.5 in both rows, as it does when the integral converges with a remainder of order Λ^{1-c} . The members with two atoms show that the threshold is read from $k(0+)$ and not from the family. All six profiles are constant or smooth near the origin, so the check does not probe the slowly varying factor of Proposition 3.7.

5. *Finite Thorin approximations* (Propositions 7.7 and 7.8). The Thorin measure of the Poisson member is $(2/\pi)d\theta$, by Proposition 5.12 at $\alpha = 1$. The quadrature with real weights has N cells: the interval from 0 to $2/N$, and the $N-1$ intervals between N points spaced geometrically from $2/N$ to $2N$. Each cell carries its mass, $2/\pi$ times its width, at its midpoint, and the part of the measure above $2N$ is carried as the Gaussian coefficient $1/(2\pi N)$. The mass below $2/N$ cannot be dropped, since without it the exponent is short by a term of order $N^{-1}\log N$. The kernel of this member is at L^1 distance 0.154 from the Poisson kernel for $N = 4$ and 1.7×10^{-3} for $N = 256$, on the window. That is the convergence the proposition states. Rounding to weight 2 is a different matter. Cells of mass 2 have the width π , their midpoints are the nodes $\pi(i - \frac{1}{2})$, and by the product formula $\cosh \omega = \prod_{i \geq 1} (1 + \omega^2/(\pi(i - \frac{1}{2}))^2)$ the members with these nodes converge to the member with transform $1/\cosh \omega$, whose kernel is $\frac{1}{2} \operatorname{sech}(\pi x/2)$. The run confirms it, for two variants that must be kept apart (Figure 10). The member realized by N lag sections alone, the finite product, is at L^1 distance 4.2×10^{-2} from that kernel at $N = 4$ and 6.2×10^{-4} at $N = 256$, a convergence of first order in $1/N$. If the part of the measure above the last node is carried as the Gaussian coefficient $1/(\pi^2 N)$, the distances are 4.7×10^{-4} and 1.9×10^{-9} ; that member is admissible, but its Gaussian factor is not realized by lag sections. For both variants the distance to the Poisson kernel tends to the distance between the two limits, which is 0.5086 on the line and 0.5070 on the window. Moving the nodes helps only in part. A simplex search from the midpoint nodes, over members with up to three nodes of weight 2 and a free Gaussian coefficient, reached the distance 0.200 from the Poisson kernel, at a single Matérn stage convolved with a Gaussian, and the same search for the Student-t member $a = \frac{3}{2}$ reached 0.027. These two numbers are upper bounds for the best distance, and a finite search cannot show that they are attained or optimal. That the best distance is positive is Proposition 7.8: the class of limits is closed under weak convergence, convergence in L^1 implies it, and the Poisson member is not in the class. All distances are L^1 norms of differences of densities at canonical scale 1, computed by inverting the transforms on 2^{20} points over $[-400, 400]$.

The last check is the numerical side of Proposition 7.8. The quadratures with real weights converge to the given member, and their stages are not rational. The members with even integer weights have rational stages and an exact cascade of lag sections, and they are members in their own right. By the proposition they cannot converge to a member whose Thorin measure is spread out, and the run gives the distances, which the proposition does not. The member with transform $1/\cosh \omega$ is itself in the closure that the proposition describes, its Thorin measure being $\sum_i 2\delta_{\pi(i-1/2)}$. How close a member realized by lag sections comes to a given member depends on the given member, and it comes closer to the Student-t member $a = \frac{3}{2}$ than to the Poisson member. The member with transform $1/\cosh \omega$ is thus the member with an exact cascade that the rounding assigns to the Poisson scale space. Its tail is exponential at the rate $\pi/(2t)$, where that of the Poisson kernel is a power, and the bottom panels of Figure 9 show both.

The recursive Gaussians. Remark 7.4 gives a test for a forward-backward recursive filter: its transfer function is the kernel of an admissible family at one scale exactly when the folded profile read off from its poles and zeros is nonnegative and nonincreasing. Figure 11 applies the test to two published designs at $\sigma = 1$, with the coefficients as printed. For a transfer function $\prod_i (1 + \omega^2/\theta_i^2)^{-1}$ with poles θ_i in the right half-plane, the identity of Proposition 5.2(1) extends

to complex θ with positive real part, by the differentiation under the integral sign used for the pair $1 \pm i$ in Remark 7.6, and gives the profile $k(x) = 2 \sum_i e^{-\theta_i x}$. The script confirms it against $-\log R$ by quadrature to 3×10^{-11} .

The design of [65] has the poles 1.1668 and $1.10783 \pm 1.40586i$, in units of $1/\sigma$ (their Table 1, p. 141), and its two passes are run in cascade. Its profile is $2e^{-1.1668x} + 4e^{-1.10783x} \cos(1.40586x)$. The complex pair decays more slowly than the real pole, so the oscillating term wins: the profile is negative from $x = 1.46\sigma$, with the minimum -0.22 at $x = 1.97\sigma$. The transfer function is therefore the transform of no infinitely divisible law: the signed density is the only candidate, the Lévy pair of a law being unique, and it is not a positive measure. This is an analytic statement and not a reading of the plot: the real pole decays faster than the pair, so at every point where the cosine equals -1 the second term is the larger, the first of them being $x = 2.23\sigma$, where $k = -0.19$. The design is the case of a dominant complex pair in the remark. The consequence for a cascade is visible in the stage from σ to 2σ , whose transfer function is the ratio of the two. It has an atom of mass $1/64$ and a continuous part with a negative lobe, of depth $1.8 \times 10^{-3}/\sigma$ and mass 5.1×10^{-3} . A pyramid built by cascading this design is thus not positivity preserving from level to level, although every level taken alone is a smoothing of the signal by a kernel that is positive to within 1.1×10^{-5} .

The fourth-order design of [13], eq. (38), p. 12, fits a sum of two damped oscillations to the Gaussian on the half-line and extends it symmetrically, the causal and the anticausal half being added. Its transfer function is rational with the poles $1.783 \pm 0.6318i$ and $1.723 \pm 1.997i$ and a cubic numerator in ω^2 . On $x \geq 0$ the fitted function is $(a_0 \cos \omega_0 x + a_1 \sin \omega_0 x) e^{-b_0 x} + (c_0 \cos \omega_1 x + c_1 \sin \omega_1 x) e^{-b_1 x}$ with the amplitudes $a_0 = 1.68$, $a_1 = 3.735$, $c_0 = -0.6803$ and $c_1 = -0.2598$, printed here to the digits of the source, of which a_0 has two decimals. The fitted function, before its normalization to unit mass, has the one-sided slope $+0.018$ at the origin, so its transform behaves like $-0.035/\omega^2$ at high frequency and is negative there, while it is positive at the origin. The transfer function therefore vanishes on the real axis, at $\omega\sigma = 4.21$, 5.08 and 9.82 for the printed coefficients. The argument does not depend on their last digits: over the whole box of coefficient vectors that round to the printed ones, the slope $a_1\omega_0 - a_0b_0 + c_1\omega_1 - c_0b_1$ lies between $+0.0068$ and $+0.0286$, most of that width coming from the two decimals of a_0 , so at least one real zero persists throughout the box. In 2000 random draws from the box all three zeros persisted, which is an observation about sensitivity and proves nothing more. The transform of an infinitely divisible law has no zero (Lemma 4.2 of the line paper), so this design is not one either. No stable filter takes its output at σ to its output at 2σ : the ratio of the two transfer functions has poles on the real axis at the three zeros, none of which is cancelled, the zeros of the numerator being at half those frequencies.

Both exclusions are statements about the continuous prototypes with the published coefficients. The recursions that are run in practice are sampled versions of them, with transfer functions in z , and sampling changes both the zeros and the profile; nothing is claimed here about a particular sampled implementation, or about other designs published under the same names.

Later designs change one or the other ingredient of the design of Young and van Vliet: [61] give a closed-form design of the poles for every order, and [60] correct the transition between the forward and the backward pass at the boundary of a finite signal. Neither is analysed here. The criterion of Proposition 7.3 applies to the poles of any such design, and the boundary rule does not enter it.

Both designs are more accurate than the real-pole cascade at the same order, and the last panel gives the price of staying inside the class. A largest error against the unit Gaussian depends on the dilation at which a kernel is read, so each number carries its gauge. At its own scale parameter the largest error of the kernel, relative to the Gaussian's peak, is 0.05% for Deriche's design and 1.04% for that of Young and van Vliet. The prototype of the latter has the variance $1.18\sigma^2$: the design is calibrated by its error and not by its variance, its error at

matched variance is 8.5%, and at its best dilation it is 0.86%. For the cascade of γ identical sections the error at matched variance is 10.8% at $\gamma = 4$ and 1.2% at $\gamma = 32$ and decays like $0.38/\gamma$. At its best dilation, which is 9% wider at $\gamma = 4$ and 1% wider at $\gamma = 32$, it is 4.1% and 0.48% and decays like $0.15/\gamma$. Best dilation against best dilation, the cascade passes the design of Young and van Vliet at $\gamma = 19$. A user who needs the Gaussian at one scale is well served by the published designs. A user who needs a scale space, with exact stages between its levels and positivity at every stage, has the Matérn members, at the price of their exponential tails.

8 Concluding remarks

The line paper [17] characterized the spatial scale spaces on the line under the hemigroup axioms: their kernels are the symmetric self-decomposable laws, each given by a Gaussian coefficient and a nonincreasing displacement profile. This paper described that class, and its results are of five kinds.

The shape of the class. The admissible exponents form a cone. Its extreme rays are the Gaussian ray and the exponents $\text{Cin}(\tau \cdot)$ of the unit-step profiles, and every member is a unique superposition of them (Proposition 3.3). The kernel of a member with no Gaussian part is bounded at the origin exactly when the value of its profile there exceeds 1. The Cin rays sit at that threshold, and so does the Matérn member at $\gamma = \frac{1}{2}$ (Corollaries 3.9 and 3.10).

The relation to the causal theory. Bochner's subordination maps the causal theory of [18] into the spatial one, family to family and corner to corner (Lemma 4.2, Proposition 4.5). The map is injective (Proposition 4.6(1)), and on the Thorin subclasses it is a bijection (Proposition 5.7). It is not onto. It misses every member with no Gaussian part whose profile is a lattice step, the Cin rays among them (Proposition 4.6). Its image is the intersection of a chain of classes indexed by dimension, the exponents whose radial extension is self-decomposable in dimension d , and the first two inclusions of the chain are strict (Proposition 4.9, proved in prose on cited facts about laws in dimension d , and Lemma 4.12, not machine-checked).

The named members. The Matérn family is the single-atom member of the Thorin subclass with no Gaussian part. Its members of integer index are exactly the families realized at every ladder of scales by a cascade of identical first-order forward-backward sections, and every member has every moment finite and exponential tails (Theorem 5.4, Corollary 5.5). The Student-t family is the image of the causal Bessel corner, with polynomial tails of a prescribed order (Proposition 5.10). The symmetric stable family is the semigroup corner, and the heavy tails of its non-Gaussian members come from the semigroup requirement (Proposition 5.12).

The evolution. Every member satisfies an evolution equation in scale whose generator is a Laplacian plus a superposition of symmetric second differences (Proposition 6.2). The generators of the named members are a differential operator, a fractional one, a bounded integral operator, mixtures of such, and for the unit-step members a single second difference (Proposition 6.4).

The algorithm. The cascade is the algorithm. Over any finite ladder of scales the cascade is exact at every knot, for a signal on the line, the sampling of the signal being a separate approximation. At integer index the Matérn stages are forward-backward first-order recursive sections. The members of the Thorin subclass with finitely many ranges and integer weights have such stages, and every member of the subclass is a limit of members with finitely many ranges. The restriction to integer weights is a real one: the limits of the families with such stages are the members whose Thorin measure consists of atoms of even integer mass, finitely many in every bounded set, possibly with a Gaussian part. Every such limit has an exponential moment, so the stable and the Student-t members are not among them (Proposition 7.8). The numerical run of §7.6 finds them at a fixed distance from the Poisson scale space.

Scope. The paper delivers the geometry of the class, the bridge, the corners, the generator form and the implementation, on the line and in the L^1 signal model. It makes no empirical

claim about natural signals. Locality of the scale evolution and of the scale space, the non-creation and non-enhancement of local extrema, and the scale-space jet are the subject of the next module. Nothing here depends on them. Dimension d , and families that are covariant only under a discrete group of dilations, remain outside, as in the line paper.

Open questions. Five are left.

1. How large the complement of the slice is. The lattice-step members with no Gaussian part are outside it, and so is every member with a compactly supported catalogue. No description of the complement is given.
2. Which admissible families have rational stages. The families realized by first-order sections are among them, and so are a family with a complex pole pair and a family with real poles and zeros only (Remark 7.6), so the Thorin subclass does not contain them all. Proposition 7.3 says which rational functions are kernels from the origin. Which are stages between two positive scales, and how the others are realized with positivity at every stage, is not attempted.
3. The formal status of three of the clauses that §1.1 lists as not formalized. The closed form of the Matérn kernel, its tail, and the moment count of the Student-t family are cited and not machine-checked (Propositions 5.3 and 5.10(4)). A machine-checked closed form would let the boundedness threshold be read from the kernel as well as from the profile.
4. The boundedness threshold in dimension d . On the line it is the value 1 of the profile at the origin, and the condition $\gamma > \frac{1}{2}$ of the Matérn member is its instance. Whether the threshold in $\mathbb{R}^d$ is the dimension is a question for the theory in dimension d , which is not begun here. If it is, an admissible aperture in the plane may be unbounded at its own centre.
5. The hemigroup weakened once more. The dilatively stable processes of [32] and the semi-selfsimilar and semistable hemigroups of [7] describe what remains of covariance when the scaling group is weakened or discretized. How much of the present geometry survives there is open.

A The proof of Lemma 4.12

The lemma is stated in §4.6. The notation is that of Definition 4.11.

Proof. Throughout $m = (d + 1)/2$, and $P(s, x) = \Gamma(s)^{-1} \int_0^x v^{s-1} e^{-v} dv$ is the regularized lower incomplete gamma function.

The datum. $k_1^{(c)}$ is nonnegative and measurable, and

$$\int_0^1 k_1^{(c)}(u) du = 2, \quad \int_1^\infty k_1^{(c)}(u) \frac{du}{u} = 2 + c \log 2,$$

both finite. So $D^{(c)}$ is a delay datum and $F^{(c)} \in \mathcal{G}$. That $\mathcal{S} \subseteq \mathcal{G}$ is Definition 4.11 read against Definition 4.1: a causally admissible datum is a delay datum, its profile being nonincreasing and hence measurable.

The profile has no nonincreasing version. Choose $\delta \in (0, \frac{1}{2})$ with $(1 - \delta)^{-1/2} - (1 + \delta)^{-1/2} < c$, which is possible because the left side tends to 0 as δ decreases to 0. For $u_1 \in (1 - \delta, 1)$ and $u_2 \in (1, 1 + \delta)$,

$$k_1^{(c)}(u_1) = u_1^{-1/2} < (1 - \delta)^{-1/2} < (1 + \delta)^{-1/2} + c < u_2^{-1/2} + c = k_1^{(c)}(u_2),$$

while $u_1 < u_2$. If a nonincreasing g agreed with $k_1^{(c)}$ almost everywhere then, both intervals having positive measure, there would be such a pair with $g(u_i) = k_1^{(c)}(u_i)$, and $g(u_1) \geq g(u_2)$ would contradict the display. So $D^{(c)}$ is not a causally admissible datum.

What the bridge computation reads. Of the four steps in the proof of Lemma 4.2, only the second reads the monotonicity of the delay profile. The finiteness of $\nu_2(x) = \int_0^\infty g_u(x) k_1(u) du/u$ off the origin uses $g_u(x)/u \leq 16/x^4$ below $u = 1$ and $g_u \leq 1$ above it, against the two integrability conditions; the Lévy condition is a Tonelli exchange against $\int_{\mathbb{R}} (1 \wedge x^2) g_u \leq 1 \wedge u$ and Lemma 2.3; and the exponent identity is a second Tonelli exchange. All three read the two integrability conditions and nothing else, so they hold for a delay datum verbatim. Hence for any delay datum (b_0, k_1) the pair $(b_0/2, 2x\nu_2(x))$ satisfies every requirement (2.3) makes except possibly the monotonicity of the profile, and its exponent is $F_1(\omega^2/2)$; so the datum's exponent is admissible as soon as that profile is nonincreasing. The substitution $u = r^2/2v$ that produces (4.3) identifies the profile with k_1 , the polar profile (4.6) at $d = 1$, the folding factor 2 being $|S^0|$.

The polar profiles in closed form. Split (4.6) at the two summands of $k_1^{(c)}$. For the power, $k_1(r^2/2v) = (r^2/2v)^{-1/2}$ turns the integral into a gamma integral and gives $2^{3/2}r^{-1}\Gamma(m)/\Gamma(d/2)$. For the bump, $\mathbf{1}_{[1,2]}(r^2/2v) = \mathbf{1}_{[r^2/4, r^2/2]}(v)$, so that summand contributes $2c[P(d/2, r^2/2) - P(d/2, r^2/4)]$. Hence

$$k_d^{(c)}(r) = \frac{2^{3/2}\Gamma(m)}{\Gamma(d/2)} \frac{1}{r} + 2c \left[P\left(\frac{d}{2}, \frac{r^2}{2}\right) - P\left(\frac{d}{2}, \frac{r^2}{4}\right) \right], \quad r > 0, \quad (\text{A.1})$$

which is smooth on $(0, \infty)$; the bracket is $\text{erf}(r/\sqrt{2}) - \text{erf}(r/2)$ at $d = 1$ and $e^{-r^2/4} - e^{-r^2/2}$ at $d = 2$. Since $\partial_x P(s, x) = x^{s-1}e^{-x}/\Gamma(s)$,

$$\frac{\partial}{\partial r} P\left(\frac{d}{2}, \frac{r^2}{2a}\right) = \frac{2}{r\Gamma(d/2)} \left(\frac{r^2}{2a}\right)^{d/2} e^{-r^2/2a},$$

so multiplying the derivative of (A.1) by $r^2\Gamma(d/2)$ and writing $z = r^2/2$ shows that it is non-positive at r if and only if

$$cz^m(e^{-z} - 2^{-d/2}e^{-z/2}) \leq \frac{1}{2}\Gamma(m), \quad z = \frac{1}{2}r^2. \quad (\text{A.2})$$

Two elementary bounds. First, $\log(1/p) \leq (1-p)/\sqrt{p}$ for $0 < p \leq 1$: with $x = 1/p \geq 1$ this reads $\log x \leq (x-1)/\sqrt{x}$, an equality at $x = 1$ where the right side has the larger derivative from then on, since $1/x \leq (x+1)/(2x^{3/2})$ is $2\sqrt{x} \leq x+1$. Second, for $\lambda \in (0, 1)$ and $m > 0$ the function $(1-p)^m(p-\lambda)$ on $[\lambda, 1]$ is maximal at $p = (m\lambda+1)/(m+1)$, with value $m^m(1-\lambda)^{m+1}/(m+1)^{m+1}$.

Where the polar profile is nonincreasing. Put $p = e^{-z/2}$ and $\lambda = 2^{-d/2}$, so that $e^{-z} - \lambda e^{-z/2} = p(p-\lambda)$, which is positive only for $p > \lambda$, and there $z = 2\log(1/p) \leq 2(1-p)/\sqrt{p}$. So for $m \leq 2$, using $p^{1-m/2} \leq 1$ and then the second bound,

$$z^m(e^{-z} - \lambda e^{-z/2}) \leq 2^m(1-p)^m p^{1-m/2}(p-\lambda) \leq 2^m(1-p)^m(p-\lambda) \leq \frac{(2m)^m(1-\lambda)^{m+1}}{(m+1)^{m+1}}.$$

At $d = 1$, where $m = 1$, $\lambda = 2^{-1/2}$ and $\Gamma(m) = 1$, the right side is $(1 - 2^{-1/2})^2/2 = (\frac{3}{2} - \sqrt{2})/2$, so (A.2) holds at every z as soon as $c(\frac{3}{2} - \sqrt{2}) \leq 1$, that is as soon as $c \leq 6 + 4\sqrt{2}$, the two numbers being reciprocal: $(\frac{3}{2} - \sqrt{2})(6 + 4\sqrt{2}) = 9 - 8 = 1$. At $d = 2$, where $m = \frac{3}{2}$, $\lambda = \frac{1}{2}$ and $\Gamma(m) = \sqrt{\pi}/2$, the right side is $3^{3/2}5^{-5/2} = (3/5)^{3/2}/5$, so at $c = 4$ the criterion holds at every z as soon as $4 \cdot 3^{3/2}5^{-5/2} \leq \sqrt{\pi}/4$, that is $(16/5)(3/5)^{3/2} \leq \sqrt{\pi}$, that is, after squaring, $6912/3125 \leq \pi$, which holds, the left side being less than 2.22. Since 4 and 8 are both below $6 + 4\sqrt{2}$, this gives that $k_1^{(c)}$ is nonincreasing for $0 < c \leq 6 + 4\sqrt{2}$ and that $k_2^{(4)}$ is nonincreasing.

Where it is not. If (A.2) fails at one $z_0 > 0$ then $k_d^{(c)}$ has a positive derivative at $r_0 = \sqrt{2z_0}$, so it is strictly increasing near r_0 ; being continuous, it agrees with no nonincreasing function

almost everywhere either. At $d = 2$ and $c = 8$ take $z_0 = \frac{1}{2}$. Then $e^{-z_0} > 3/5$ and $e^{-z_0/2} < 4/5$ — the first is $e < 25/9$, the second $e > (5/4)^4 = 625/256$ — so $e^{-z_0} - \frac{1}{2}e^{-z_0/2} > 3/5 - 2/5 = 1/5$; and $z_0^{3/2} = 2^{-3/2} > 7/20$, since $400 > 392$. The left side of (A.2) therefore exceeds $8 \cdot \frac{7}{20} \cdot \frac{1}{5} = \frac{14}{25}$, while its right side $\sqrt{\pi}/4$ is below 0.45 because $\pi < 3.24$. At $d = 3$ and $c = 4$ take $z_0 = 1$, where $\Gamma(m) = \Gamma(2) = 1$. Then $e^{-1} > 9/25$, again because $e < 25/9$, while $2^{-3/2} < 9/25$ because $625 < 648$ and $e^{-1/2} < 61/100$ because $e > (100/61)^2$. So

$$e^{-1} - 2^{-3/2}e^{-1/2} > \frac{9}{25} - \frac{9}{25} \cdot \frac{61}{100} = \frac{351}{2500},$$

and the left side of (A.2) exceeds $4 \cdot 351/2500 > \frac{1}{2}$. So $k_2^{(8)}$ and $k_3^{(4)}$ are not nonincreasing, which completes clause (2).

No causal ancestor. Suppose $F^{(c)} \in \mathcal{S}$. By Proposition 4.6(1) the function $\sigma \mapsto F^{(c)}(\sqrt{2}\sigma) = F_1^{(c)}(\sigma)$ is causally admissible, so there are $b'_0 \geq 0$ and a nonincreasing k' satisfying the two integrability conditions whose exponent is $F_1^{(c)}$ on $[0, \infty)$. The differentiation step in the proof of Proposition 4.6(2) dominates the σ -derivative of the integrand by $e^{-\sigma u/2}k_1(u)$ and reads the two integrability conditions and nothing else, so it applies to both data and gives

$$b'_0 + \int_0^\infty e^{-\sigma u} k'(u) du = \int_0^\infty e^{-\sigma u} k_1^{(c)}(u) du, \quad \sigma > 0.$$

Both integrals are finite and tend to 0 as σ grows: for $\sigma \geq 1$ the integrand is at most k_1 on $(0, 1)$ and at most $k_1(u)u^{-1}$ on $(1, \infty)$, since $e^{-u} \leq u^{-1}$ there, and it tends to 0 at every $u > 0$, so dominated convergence applies. Hence $b'_0 = 0$, and the measures $k'(u) du$ and $k_1^{(c)}(u) du$ on $(0, \infty)$ have equal finite Laplace transforms on $(0, \infty)$. By the uniqueness of the Laplace transform of a locally finite measure on the half-line, proved in the line paper for the Thorin representation (Proposition 2.9), they are equal, so k' is a nonincreasing version of $k_1^{(c)}$, which the second step excludes. So $F^{(c)} \notin \mathcal{S}$ for every $c > 0$; for $0 < c \leq 6 + 4\sqrt{2}$ the fifth step makes $F^{(c)}$ admissible with Gaussian coefficient 0 and profile $k_1^{(c)}$, and clause (1) follows.

The polar profile is the polar profile. Clause (3) reads clause (2) through the polar criterion, and the reading needs two steps. The first is that $k_d^{(c)}$ is the polar profile of a law on $\mathbb{R}^d$ with radial exponent $\xi \mapsto F^{(c)}(|\xi|)$, which is the computation of Proposition 4.9(2) carried out at a datum that need not be monotone. That computation builds $\nu_d(dx) = (\int_0^\infty g_u^{(d)}(x) k_1(u) du/u) dx$ and establishes, in this order: that ν_d is finite off the origin and satisfies the Lévy condition, by a Tonelli exchange against $\int_{\mathbb{R}^d} (1 \wedge |x|^2) g_u^{(d)} \leq 1 \wedge du$; that $F_1(|\xi|^2/2) = \frac{1}{2}\xi \cdot (b_0 I)\xi + \int (1 - \cos \xi \cdot x) \nu_d(dx)$, by a second Tonelli exchange; that this is therefore the exponent of an infinitely divisible law, isotropic because its transform is, whose Lévy measure is ν_d ; and that the polar profile of ν_d , namely $|S^{d-1}|r^d f_d(r)$ with f_d the radial density, is (4.5), by the substitution $u = r^2/2v$. Each of those steps reads the two integrability conditions of the datum and nothing else. Monotonicity of k_1 enters only in the last line of that proof, where the polar profile is concluded to be nonincreasing. So the computation applies verbatim to $D^{(c)}$ and gives an isotropic infinitely divisible law $\mu_d^{(c)}$ on $\mathbb{R}^d$ with radial exponent $F^{(c)}(|\xi|)$ and Lévy measure of polar profile $k_d^{(c)}$, (4.6) being (4.5).

A rotation-invariant Lévy measure of a self-decomposable law has a nonincreasing radial profile. This is the second step, and it is the polar criterion averaged over the sphere. Let μ be self-decomposable on $\mathbb{R}^d$ with Lévy measure ν . The criterion gives a finite measure λ on the unit sphere and, for λ -almost every direction ξ , a nonincreasing $k_\xi : (0, \infty) \rightarrow [0, \infty)$ with $\nu(B) = \int_{S^{d-1}} \lambda(d\xi) \int_0^\infty \mathbf{1}_B(r\xi) k_\xi(r) dr/r$. Take B of the form $\{x : |x| \in A\}$ with $A \subseteq (0, \infty)$ Borel and put $K(r) := \int_{S^{d-1}} k_\xi(r) \lambda(d\xi)$, a function with values in $[0, \infty]$ that is nonincreasing, each integrand being so. Then $\nu\{|x| \in A\} = \int_A K(r) dr/r$ for every such A ; and if ν is rotation invariant with radial density f , the same quantity is $|S^{d-1}| \int_A f(r) r^{d-1} dr = \int_A |S^{d-1}| r^d f(r) dr/r$. Two

measures on $(0, \infty)$ agreeing on every Borel set are equal, so $|S^{d-1}|r^d f(r) = K(r)$ for almost every r : the polar profile agrees almost everywhere with a nonincreasing function.

Clause (3). Both $F^{(4)}$ and $F^{(8)}$ are admissible, 4 and 8 being below $6 + 4\sqrt{2}$, so both lie in $\mathcal{A}_1$, which is the whole admissible cone (Proposition 4.9(1)). For $F^{(4)} \in \mathcal{A}_2$: the first step gives an infinitely divisible law on $\mathbb{R}^2$ with radial exponent $F^{(4)}(|\xi|)$ whose Lévy measure has polar profile $k_2^{(4)}$, which clause (2) says is nonincreasing, so the law is self-decomposable by the polar criterion. For the two exclusions, suppose $\xi \mapsto F^{(c)}(|\xi|)$ were the exponent of a self-decomposable law μ on $\mathbb{R}^d$, with $(c, d) = (8, 2)$ or $(4, 3)$. An exponent determines the law, so μ is the $\mu_d^{(c)}$ of the first step, whose Lévy measure is rotation invariant with radial density $f_d^{(c)}$ and polar profile $k_d^{(c)}$. By the second step $k_d^{(c)}$ agrees almost everywhere with a nonincreasing function; but it is continuous and strictly increasing on an interval, by the paragraph *where it is not* above, so it agrees with no nonincreasing function almost everywhere — given $r < s$, points $u_n \downarrow r$ and $v_n \uparrow s$ of the conull agreement set give $k_d^{(c)}(u_n) \geq k_d^{(c)}(v_n)$, and continuity would force $k_d^{(c)}(r) \geq k_d^{(c)}(s)$ at every such pair. Hence $F^{(8)} \notin \mathcal{A}_2$ and $F^{(4)} \notin \mathcal{A}_3$. The last sentence of the clause is the nesting $\mathcal{A}_\infty \subseteq \mathcal{A}_3$ of Proposition 4.9(1). $\square$

Author contribution and use of AI

This article was written in close collaboration with generative AI, and the collaboration went beyond editing assistance at every stage. I write this section in the first person because it is my own declaration; the article itself is written in the neutral third person.

Roles. The research programme and the question are mine, as for the line paper [17]. This article is the second of three modules cut from one working draft, and its mathematics and its formal development were produced in the same campaign as the line paper’s, so that paper’s statement and process note cover how they came about. The decisions of scope, method and policy for this module are mine: the cut between this module and the next along the dependency graph of the statements; the admission of the three interface names for the behaviour of a kernel at the origin; that the two constructions behind the dimension filtration be stated and proved and not only remarked; and the form of the article, which I set in a review of every section. The article restates what it uses of the line paper, so that it can be read alone. Its technical sections open with a map. A term is introduced where it is first used, with its literature. The account of the machine checking is in §1.1, and the body says of a step that it is machine-checked only inside a few proofs. The numerical run of §7.6 was my request.

The prose, the code, the figures and the numerical experiments were written by AI agents throughout, in dialogue with me; I worked from the conversation and from the built article rather than from the editor. The Lean 4 development is theirs essentially in full, and none of the proof terms are mine. So is mathematical content. The admissible cone with its extreme rays, the bridge from the causal theory and the Matérn family as the corner with exact recursive stages were all named in an agent’s reply to my question on the day I asked it, before any plan existed. The boundedness threshold at the origin and the dimension filtration were proposed by an agent reading the released line paper in the role of a postdoctoral colleague. The design of the numerical run is an agent’s. During the writing the agents also found and repaired errors in the statements of record: two while transcribing the statements into the article, and two in a blind review, a hypothesis that a restating clause had dropped and a uniqueness claim in the Matérn corollary made against too wide a class of realizations. The numerical run then showed that the text claimed an approximation property for the families with rational stages which the proposition behind it does not state and which fails. Corrections that make the article claim less are delegated to the agents by a standing rule of the project, each one recorded and reported to me, and I keep the veto. The technical decisions of the formalization were again the agents’

recommendations, which I read with their consequences and followed.

Verification and review. What is machine-checked, and the axioms on which the verified development rests, are set out in §1.1. Review by agents that had no part in the writing was a working method, as for the line paper: every section was read blind against the writing standard, one reviewer to a section, and each reviewer restated the claims from the statements and compared them with the prose, which is how the two errors above were found. The manuscript was then read by an AI system of another vendor, OpenAI's GPT-6 Astra, in a referee-style review that returned a verdict of major revision and in a presentation review. Every finding was verified against the text, the sources and the formal development before it was acted on, and the substantive ones held. The review found what the writing and the internal reviews had missed: a cited fact on the density at the origin that had been admitted in a false form, because the record of the source had paraphrased its formula; a hypothesis missing from the printed statement of the Matérn theorem; the realization corollary stated without its restriction to integer index; a converse the text expected and that is false; and errors of the numerical example reported under the wrong description. It also proved a statement the paper had listed as open, which is now clause (4) of Proposition 4.9; the argument is that system's. A second round followed the revision. No other vendor's system could be used for it, so it was run as five referees that were fresh instances of the models that had worked on the article (Claude Opus 5 and Claude Fable 5.1). Each was given the PDF alone and was forbidden the repository and the first review, three read the proofs section by section, with the cited theorems read in the sources, one reproduced the numerical example from its printed description with code of its own, and one compared the headline claims with the statements and searched the literature. This round is less independent than the first, since its readers share the blind spots of the writers, and the false cited fact of the first round was found by the other vendor's system. It found no false numbered statement. It found an imprecise domain in one statement, summaries that again claimed more than their statements, a dozen steps that were true and not on the page, two wrong descriptions in the numerical example and a comparison made in mixed scales, and prior work that was not cited, the Pólya frequency functions and Lindeberg's time-causal limit kernel among it. Several statements were added or widened in response, all proved in prose. The reviews of both rounds are archived verbatim in the repository beside the plans that record the disposition of each finding. The statements that are not machine-checked, which §1.1 lists and which include the whole of §7, were written by the agents and have been read by me and by these reviewers, and by no other mathematician. No human reader beyond me has yet reviewed the article.

The record. The sessions were archived as the work went on, so the division of labour can be stated rather than estimated. It is stated, with its dates and its caveats, in the repository's process note (`notes/PROCESS.md`), together with the chronology and the places where the record qualifies a simple account of who did what. The short form: the work specific to this article took six days on top of the line paper's campaign, and in it the agents' words outnumber mine by about twenty-four to one. Those figures are lower bounds. I would not, unaided and as a spare-time activity, have written this article.

Responsibility. I directed the work throughout and have satisfied myself of the mathematical content. The cited results were read at their page anchors from held copies by an agent acting as librarian. Responsibility for this article, including for its errors, is mine alone; the validity of the machine-checked results depends on the Lean kernel and not on how, or by whom, the proof terms were produced.

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